

Central Force Motion: The Kepler Problem

The orbital motion of a planet around the Sun was one of the first important problems to be analyzed in terms of Newton's laws of motion. The gravitational force attracting a planet to the Sun is a *central force*, that is, a force directed towards a fixed point. The motion of a planet is a prime example of the more general problem of the behavior of a particle acted upon by a central force.

Although we shall be primarily concerned with the motion of planets and satellites, the techniques you will learn in this chapter are applicable to any kind of central force. In this chapter, as well as learning the laws governing the motion of celestial bodies, you will be exposed to the concept of *effective potential energy*, and how constants of the motion are used in solving physics problems.

Historically, the quantitative analysis of orbital motion began with Kepler's realization that the motion of planets can be described by three empirical laws. An important point made in this chapter is that Kepler's laws of planetary motion can be derived theoretically from Newton's laws of motion. Additionally, Newton's laws give us a much deeper understanding of Kepler's laws. This application of Newton's ideas amazed and fascinated the "natural philosophers" of his era and was one of the most important events in the history of science.¹

10.1 Johannes Kepler (Historical Note)

Johannes Kepler lived from 1571 to 1630. He was born into a poor family in a small town in Germany. His father was a professional soldier who spent much of his time away from home and his mother was a quarrelsome woman who was accused of witchcraft in her old age. It would be fair to assume that Kepler had an unhappy childhood. Nevertheless, he was very intelligent and an excellent student. The prince who ruled the region sent him to study at a Lutheran seminary. Eventually, Kepler graduated from the University of Tübingen. Originally he planned to study divinity, but before he was ordained, the authorities in the seminary convinced him that he was not cut out for the clergy. With the help of his advisers, he was appointed to a position teaching mathematics in Graz, Austria. He was a quiet, introspective person and not greatly interested in teaching; in fact, he was probably a terrible teacher for he had the reputation of interrupting his own lectures to silently mull over some idea that had just occurred to him.

In Graz, Kepler had an idea that changed the course of his life. This idea, which became an obsession with him, was (he thought) a glimpse into the mind of God: a vision of the basic

¹ Much of the material in this chapter is an introduction to celestial mechanics. Many excellent celestial mechanics books are available, including A. E. Roy, *Orbital Motion*, Adam Hilger Press, Bristol, 1988, and J. M. A. Danby, *Fundamentals of Celestial Mechanics*, William-Bell Inc., Richmond, VA, 1992. A more recent book that is particularly well written is *Solar System Dynamics* by C. D. Murray and S. F. Dermott, Cambridge University Press, Cambridge, 1999.

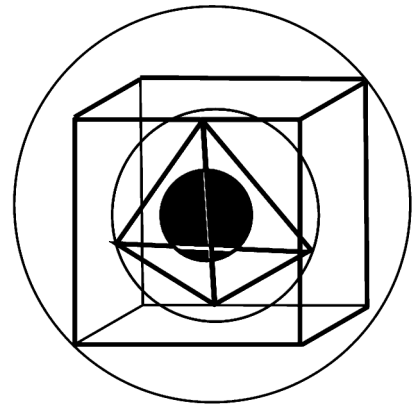


Figure 10.1 A nested tetrahedron and cube with inscribed and circumscribed spheres. In Kepler’s model the nested perfect solids (from innermost outwards) were an octahedron, an icosahedron, a dodecahedron, a tetrahedron, and a cube.

structure of the universe. It was, so to speak, revealed to him why there are only five planets, and why they are in their particular orbits around the Sun. Kepler’s mind-boggling inspiration was this: There are only five planets (besides Earth) because there are only five “perfect” solids, and the orbits of the planets correspond to spheres inscribed and circumscribed about the perfect solids when they are nested, one inside the other.

The “perfect” solids (or “simple polyhedrons”) are the geometrical figures formed from regular polygons. A regular polygon is one with equal sides. Thus, for example, a tetrahedron (or equilateral pyramid) is made up of four equilateral triangles. A cube is made up of six squares. Similarly, an octahedron is an eight-sided solid made of equilateral triangles, an icosahedron is made of twenty equilateral triangles, and the dodecahedron is composed of twelve pentagons. Many other solids can be constructed from polygons; for example, a “soccer ball” shape can be made up of pentagons and hexagons. There are, however, only *five* “perfect” solids whose faces are a *single* type of regular polygon. There is a very neat proof that there can *only* be five such solids; this proof can be traced back to the ancient Greeks. If you are interested, it is reproduced in the book *Cosmos*, by Carl Sagan.²

Kepler’s idea was that if the perfect solids were placed one inside the other, and each was circumscribed with a sphere, then the Sun would be at the center of the system and each of the five planets would orbit the Sun in a circular orbit whose radius would be equal to the radius of the corresponding circumscribed sphere. Kepler built models of these nested solids and their circumscribed spheres, but he could not prove his theory because he did not have enough information on the distances of the five (known) planets to the Sun. [Figure 10.1](#) illustrates the idea behind Kepler’s model.

At that time, the best astronomical data in Europe were in the observatory of an eccentric Danish astronomer named Tycho Brahe.³ Kepler went to visit Brahe to get his data. He was certain that Brahe would be overwhelmed by his wonderful new theory. Brahe was not overwhelmed. In fact, at first Brahe would not even let Kepler see the data! However, Brahe soon realized that Kepler was an excellent mathematician and he offered to let him work on a small portion of the data to calculate the orbit of Mars. This was certainly not what Kepler had in mind, but he grudgingly agreed.

² Carl Sagan, *Cosmos*, Random House, New York, 1980 (Appendix 2).

³ Tycho Brahe (1546–1601) set up an observatory at Uraniborg on the Danish island of Hven. A stream of young assistants from all over Europe came to Uraniborg where they carried out experiments in chemistry during the day and observed the heavens at night. Uraniborg has been described as the first research institution involved in “big science.” For the full story read the book by John Robert Christianson, *On Tycho’s Island: Tycho Brahe and his Assistants, 1570–1601*, Cambridge University Press, Cambridge, 2000. At the time Kepler went to visit him, Brahe had moved to Prague.

This was, perhaps, the original “graduate student–professor” relationship in science. To this day the same pattern exists. A young, aspiring scientist makes the pilgrimage to the laboratory of the established professor with the hope of being allowed to share in the professor’s knowledge and data. You yourself may be doing this a few years from now. I hope your relationship with your graduate advisor will be less tempestuous than that of Kepler and Tycho. They did not get along at all. Tycho was a man who loved a party, who spent his evenings eating, drinking, and carousing, whereas Kepler was somber and rather puritanical and did not at all approve of Tycho’s lifestyle.

Some years after the collaboration began, Tycho died. (Rumor has it that he died of a burst bladder while on a drinking spree.) Kepler inherited all of Tycho’s data as well as Tycho’s position as court astronomer. After a great deal of analysis, much to his dismay, Kepler found that Tycho’s data did not support his grandiose theory. In fact, the data showed that the orbits of planets were not even circles, but rather ellipses! Kepler tried to modify his theory by slipping elliptical orbits between the inscribed perfect solids, but it did not quite work. Kepler never did know that there are more than five planets, as the discoveries of the planets Uranus and Neptune came many years after his death. By the time these were discovered, Kepler’s theory on the inscribed orbits of the planets was no more than a historical oddity.

Kepler’s life is full of instructive incidents for physicists. His most valuable contribution to science was his analysis and synthesis of the observations of Tycho Brahe. It is interesting to consider that he was obsessed by a beautiful theory that did not agree with experiment. No matter how beautiful a theory may be, if it does not agree with experimental measurements, it must be discarded! Physics is the study of the physical universe and it is Nature that determines the way things behave. It is to Kepler’s credit that he respected and believed the data, even though the data did not agree with his theory.

10.2 Kepler’s Laws

Kepler’s synthesis of Tycho’s planetary data can be expressed as three statements that are now called Kepler’s laws of planetary motion. They are:

- (1) The orbit of a planet is an ellipse with the Sun at one focus.
- (2) The radius vector of a planet (the Sun–planet line) sweeps out equal areas in equal times.
- (3) The period of a planet squared is proportional to the cube of the semimajor axis of its elliptical orbit.

These laws express three facts about the motion of a planet. They are “empirical laws,” that is, they were obtained from the data but they had no theoretical basis. Kepler’s laws describe the motion of planets but neither Kepler nor anyone else living at that time could give a reasonable explanation for this behavior. About 70 years later, Newton applied the law of universal gravitation and his laws of motion to the problem and succeeded brilliantly in showing that Kepler’s laws are a consequence of some very basic physical relations. It is no wonder that other scientists of his era were in such awe of him. In this chapter, I will show you what Newton did, but of course I will use modern methods. Near the end of the chapter I will return to Kepler’s laws: by that time you should have a deeper appreciation for them.

10.3 Central Forces

Any force that is directed toward or away from a fixed point (often taken as the origin of coordinates) and whose magnitude is a function *only* of the distance to the fixed point is called a *central force*. In general, a central force will have the form

$$\mathbf{F} = f(r)\hat{\mathbf{r}},$$

where $f(r)$ is the magnitude of the force (a function only of r) and $\hat{\mathbf{r}}$ is the radial unit vector. The direction of the force is along the line joining the particle and the origin. Two important central forces are the gravitational force between masses and the electrostatic force between charges. Assuming one of the bodies is fixed at the origin, the gravitational force is a central force given by

$$\mathbf{F} = -\frac{Gm_1m_2}{r^2}\hat{\mathbf{r}},$$

and the electrostatic force is

$$\mathbf{F} = \frac{Q_1Q_2}{4\pi\epsilon_0r^2}\hat{\mathbf{r}}.$$

We can also *imagine* other central forces such as, for example,

$$\mathbf{F} = \frac{k}{r^5}\hat{\mathbf{r}}.$$

Such forces may or may not exist in nature, but we can analyze them mathematically anyway. Although this seems like a useless theoretical exercise, such studies may have a practical outcome. For example, to a first approximation, the force between molecules can be expressed as the sum of two central forces. This “Lennard-Jones” force can be expressed in terms of the potential energy as

$$V(r) = -\frac{a}{r^6} + \frac{b}{r^{12}}.$$

Here a and b are constants that depend on the properties of the particular molecules involved in the interaction. The force itself is obtained from $\mathbf{F} = -\nabla V$.

We are interested in determining the motion of a particle subjected to the gravitational force of another body (which we also treat as if it were a particle). For the sake of simplicity let us assume the more massive body is at rest. Strictly speaking, this cannot be true; the two objects actually orbit about their center of mass. But if one particle is much more massive than the other, then the more massive particle has a much smaller acceleration and in the limit, as the ratio of the masses goes to infinity, the more massive particle can be considered to be at rest.

To appreciate this, imagine a binary star system in which both stars have the same mass. The two stars will orbit around their center of mass, which is at a point halfway between them. But suppose one of the stars is more massive than the other. The center of mass will be closer to the more massive star. If one star is infinitely more massive than the other, the center of mass will be at the center of the larger star. Similarly, the International Space Station and the Earth are orbiting around their center of mass, but for all intents and purposes, the center of mass of this system lies at the center of the Earth and we are perfectly justified in considering the Earth to be at rest.⁴

In considering the motion of a particle of mass m in a central force field, the first thing to note is that *the particle moves in a plane*. The proof of this statement is given by the following argument. At any instant in time the particle is in the plane defined by the position vector and the velocity vector. The only way the particle can move out of this plane is if its acceleration has a component perpendicular to the plane. But for a central force, the force (and hence the acceleration) lies along $\mathbf{r}$, and therefore there is no component of the acceleration perpendicular to the plane.

⁴ Recall the discussion of reduced mass, defined as $\mu = m_1m_2/(m_1 + m_2)$. See Equation (6.20). If the central force problem is treated in terms of the reduced mass, then no approximation is made. The relations obtained are the same as those derived here except that m is replaced by μ and $\mathbf{r}$ is the position of one body relative to the other rather than the distance from the center of mass.

Perhaps the most important aspect of the motion of a particle under the action of a central force is that the moving particle *has a constant angular momentum*. By definition, the angular momentum of a particle is $\mathbf{l} = \mathbf{r} \times \mathbf{p}$, where $\mathbf{r}$ is the position of the particle and $\mathbf{p}$ is its linear momentum. Recall from [Section 7.2.2](#) that the time derivative of the angular momentum is

$$\frac{d\mathbf{l}}{dt} = \frac{d}{dt} (\mathbf{r} \times \mathbf{p}) = \mathbf{r} \times \mathbf{F}.$$

In the case of a central force, $\mathbf{F} = f(r)\hat{\mathbf{r}}$. Hence,

$$\begin{aligned} \frac{d\mathbf{l}}{dt} &= \mathbf{r} \times f(r)\hat{\mathbf{r}} = r\hat{\mathbf{r}} \times f(r)\hat{\mathbf{r}} \\ &= rf(r)(\hat{\mathbf{r}} \times \hat{\mathbf{r}}) = 0. \end{aligned}$$

Consequently, for a central force, the time derivative of the angular momentum is zero. This means that the angular momentum is constant. Also recall that the time rate of change of angular momentum is equal to the torque. A central force does not exert a torque on a particle.

These simple physical arguments lead us to an important conservation law concerning the motion of *any* particle under the action of *any* central force:

The angular momentum is constant.

This means that both the *magnitude* and the *direction* of the angular momentum are constant.

For a more sophisticated proof of the constancy of angular momentum, you can write the Lagrangian for a particle in a central force field. In polar coordinates the kinetic energy is $T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2)$ and the potential energy is $V = -\int_{r_0}^r f(r)dr$. The Lagrangian is

$$L = T - V = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 + \int_{r_0}^r f(r)dr.$$

Since θ is ignorable, $\partial L / \partial \dot{\theta} = \text{constant}$. That is,

$$\frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} = l = \text{constant}.$$

We can use the constancy of angular momentum to generate another proof that a particle in a central force field moves in a plane. Assume the particle is at position $\mathbf{r}$ and has velocity $\mathbf{v}$. These two vectors define a plane and, of course, the particle lies in this plane. By the definition of the cross product, the vector $\mathbf{r} \times \mathbf{v}$ is perpendicular to the plane containing the vectors $\mathbf{r}$ and $\mathbf{v}$. But $\mathbf{l} = m\mathbf{r} \times \mathbf{v}$, so the vector $\mathbf{l}$ is perpendicular to the plane containing $\mathbf{r}$ and $\mathbf{v}$. Because $\mathbf{l} = \text{constant}$, the perpendicular to the plane containing $\mathbf{r}$ and $\mathbf{v}$ is constant. Therefore, the particle moves in a constant plane. See [Figure 10.2](#).

Because the motion of the particle lies in a plane, two coordinates are sufficient to specify its position. These can be x and y or r and θ . The origin of the coordinate system is usually placed

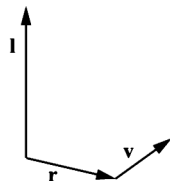


Figure 10.2 The angular momentum vector is perpendicular to the plane containing $\mathbf{r}$ and $\mathbf{v}$. Because the angular momentum is constant, the plane is invariant.

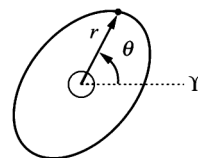


Figure 10.3 The position of a planet referenced to “The First Point in Aries.”

at the primary (assumed to be at rest).⁵ To orient the coordinates, it is necessary to specify a fixed direction. Astronomers pick an imaginary line from the center of the Earth towards a position called “The First Point in Aries” which is the position of the Sun at the vernal equinox.

In [Figure 10.3](#) the symbol Υ indicates the fixed line in space. In polar coordinates the angle θ is measured from Υ . In Cartesian coordinates one usually defines the x -axis along this fixed line. The y -axis is selected in the plane of the orbit and perpendicular to the x -axis. The z -axis is then perpendicular to the orbit and along the angular momentum vector. For many problems it is safe to assume that this coordinate system is an inertial system.

Five thousand years ago the position of the Sun at noon on the vernal equinox was in the constellation Aries (“the ram”). Because the Earth’s axis of rotation precesses with a period of about 25 000 years, the position of the Sun at noon on the vernal equinox has changed and it is presently in the constellation Pisces; within a few hundred years it will enter into the constellation Aquarius. However, the name “First Point in Aries” and the symbol Υ , representing ram’s horns, are still used to represent this arbitrary fixed line in space. It is amusing to note that due to the precession of the Earth’s axis, the positions of the zodiacal constellations have shifted, but astrologers still use the values of 5000 years ago. Thus, people born when the Sun was in the constellation Pisces think they are Aries, those who were born when the Sun was in Aries think they are Taurus, and so on. If astrology had any validity, this horrible mix-up in the zodiacal signs would be serious indeed!

Worked Example 10.1 A particle is in a circular orbit under the action of an inverse cubed attractive central force given by $f(r) = -k/r^3$. Obtain an expression for the angular momentum and show that it is constant.

Solution By Newton’s second law, $m\ddot{\mathbf{r}} = -f(r)\hat{\mathbf{r}} = -\frac{k}{r^3}\hat{\mathbf{r}}$. Recall from our discussion of plane polar coordinates in [Chapter 2](#) that the acceleration in polar coordinates is

$$\ddot{\mathbf{r}} = (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\boldsymbol{\theta}}.$$

For a circular orbit, the magnitude r is constant, so $\dot{r} = 0$ and $\ddot{r} = 0$ so

$$m\ddot{\mathbf{r}} = m(-r\dot{\theta}^2\hat{\mathbf{r}} + r\ddot{\theta}\hat{\boldsymbol{\theta}}) = -f(r)\hat{\mathbf{r}}.$$

Therefore, for an inverse cube force law, $m\ddot{\mathbf{r}} = -f(r)\hat{\mathbf{r}}$ yields two relations, namely

$$-mr\dot{\theta}^2 = -f(r) = -\frac{k}{r^3},$$

and

$$mr\ddot{\theta} = 0.$$

⁵ In the language of astronomy, if one body is much more massive than the other, the massive body is called the “primary.”

The second equation boils down to $\ddot{\theta} = 0$, telling us that the angular velocity, $\dot{\theta}$ is constant. The first equation yields

$$mr^4\dot{\theta}^2 = k.$$

But because $l = mr^2\dot{\theta}$ we see that the left-hand side is l^2/m and consequently

$$l^2 = mk,$$

and hence the angular momentum is constant. This result also follows from $r = \text{constant}$ and $\dot{\theta} = \text{constant}$.

Worked Example 10.2 Consider a particle in an attractive force field whose potential energy is of the form $V(r) = kr^{n+1}$. Show that for a periodic orbit the average kinetic energy is related to the average potential energy by

$$\langle T \rangle = \frac{n+1}{2} \langle V \rangle.$$

Apply to the gravitational force. (This is a special case of the virial theorem.)

Solution This problem asks us to determine the relationship between the average values of the kinetic and potential energy. The time average of any quantity (such as the kinetic energy) is defined as:

$$\langle T \rangle = \frac{1}{\tau} \int_0^\tau T(t) dt.$$

For this problem, it is reasonable to let τ be the orbit period.

If a particle is in orbit, after one period it will have returned to its original location and have the same velocity as it did initially. Therefore, the quantity $\mathbf{p} \cdot \mathbf{r}$ repeats periodically. Let us define the periodic function $G(t) = \mathbf{p} \cdot \mathbf{r}$. Consider the time average of the derivative of G with respect to time:

$$\left\langle \frac{dG}{dt} \right\rangle = \frac{1}{\tau} \int_0^\tau \frac{dG}{dt} dt = \frac{1}{\tau} \int_0^\tau dG = \frac{1}{\tau} [G(\tau) - G(0)] = 0.$$

But

$$\begin{aligned} \frac{dG}{dt} &= \frac{d}{dt} (\mathbf{p} \cdot \mathbf{r}) = \mathbf{p} \cdot \dot{\mathbf{r}} + \dot{\mathbf{p}} \cdot \mathbf{r} \\ &= m\mathbf{v} \cdot \mathbf{v} + \mathbf{F} \cdot \mathbf{r} = 2T - \nabla V \cdot \mathbf{r} \\ &= 2T - \frac{dV}{dr} r, \end{aligned}$$

because $\nabla V \cdot \mathbf{r} = \frac{dV(r)}{dr} r$. Therefore,

$$\left\langle \frac{dG}{dt} \right\rangle = \frac{1}{\tau} \int_0^\tau \left(2T - \frac{dV}{dr} r \right) dt = 0,$$

and

$$\begin{aligned} 2\langle T \rangle - \left\langle \frac{dV}{dr} r \right\rangle &= 0, \\ \langle T \rangle &= \frac{1}{2} \left\langle \frac{dV}{dr} r \right\rangle. \end{aligned}$$

Since $V = kr^{n+1}$ we have

$$\frac{dV}{dr}r = (n+1)kr^n r = (n+1)kr^{n+1} = (n+1)V,$$

and

$$\left\langle \frac{dV}{dr}r \right\rangle = (n+1) \langle V \rangle,$$

so

$$\langle T \rangle = \frac{n+1}{2} \langle V \rangle.$$

The gravitational potential has the form $V = -\frac{k}{r}$ so $n = -2$ and

$$\langle T \rangle = -\frac{1}{2} \langle V \rangle.$$

This relationship between kinetic and potential energy is useful when solving orbital mechanics problems. It might be mentioned that the virial theorem has important applications in thermodynamics and statistical mechanics.

Exercise 10.1

Use $\mathbf{F} = -\nabla V$ to obtain an expression for the Lennard-Jones force. Determine the value of r where the force changes from attractive to repulsive. Answer: $\mathbf{F} = -\hat{\mathbf{r}} \left(\frac{6a}{r^7} - \frac{12b}{r^{13}} \right)$. ■

Exercise 10.2

Use $V = -\int_{r_0}^r F(r)dr$ to obtain the potential energy for the gravitational force, the electrostatic force, and the force exerted by a spring. Select r_0 appropriately. ■

Exercise 10.3

A planet of mass m is in a circular orbit of radius r_0 about a star of mass M . Determine its angular momentum. Answer: $l = m\sqrt{GM r_0}$. ■

Exercise 10.4

Assume that at some initial moment the radius vector $\mathbf{r}$ and the velocity vector $\mathbf{v}$ are perpendicular to one another – however, the angle between them is changing with time. Explain why the two vectors cannot ever be parallel to one another. Answer: Because angular momentum is constant. ■

10.4 The Equation of Motion

Our next task is to determine the equation of motion for a planet. We will do this in two ways: first by writing down Newton's second law, and second by applying the Lagrangian technique.

The force acting on a particle (planet) of mass m is given by Newton's law of universal gravitation

$$\mathbf{F} = -\frac{GMm}{r^2}\hat{\mathbf{r}},$$

where M is the mass of the larger attracting body, assumed fixed at the origin of coordinates. The equation of motion of the satellite (mass m) is

$$m\ddot{\mathbf{r}} = -G\frac{Mm}{r^2}\hat{\mathbf{r}},$$

or

$$\ddot{\mathbf{r}} = -\frac{GM}{r^2}\hat{\mathbf{r}}. \quad (10.1)$$

We evaluated the acceleration $\ddot{\mathbf{r}}$ in polar coordinates in [Chapter 2](#) – see [Equation \(2.14\)](#) – obtaining

$$\ddot{\mathbf{r}} = (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\boldsymbol{\theta}}.$$

Substituting this expression into [Equation \(10.1\)](#), yields

$$(\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\boldsymbol{\theta}} = -\frac{GM}{r^2}\hat{\mathbf{r}}.$$

Separating the radial and angular components leads to the following two scalar equations

$$\begin{aligned} \ddot{r} - r\dot{\theta}^2 &= -\frac{GM}{r^2}, \\ r\ddot{\theta} + 2\dot{r}\dot{\theta} &= 0. \end{aligned} \quad (10.2)$$

These equations are a pair of *coupled* second-order ordinary differential equations. The second equation, the “ θ -equation,” is easy to analyze by going back to the definition of angular momentum and recalling that

$$\mathbf{l} = \mathbf{r} \times \mathbf{p} = \mathbf{r} \times m\mathbf{v}.$$

The velocity in plane polar coordinates is $\mathbf{v} = \dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\boldsymbol{\theta}}$, so

$$\mathbf{l} = mr\hat{\mathbf{r}} \times (\dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\boldsymbol{\theta}}).$$

Now carry out the cross products, using the facts that $\hat{\mathbf{r}} \times \hat{\mathbf{r}} = 0$ and $\hat{\mathbf{r}} \times \hat{\boldsymbol{\theta}} = \hat{\mathbf{k}}$, where $\hat{\mathbf{k}}$ is perpendicular to the plane of motion. You obtain

$$\mathbf{l} = mr(r\dot{\theta})\hat{\mathbf{k}} = mr^2\dot{\theta}\hat{\mathbf{k}}. \quad (10.3)$$

This, in itself, is a useful equation. But it becomes even more useful if you take its time derivative,

$$\frac{d\mathbf{l}}{dt} = \frac{d}{dt}(mr^2\dot{\theta})\hat{\mathbf{k}}.$$

For central forces, the angular momentum is constant, so $d\mathbf{l}/dt = 0$ and the left-hand side of this equation is zero. Therefore,

$$\begin{aligned} 0 &= \frac{d}{dt}(mr^2\dot{\theta}) = m(2r\dot{r}\dot{\theta} + r^2\ddot{\theta}), \\ &= mr(2\dot{r}\dot{\theta} + r\ddot{\theta}). \end{aligned}$$

Since neither m nor r is equal to zero, this implies that

$$r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0.$$

But this is just the θ equation! Therefore, we see that the θ equation of motion, the second of [Equations \(10.2\)](#), is a statement that angular momentum is constant. Consequently, that equation can be replaced by the equivalent relation

$$l = mr^2\dot{\theta} = \text{constant}.$$

This equation, in turn, gives a nice expression for $\dot{\theta}$, namely,

$$\dot{\theta} = \frac{l}{mr^2}. \quad (10.4)$$

Replacing $\dot{\theta}$ in the r equation, the first of [Equations \(10.2\)](#), by expression (10.4) yields the following equation for the radial motion of mass m :

$$\ddot{r} - \frac{l^2}{m^2 r^3} = -\frac{GM}{r^2}. \quad (10.5)$$

This equation involves only r . Thus, conservation of angular momentum *decouples* the equations of motion. The equation has only one variable (r), so it is often called a “one-dimensional equation.” But you should always keep in mind that the motion takes place in *two* dimensions. Once [Equation \(10.5\)](#) has been solved for $r = r(t)$, you can use it in [Equation \(10.4\)](#) to determine $\theta = \theta(t)$.

We shall now use the Lagrangian technique to determine the equations of motion. (We had better obtain the same result!)

As you know, the Lagrangian is $L = T - V$. In this problem V is the potential energy for a particle of mass m attracted gravitationally to a body of mass M . According to [Equation \(9.6\)](#) this is

$$V(r) = -\frac{GMm}{r}.$$

In polar coordinates, the kinetic energy ($\frac{1}{2}mv^2 = \frac{1}{2}m\mathbf{v} \cdot \mathbf{v}$) is

$$T = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2.$$

Therefore the Lagrangian is

$$L = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 + \frac{GMm}{r}.$$

Recall that the Lagrange equations of motion have the form

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0,$$

where the q_i s are now r and θ . So we have two equations, namely,

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{r}} - \frac{\partial L}{\partial r} = 0,$$

and

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0.$$

The partial derivatives are easily evaluated. You should prove for yourself that the two equations of motion are

$$\frac{d}{dt}(m\dot{r}) - mr\dot{\theta}^2 + \frac{GMm}{r^2} = 0, \quad (10.6)$$

and

$$\frac{d}{dt}(mr^2\dot{\theta}) = 0, \text{ or } \frac{dl}{dt} = 0. \quad (10.7)$$

The second of these equations gives

$$\dot{\theta} = \frac{l}{mr^2}.$$

Using this expression, Equation (10.6) leads to

$$\ddot{r} - \frac{l^2}{m^2r^3} = -\frac{GM}{r^2}. \quad (10.8)$$

These are, of course, the same as the equations of motion obtained using Newton's second law, namely Equations (10.4) and (10.5). Note, however, how much easier it is to use the Lagrangian than Newton's second law.

Exercise 10.5

Carry out the steps to show that $T = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2$. ■

Exercise 10.6

Obtain Equations (10.6) and (10.7). ■

Exercise 10.7

Suppose the force between a particle of mass m and a fixed point is given by $\mathbf{F} = -kr\hat{\mathbf{r}}$, where k is a constant. Obtain the Lagrangian and the equations of motion. Is angular momentum conserved for this system? Answer: $m\ddot{r} - mr\dot{\theta}^2 + kr = 0$; $\frac{d}{dt}(mr^2\dot{\theta}) = 0$. ■

10.5 Energy and the Effective Potential Energy

We have (twice!) obtained the equation of motion for the radial coordinate – Equation (10.5) and Equation (10.8). You probably expect to proceed by integrating it to get the value of r as a function of time and initial conditions. Indeed, we shall do that in Section 10.6, but first let us consider what the conservation of energy principle tells us about our problem. Since gravity is a conservative force (and there are no other forces), the total mechanical energy of our system is constant. Thus,

$$\begin{aligned} E &= \text{constant} = T + V \\ &= \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - G\frac{Mm}{r}. \end{aligned}$$

But $l = mr^2\dot{\theta}$, so we can replace $\dot{\theta}$ by l/mr^2 and write

$$E = \frac{1}{2}m\dot{r}^2 + \frac{m}{2}r^2 \left(\frac{l^2}{m^2r^4} \right) - \frac{GMm}{r},$$

or

$$E = \frac{1}{2}m\dot{r}^2 + \frac{l^2}{2mr^2} - \frac{GMm}{r}. \quad (10.9)$$

The first term depends on the square of the radial speed ($\dot{r}^2$) and the other terms depend only on position, so the first term *looks like* kinetic energy and the remaining terms *look like* potential energy. So it seems reasonable to associate a “radial kinetic energy” with the term $\frac{1}{2}m\dot{r}^2$ and an “effective potential energy” V_{eff} with the remaining two terms. The effective potential energy V_{eff} is given by⁶

$$V_{eff} = \frac{l^2}{2mr^2} - \frac{GMm}{r}. \quad (10.10)$$

Although V_{eff} looks and acts like a potential energy and is a function only of position, it is definitely *not* a potential energy because it actually contains a kinetic energy term, namely $l^2/(2mr^2) = \frac{1}{2}m\dot{\theta}^2$.

It is instructive to draw an energy diagram in terms of the effective potential energy.⁷ Note that V_{eff} is the sum of two terms, one positive and the other negative. For $r \rightarrow \infty$, the negative term in V_{eff} is the dominant term because

$$\left. \frac{1}{r} \right|_{r \rightarrow \infty} > \left. \frac{1}{r^2} \right|_{r \rightarrow \infty}.$$

As $r \rightarrow 0$, the positive term dominates because

$$\left. \frac{1}{r} \right|_{r \rightarrow 0} < \left. \frac{1}{r^2} \right|_{r \rightarrow 0}.$$

Therefore the plot of V_{eff} vs. r must have the general shape shown in [Figure 10.4](#). Study this figure carefully and convince yourself it is qualitatively correct, specifically that V_{eff} is positive as $r \rightarrow 0$ and negative as $r \rightarrow \infty$. Notice particularly that the $l^2/2mr^2$ term bends much more sharply than $-GMm/r$. Also, don't forget that r can only take on positive values.

[Figure 10.5](#) is also a plot of $V_{eff}(r)$ vs. r . In this plot the effective potential energy does not have quite the right shape because I drew it to make it easy for you to appreciate various aspects of the effective potential that are hard to see on a more accurate plot, such as [Figure 10.4](#). In [Figure 10.5](#) you see four possible values for the total energy, denoted E_0 , E_1 , E_2 , and E_3 . Consider first a particle with energy E_1 . From [Equations \(10.9\) and \(10.10\)](#), we can write

$$\frac{1}{2}m\dot{r}^2 = E - V_{eff}. \quad (10.11)$$

⁶ The terms “effective potential energy” and “effective potential” are used interchangeably, even though a “potential” is actually potential energy per unit mass. Some books use the term “fictitious potential.” An older term is “centrifugal potential.” In general, the effective potential energy is defined as the sum of $l^2/2mr^2$ and the potential energy. Thus for an electron orbiting about a proton, the effective potential would be

$$V_{eff} = \frac{l^2}{2mr^2} - \frac{e^2}{4\pi\epsilon_0 r}.$$

⁷ At this time you may wish to review the material on energy diagrams in [Section 5.6](#).

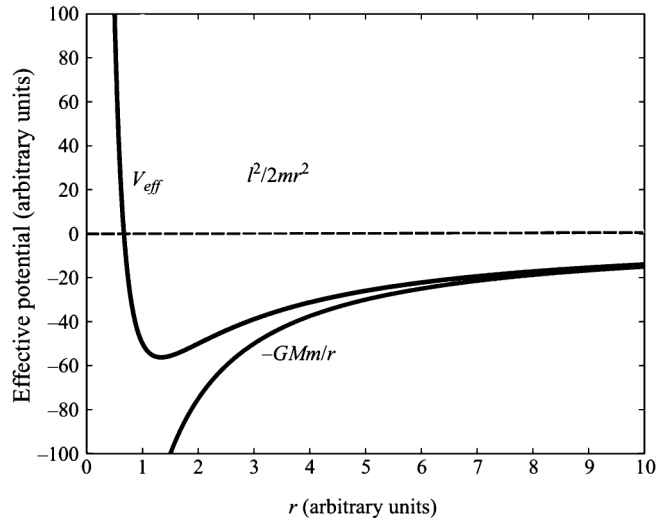


Figure 10.4 The effective potential energy $V_{eff}(r)$ is the sum of two terms, one positive and one negative.

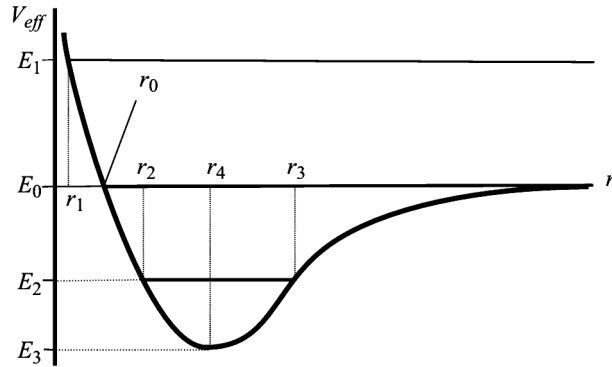


Figure 10.5 Energy diagram for the effective potential. The turning points for various values of total energy are indicated.

Because $\frac{1}{2}m\dot{r}^2$ can never be negative, the particle cannot be located at values of r for which $E < V_{eff}$. From Figure 10.5 this implies that if the energy is E_1 , then $r \geq r_1$, where r_1 is the point of closest approach. You can imagine the particle starting at $r = \infty$, coming closer and closer to the primary until it reaches r_1 (the turning point), and then moving back out to infinity.

Note that according to Equation (10.11), the radial component of the velocity $\dot{r}$ is given by

$$\dot{r} = \sqrt{\frac{2}{m} [E - V_{eff}(r)]}. \quad (10.12)$$

Thus, the square of the radial speed is proportional to $E - V_{eff}$. (Note that $E - V_{eff}$ is not the kinetic energy, only the radial part of the total kinetic energy.) On the energy diagram (Figure 10.5) the distance between the horizontal line at E_1 and the heavy line representing V_{eff} is proportional to $\dot{r}^2$. At r_1 the value of V_{eff} is E_1 , so $\dot{r} = 0$. That is, at the turning point the particle has zero radial velocity, which makes perfect sense. At the turning point, the angular



Figure 10.6 A comet or some other celestial object in a hyperbolic path.

component of the velocity is a maximum because $r\dot{\theta} = (r)(l/mr^2) = l/mr$ is greatest when r is smallest.

Let me emphasize that this not a complete description of the motion; it is only a description of the *radial* motion. Meanwhile the particle is also moving in θ with a velocity given by $\dot{\theta} = l/mr^2$. The angular velocity increases as the radial distance decreases, in agreement with Kepler's second law. As we shall see, for positive values of the energy, the path of the particle is a hyperbola as illustrated in [Figure 10.6](#). This could be the path of a comet coming in from infinity, speeding up as it approaches the Sun, swooping around the Sun, and then moving out to infinity. (Here "infinity" is a place far from the Sun, such as the Oort Cloud from which many comets are believed to originate.)

Next consider a particle with zero total energy ($E = E_0 = 0$; see [Figure 10.5](#)). This means that the positive "radial kinetic energy" $\frac{1}{2}m\dot{r}^2$ is equal in magnitude to the negative effective potential energy V_{eff} . The motion of the particle as it comes from $r = \infty$ to the turning point at r_0 and then goes back out to $r = \infty$ is similar to the motion of the particle with energy E_1 . As we shall see, the main difference is that the trajectory for energy $E_1 > 0$ is a hyperbola and the trajectory for energy $E_0 = 0$ is a parabola. For the parabolic orbit, the radial speed of the particle $\dot{r}$ is zero at infinity as well as at the turning point. As the particle comes in from infinity, $\dot{r}$ increases, reaching a maximum at r_4 where V_{eff} reaches its greatest negative value, then decreases to zero at r_0 . The angular velocity is given by $\dot{\theta} = l/mr^2$. (You should be able to describe the angular velocity as the particle comes in from infinity to the point of closest approach and then moves back out to infinity.)

If the particle has *negative* total energy, as indicated by E_2 in [Figure 10.5](#), the motion is quite different; there are now *two* turning points and the particle can neither reach $r = 0$ nor move out to $r = \infty$. That is, the motion is *bounded*. The particle is trapped in a potential well. As it moves back and forth radially between the two points denoted by r_2 and r_3 , it is also moving azimuthally with a varying angular velocity $\dot{\theta}$. As we shall see shortly, this combination of radial and angular motion represents a trajectory which is an ellipse.

Finally, if the particle has energy E_3 , the minimum possible total energy, the value of $\dot{r}$ is zero at all times and the particle is at a constant radial position $r = r_4$. The path is a circle. The angular velocity is $\dot{\theta} = l/mr_4^2 = \text{constant}$. The particle is therefore moving with constant angular velocity in a circular path.

Depending on the value of the energy, the trajectory or orbit of the particle is a hyperbola, a parabola, an ellipse, or a circle. These are called conic sections because they can be generated by cutting a cone in various different ways, as shown in [Figure 10.7](#).

We have been considering the motion of massive bodies interacting gravitationally, but the ideas and methods developed here are quite general and can be easily adapted to any central force problem, such as the motion of an electron in orbit around a proton in the Bohr model of the hydrogen atom.

Exercise 10.6

A particle is in a parabolic orbit. Where is its turning point? Answer: $r_0 = l^2/2GMm^2$. ■

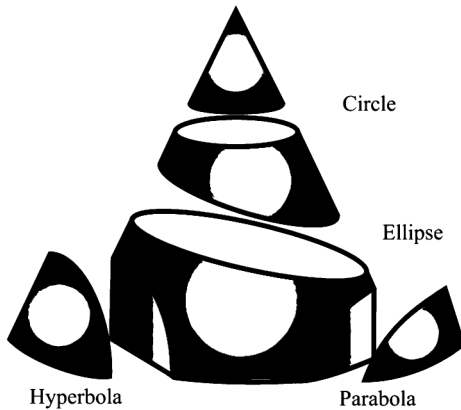


Figure 10.7 The conic sections. If the cone is cut as shown, the cross sections of the cuts will be a circle, an ellipse, a parabola, and one branch of a hyperbola.

Exercise 10.9

In the Bohr model of the hydrogen atom, an electron in the lowest orbit has angular momentum $l = \hbar$, where $\hbar$ is Planck's constant divided by 2π . The electrostatic potential energy of the electron and proton is $V = -e^2/4\pi\epsilon_0 r$. Show that the total energy (mechanical plus electrostatic) of the lowest orbit is $-(1/2)me^4/(4\pi\epsilon_0\hbar)^2$. Assume a circular orbit. ■

10.6 Solving the Radial Equation of Motion

The radial equation of motion, [Equation \(10.5\)](#), is a second-order ordinary differential equation of the form

$$\frac{d^2 r}{dt^2} = \frac{l^2}{m^2 r^3} - \frac{GM}{r^2}.$$

In previous chapters we learned two different ways of solving such an equation. In [Section 3.6](#) we integrated directly by setting $\frac{d^2 r}{dt^2} = v \frac{dv}{dr}$. In [Section 5.7](#) we solved for the motion by using the energy integral in the form of [Equation \(5.15\)](#), namely,

$$\int_{x_0}^{x(t)} \frac{dx}{\sqrt{E - V(x)}} = \sqrt{\frac{2}{m}} t.$$

Applying the energy approach to the present problem, we see that [Equation \(10.9\)](#) leads to

$$\frac{dr}{dt} = \left[\frac{2}{m} \left(E - \frac{l^2}{2mr^2} + \frac{GMm}{r} \right) \right]^{\frac{1}{2}}. \quad (10.13)$$

This equation yields the following definite integral

$$\int_{r_0}^{r(t)} \frac{dr}{\sqrt{\frac{2}{m} \left(E - \frac{l^2}{2mr^2} + \frac{GMm}{r} \right)}} = \int_0^t dt = t.$$

Integrating gives

$$t = t(r, r_0, E, l),$$

which can, in principle, be inverted to yield

$$r = r(t, r_0, E, l).$$

In this solution, the total energy E and angular momentum l are arbitrary constants. Note the use of the term *arbitrary* constant. Such constants are arbitrary in the sense that the differential equation is satisfied by the solution no matter what values the constants happen to have. But for a particular problem, these constants are anything but arbitrary! In simple kinematics problems the “arbitrary constants” are usually the initial position and initial velocity. In more complex problems such as the one we just solved, they tend to be other constants of the motion such as the energy and angular momentum.

Exercise 10.10

Obtain $t = t(r, r_0, E, l)$ for a parabolic orbit (for which $E = 0$). (You will have to look up the integral.) Note that it would be very tedious to invert the expression to obtain $r = r(t)$. Answer:

$$t = \frac{\sqrt{2m}}{3(GMm)^2} \left(GMmr + \frac{l^2}{m} \right) \sqrt{GMmr - \frac{l^2}{2m}} - \frac{\sqrt{2m}}{3(GMm)^2} \left(GMmr_0 + \frac{l^2}{m} \right) \sqrt{GMmr_0 - \frac{l^2}{2m}}. \quad \blacksquare$$

10.7 The Equation of the Orbit

Many physicists work in the space program. In a few years you may be involved in planning a Moon shot or a planetary probe to study the clouds of Titan. If that happens, you will need to know the position of the spacecraft as a function of time. You will have to calculate $\mathbf{r} = \mathbf{r}(t)$ and you could use the technique described in the previous section.⁸

On the other hand, if you wish to describe the *orbit* of a satellite, planet or comet, you are not interested in its position at a particular time. Rather, you want a description of the *path* followed by the celestial object. Mathematically, this is given by the *equation of the orbit*, $r = r(\theta)$. Once you have this equation, you can determine r for every value of θ and thus trace out the path.

I will first present a “brute force” method and then a “sophisticated technique” for obtaining the orbit.

Brute Force Method

To determine the equation of the orbit, $r = r(\theta)$, start with the expression for $\frac{dr}{dt}$ obtained using the energy equation, Equation (10.13). Using the chain rule,

$$\frac{dr}{dt} = \frac{dr}{d\theta} \frac{d\theta}{dt} = \frac{dr}{d\theta} \dot{\theta}.$$

But recall that the angular momentum is $l = mr^2\dot{\theta}$, so

$$\frac{dr}{dt} = \frac{l}{mr^2} \frac{dr}{d\theta}. \quad (10.14)$$

⁸ There are other ways to determine the position of a space probe. These are described in engineering books on “astronautics.”

Substituting into Equation (10.13) yields

$$\begin{aligned}\frac{dr}{d\theta} &= \frac{mr^2}{l} \left[\frac{2}{m} \left(E - \frac{l^2}{2mr^2} + \frac{GMm}{r} \right) \right]^{\frac{1}{2}} \\ &= \left[\frac{2mE}{l^2} r^4 - r^2 + \frac{2GMm^2}{l^2} r^3 \right]^{\frac{1}{2}} \\ &= [\alpha r^2 + \beta r^3 + \gamma r^4]^{\frac{1}{2}},\end{aligned}$$

where I used the Greek letters α, β , and γ to illustrate the form of this equation. Rewriting and integrating the last equation gives

$$\int_{r_0}^r \frac{dr}{r(\alpha + \beta r + \gamma r^2)^{1/2}} = \int_{\theta_0}^{\theta} d\theta. \quad (10.15)$$

This integral can be found in tables of integrals.

Thus, in principle, the problem of determining the equation of the orbit is solved. The only things left to do are: (1) integrate Equation (10.15), and (2) invert the result to obtain $r = r(\theta)$.

Sophisticated Technique

There is a different way to obtain the equation of the orbit that cleverly avoids evaluating the complicated integral in Equation (10.15) and then carrying out the inversion to obtain $r = r(\theta)$. Bear with me for a little while because the math is a bit involved (but not difficult).

We will begin with Equation (10.2), the radial equation of motion:

$$\ddot{r} - r\dot{\theta}^2 = -\frac{GM}{r^2}.$$

Using $l = mr^2\dot{\theta}$ to eliminate $\dot{\theta}$ gives Equation (10.8) which is repeated here for convenience:

$$\ddot{r} - \frac{l^2}{m^2 r^3} = -\frac{GM}{r^2}.$$

Now let us introduce a new variable, u , defined as the inverse of r . That is,

$$u \equiv \frac{1}{r}.$$

Then $r = u^{-1}$ and $dr = -(1/u^2)du$. Therefore,

$$\frac{dr}{dt} = -\frac{1}{u^2} \frac{du}{dt} = -\frac{1}{u^2} \frac{du}{d\theta} \frac{d\theta}{dt},$$

where the last step uses the chain rule and the fact that $u = u(\theta)$. So,

$$\frac{dr}{dt} = -\frac{1}{u^2} \frac{du}{d\theta} \dot{\theta} = -r^2 \dot{\theta} \frac{du}{d\theta}.$$

But $r^2 \dot{\theta} = l/m$. Replacing,

$$\frac{dr}{dt} = -\frac{l}{m} \frac{du}{d\theta}.$$

Taking the derivative with respect to time again,

$$\begin{aligned}\frac{d}{dt} \left(\frac{dr}{dt} \right) &= \frac{d}{dt} \left(-\frac{l}{m} \frac{du}{d\theta} \right), \\ \ddot{r} &= -\frac{l}{m} \frac{d}{dt} \left(\frac{du}{d\theta} \right) = -\frac{l}{m} \frac{d}{d\theta} \left(\frac{du}{d\theta} \right) \frac{d\theta}{dt}, \\ &= -\frac{l}{m} \frac{d^2 u}{d\theta^2} \left(\frac{l}{mr^2} \right) = -\frac{l^2}{m^2 r^2} \frac{d^2 u}{d\theta^2}.\end{aligned}$$

But $1/r^2 = u^2$ so

$$\ddot{r} = -\frac{l^2 u^2}{m^2} \frac{d^2 u}{d\theta^2}.$$

Substituting this into Equation (10.8) and using $1/r^3 = u^3$ we have

$$-\frac{l^2 u^2}{m^2} \frac{d^2 u}{d\theta^2} - \frac{l^2}{m^2} u^3 = -GMu^2,$$

or

$$\frac{d^2 u}{d\theta^2} + u = \frac{GMm^2}{l^2}. \quad (10.16)$$

This is a very interesting equation. It looks like the equation for simple harmonic motion except for the additional constant term on the right-hand side. It can be made to look *exactly* like the simple harmonic motion equation by defining a new variable

$$w = u - \frac{GMm^2}{l^2}.$$

Then, since the last term is a constant,

$$\frac{dw}{d\theta} = \frac{du}{d\theta},$$

and

$$\frac{d^2 w}{d\theta^2} = \frac{d^2 u}{d\theta^2}.$$

Equation (10.16) can now be written in the form of the SHM equation:

$$\frac{d^2 w}{d\theta^2} + w = 0.$$

As noted in Section 3.7, the solution is sinusoidal and we can write

$$w = A \cos(\theta - \theta_0),$$

where A and θ_0 are integration constants. Consequently,

$$u - \frac{GMm^2}{l^2} = A \cos(\theta - \theta_0).$$

Now u is just the inverse of r , so the derivation finally gives an equation for r in terms of θ :

$$r = \frac{1}{u} = \frac{1}{\frac{GMm^2}{l^2} + A \cos(\theta - \theta_0)}.$$

Dividing top and bottom by GMm^2/l^2 puts this in a nicer form:

$$r = \frac{l^2/GMm^2}{1 + \frac{Al^2}{GMm^2} \cos(\theta - \theta_0)}. \quad (10.17)$$

From a mathematical point of view, the problem is now solved because r has been expressed in terms of θ and other known quantities (such as the angular momentum) and two constants of integration, A and θ_0 .

Equation (10.17) describes all the possible orbits for the two-body problem, i.e., circles, ellipses, parabolas, and hyperbolas. Because elliptical motion is of particular interest, let us assume that the total energy E is negative. Then the motion is bounded and the particle (planet) oscillates radially between turning points r_2 and r_3 as illustrated in Figure 10.5 for energy E_2 .

Equation (10.17) contains two constants of integration, A and θ_0 . We now consider their physical significance, beginning with A . At the turning points, the effective potential energy is equal to the total energy, $V_{eff}^{TP} = E$. This fact allows us to evaluate r_2 and r_3 which we write, in general, as r_{tp} . From Equation (10.10) we have

$$V_{eff}^{TP} = \frac{l^2}{2mr_{tp}^2} - \frac{GMm}{r_{tp}} = E.$$

Therefore, in terms of u ,

$$\frac{l^2}{2m}u_{tp}^2 - GMmu_{tp} - E = 0.$$

The two solutions of this quadratic equation for u_{tp} are:

$$u_{\pm} = \frac{m}{l^2} \left(GMm \pm \sqrt{(GMm)^2 + \frac{4l^2 E}{2m}} \right). \quad (10.18)$$

(Note that $1/u_{\pm}$ are the r_2 and r_3 turning points of Figure 10.5.)

But we have seen that

$$u = A \cos(\theta - \theta_0) + \frac{GMm^2}{l^2}.$$

The maximum and minimum values of this expression occur when $\cos(\theta - \theta_0) = \pm 1$. That is,

$$u_+ = A + \frac{GMm^2}{l^2},$$

and

$$u_- = -A + \frac{GMm^2}{l^2}.$$

Using u_+ and equating the expression above to the relation given by Equation (10.18) leads to

$$A + \frac{GMm^2}{l^2} = \frac{GMm^2}{l^2} + \frac{m}{l^2} \sqrt{(GMm)^2 + \frac{2l^2 E}{m}},$$

or

$$A = \left[\frac{(GMm)^2 m^2}{l^4} + \frac{2Em}{l^2} \right]^{\frac{1}{2}}. \quad (10.19)$$

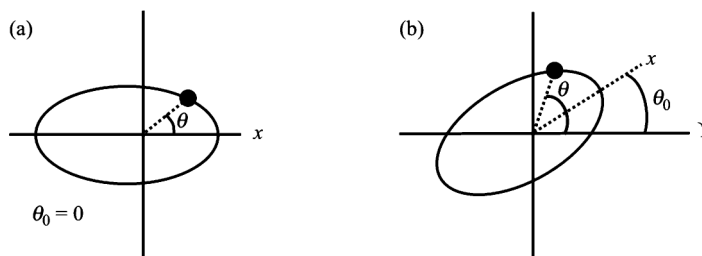


Figure 10.8 The position of a satellite relative to the major axis is given by the angle θ . (a) The major axis can be defined to be the x -axis. (b) The angle θ_0 gives the orientation of the elliptical orbit relative to a fixed line in space.

Thus A is related to the total energy and the angular momentum in a rather complicated way.

Let us now turn our attention to θ_0 , the other constant of integration. Considering [Equation \(10.17\)](#) we note that when $\cos(\theta - \theta_0) = +1$, the denominator is maximized and r reaches its minimum value. If M is the mass of the Sun, the place where r is a minimum is called the *perihelion*. The x -axis is usually taken to be a line from M through the perihelion, as shown in [Figure 10.8](#). If θ is measured from the x -axis, then $\theta_0 = 0$.

On the other hand, if we measure θ from a fixed direction in space, such as the “first point in Aries” (Υ) then θ_0 is the angle between Υ and the x -axis and gives the orientation of the ellipse relative to the fixed direction.

We have, then, determined A in terms of E and l and given the physical interpretation of θ_0 as a description of the orientation of the ellipse with respect to inertial axes. Now the problem is indeed fully solved. Unfortunately, as you have no doubt noticed, our expressions are rather unwieldy. In a moment I will write them in a neater form. But first we need to consider some properties of conic sections, particularly ellipses.

10.8 The Equation of an Ellipse

You probably remember the grade school method for drawing an ellipse. You place a sheet of paper on a cork board and stick two tacks in it. Then you take a piece of string, tie the ends together, and loop it around the tacks. Finally you place a pencil in the loop of string and move it around the tacks, keeping the string taut, and if you are very careful you will draw an ellipse on the paper. See [Figure 10.9](#).

As the pencil traces out the ellipse, the distance from the pencil to either tack varies. But the sum of the distances to the two tacks remains the same because the string is of constant length and the distance between the two tacks does not change. Therefore the distance from the first tack to the pencil *plus* the distance from the pencil to the second tack is constant. This fact is used for the mathematical definition of an ellipse.

An ellipse is the locus of points whose distances from two fixed points sum to a constant.

That is a bit complicated to say, but it is expressed quite simply as an equation. First choose the two fixed points, call them focal points and denote them F and F' . The distance from a point on the ellipse to F will be called r and the distance from that same point to F' will be r' . See [Figure 10.10](#). Then an ellipse is defined as the locus of points such that $r + r' = \text{constant}$.

It is easy to obtain an equation for the ellipse in polar coordinates by introducing the angle θ between r and the major axis of the ellipse. The length of the semimajor axis ($\frac{1}{2}PP'$) will be

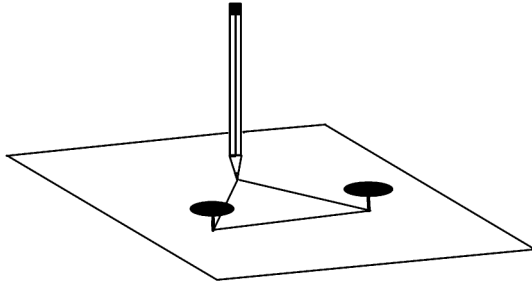


Figure 10.9 How to draw an ellipse.

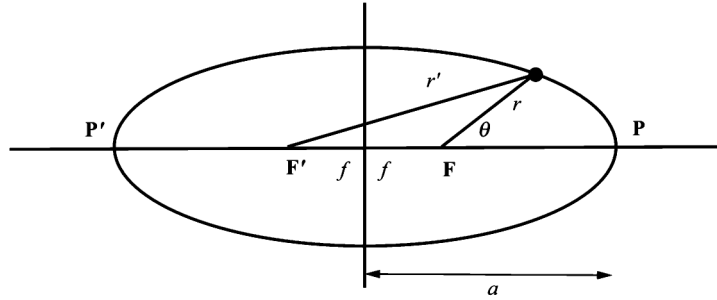


Figure 10.10 An ellipse. The total distance $r + r' + \overline{F'F}$ is a constant, so $r + r'$ must be constant. Points P' and P are called the apsides.

denoted a , and the semiminor axis is denoted b . The distance from the center of the ellipse to either focal point will be $f = ea$, where e is a number less than one, called the *eccentricity*. The eccentricity e and the semimajor axis a determine the size and shape of the ellipse.

It turns out that

$$r + r' = 2a.$$

This relation may surprise you, but it is easily proved. Consider point P . There $r = \overline{FP}$ and $r' = \overline{F'P}$. By symmetry, $\overline{F'P} = \overline{FP'}$ and therefore $r + r' = \overline{FP} + \overline{FP'} = 2a$. Since $r + r'$ is a constant, if it is equal to $2a$ at one point, it is equal to $2a$ at any point.

Now consider the triangle formed by r , r' , and $2f$, where $f = ea$. See Figure 10.10. According to the law of cosines, if θ is the “external” angle,

$$r'^2 = r^2 + (2ae)^2 + 2r(2ae) \cos \theta. \quad (10.20)$$

Substituting $r' = 2a - r$ into Equation (10.20) gives

$$(2a - r)^2 = r^2 + 4a^2e^2 + 4aer \cos \theta,$$

and a little bit of algebra leads to

$$r = \frac{a(1 - e^2)}{1 + e \cos \theta}. \quad (10.21)$$

This is the equation for an ellipse in plane polar coordinates. If we were to go through the same process for hyperbolas and parabolas, we would obtain similar equations for $r = r(\theta)$. In fact, the equation for any conic section can be expressed in the form

$$r = \frac{p}{1 + e \cos \theta}, \quad (10.22)$$

where $p = a(1 - e^2)$ for an ellipse, $p = a(e^2 - 1)$ for a hyperbola,⁹ and $p = a$ for a parabola¹⁰ (a is a characteristic constant for each type of curve and, in general, $p = l^2/GMm^2$). The various curves correspond to different choices of the eccentricity e . Specifically,

$$\begin{aligned} e = 0 & \text{ gives a circle,} \\ e < 1 & \text{ gives an ellipse,} \\ e = 1 & \text{ gives a parabola,} \\ e > 1 & \text{ gives a hyperbola.} \end{aligned}$$

The semimajor axis a determines the *size* of an ellipse and the eccentricity determines its *shape*. At zero eccentricity, the ellipse degenerates into a circle. As the eccentricity gets closer and closer to one, the ellipse gets flatter and flatter. That is, as e approaches unity, the ratio of the semiminor axis to the semimajor axis approaches zero. As you will prove in [Exercise 10.12](#),

$$\frac{b}{a} = (1 - e^2)^{1/2}. \quad (10.23)$$

You probably know from geometry that the area of an ellipse is given by $S = \pi ab$. It is sometimes convenient to write this in terms of a only. Using [Equation \(10.23\)](#),

$$S = \pi a^2 (1 - e^2)^{1/2}. \quad (10.24)$$

Having considered the properties of conics, let's go back to the motion of planets. Comparing the equation of an ellipse ([10.21](#)) with the equation of the orbit ([10.17](#)), you can see that the orbit is an ellipse (or, more generally, a conic section). A term-by-term comparison of these two equations shows that

$$\frac{l^2}{GMm^2} = a(1 - e^2), \quad (10.25)$$

and that

$$\frac{Al^2}{GMm^2} = e.$$

But we showed in [Equation \(10.19\)](#) that

$$A = \left[\frac{(GMm)^2 m^2}{l^4} + \frac{2Em}{l^2} \right]^{\frac{1}{2}},$$

so

$$e^2 = \frac{(GMm)^2 m^2}{l^4} \frac{l^4}{(GMm)^2 m^2} + \frac{2Em}{l^2} \frac{l^4}{(GMm^2)^2}.$$

Therefore,

$$e^2 = 1 + \frac{2El^2}{m(GMm)^2}. \quad (10.26)$$

⁹ A hyperbola has two branches and is described by the equation

$$r = \frac{a(e^2 - 1)}{\pm 1 + e \cos \theta},$$

where the “plus branch” corresponds to an attractive force and the “minus branch” corresponds to a repulsive force.

The quantity a is one half the distance between the vertices.

¹⁰ For a parabola $a = 2r_p$, where r_p is the distance from the focus to the nearest point on the parabolic curve.

This gives us an expression for the eccentricity in terms of constants of the motion. Note that for $E > 0$ we get $e > 1$, a hyperbola. For $E = 0$ we get $e = 1$, a parabola. For $E < 0$, we get $e < 1$, an ellipse. Finally, for a circle, $e = 0$ so $2El^2/m(GMm)^2 = -1$.

Plugging Equation (10.26) into (10.25) gives the following expression for the semimajor axis:

$$a = -\frac{GMm}{2E}. \quad (10.27)$$

The minus sign is necessary because the total energy is negative for an ellipse. Equation (10.27) shows that the semimajor axis depends only on the energy.

Worked Example 10.3 A comet of mass m starts from infinity with velocity v_0 and impact parameter b . (If undeflected, the path of the comet would be a straight line passing the Sun at a distance b .) (a) Show that the distance of closest approach to the Sun is approximately $b^2 v_0^2 / 2GM$. (b) Write the equation of the orbit in polar coordinates in terms of M , v_0 , and b , where M is the mass of the Sun.

Solution (a) Let the Sun–comet distance at perihelion be denoted d . At that point, the speed of the comet is v_d and its velocity is perpendicular to the Sun–comet line.

By conservation of angular momentum $l_\infty = l_d$, and

$$mv_0 b = mv_d d \Rightarrow \frac{b}{d} = \frac{v_d}{v_0}.$$

By conservation of energy

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_d^2 - \frac{GMm}{d},$$

so dividing by $mv_0^2/2$,

$$1 = \left(\frac{b}{d}\right)^2 - \frac{2GM}{dv_0^2}.$$

Multiply by d^2 and obtain the quadratic

$$d^2 + \frac{2GM}{v_0^2}d - b^2 = 0,$$

$$d = -\frac{GM}{v_0^2} \pm \sqrt{\left(\frac{GM}{v_0^2}\right)^2 + b^2}.$$

Since $d > 0$, we use the positive sign and write

$$d = \frac{GM}{v_0^2} \left(-1 + \left[1 + \frac{b^2 v_0^4}{G^2 M^2} \right]^{1/2} \right).$$

Apply the binomial expansion to get

$$\begin{aligned} d &= \frac{GM}{v_0^2} \left(-1 + \left[1 + \frac{1}{2} \frac{b^2 v_0^4}{G^2 M^2} + \cdots \right] \right), \\ d &\simeq \frac{GM}{v_0^2} \frac{1}{2} \frac{b^2 v_0^4}{G^2 M^2} = \frac{b^2 v_0^2}{2GM}. \end{aligned}$$

(b) In polar coordinates the hyperbolic orbit under an attractive force is

$$r = \frac{a(e^2 - 1)}{1 + e \cos \theta}.$$

The problem requires us to express a and e in terms of the given parameters, namely v_0, b , and constants. Taking the derivative of r with respect to time we have

$$\dot{r} = \frac{a(e^2 - 1)}{(1 + e \cos \theta)^2} e \sin \theta \dot{\theta},$$

Now $l = mr^2\dot{\theta}$ so

$$\dot{\theta} = \frac{l}{mr^2} = \frac{l}{m} \frac{(1 + e \cos \theta)^2}{a^2(e^2 - 1)^2},$$

and we can write

$$\dot{r}^2 = \frac{e^2 l^2 / m^2}{[a(e^2 - 1)]^2} \sin^2 \theta.$$

The square of the speed of the comet in polar coordinates is

$$v^2 = \dot{r}^2 + r^2 \dot{\theta}^2.$$

The second term in v^2 is $r^2 \dot{\theta}^2$. But,

$$\begin{aligned} r \dot{\theta} &= \frac{1}{r} \frac{l}{m} = \frac{l}{m} \frac{1 + e \cos \theta}{a(e^2 - 1)}, \\ \therefore r^2 \dot{\theta}^2 &= \frac{l^2}{m^2} \frac{1 + 2e \cos \theta + e^2 \cos^2 \theta}{[a(e^2 - 1)]^2}, \end{aligned}$$

so

$$\begin{aligned} v^2 &= \frac{l^2(1 + 2e \cos \theta + e^2(\sin^2 \theta + \cos^2 \theta))}{m^2 [a(e^2 - 1)]^2} \\ &= \frac{l^2}{m^2 a(e^2 - 1)} \left(\frac{2 + 2e \cos \theta + e^2 - 1}{a(e^2 - 1)} \right) \\ &= \frac{l^2}{m^2 a(e^2 - 1)} \left(2 \frac{1 + e \cos \theta}{a(e^2 - 1)} + \frac{e^2 - 1}{a(e^2 - 1)} \right) \\ &= \frac{l^2}{m^2 a(e^2 - 1)} \left[\frac{2}{r} + \frac{1}{a} \right]. \end{aligned}$$

Comparing $r = \frac{a(e^2 - 1)}{1 + e \cos \theta}$ for a hyperbolic orbit with the general form of the equation of the orbit, [Equation \(10.17\)](#), we appreciate that

$$\frac{l^2}{GMm^2} = a(e^2 - 1).$$

Consequently,

$$v^2 = \frac{GMm^2 a(e^2 - 1)}{m^2 a(e^2 - 1)} \left[\frac{2}{r} + \frac{1}{a} \right] = GM \left[\frac{2}{r} + \frac{1}{a} \right].$$

At $r = \infty$, the velocity is v_0 and we see that

$$v_0^2 = \frac{GM}{a},$$

so

$$a = \frac{GM}{v_0^2}.$$

Also

$$a(e^2 - 1) = \frac{l^2}{GMm^2}.$$

Solving for e we obtain

$$e = \left(1 + \left[\frac{v_0^2 b}{(GM)} \right]^2 \right)^{1/2},$$

and finally

$$r = \frac{v_0^2 b^2 / GM}{1 + \sqrt{1 + \left(\frac{bv_0^2}{GM} \right)^2} \cos \theta}.$$

Exercise 10.11

Starting with Equation (10.20), obtain Equation (10.21). ■

Exercise 10.12

Show that for an ellipse $b/a = (1 - e^2)^{1/2}$. (Hint: Apply the Pythagorean theorem to the triangle FOP in Figure 10.11.) ■

Exercise 10.13

A planet is in an elliptical orbit with semimajor axis a . By averaging the largest and smallest values of r , show that it leads to an “average” value of the potential energy equal to $-GMm/a$. (Compare with the virial theorem discussed in Worked Example 10.2.) ■

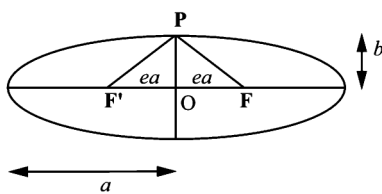


Figure 10.11 An ellipse. Note that the distance from the center to the focal point is ea and that the sum of the distances FP and $F'P$ is $2a$.

Exercise 10.14

Plot Equation (10.22) for $e = 2$, $e = 1.0$, $e = 0.5$, and $e = 0$. ■

10.9 Kepler's Laws Revisited

Having applied Newton's second law and Newton's law of universal gravitation to the problem of planetary motion, let us now return to Kepler's laws that were given in Section 10.2. You will find that Kepler's laws are simply descriptions of some aspects of the theory we have developed.

Recall that Kepler's first law states:

The orbit of a planet is an ellipse with the Sun at one of the focal points.

We applied Newton's second law to a particle of mass m in the gravitational field of M , writing $m\ddot{\mathbf{r}} = (-GMm/r^2)\hat{\mathbf{r}}$. After a significant amount of math, this led to the equation of the orbit $r = r(\theta)$, which we recognized as the equation of an ellipse. Thus Kepler's first law is a consequence of Newton's law of universal gravitation. Actually, the proof is even more general than Kepler's first law, for we have seen that objects in the solar system acting under the inverse square gravitational force will move in a trajectory which may be an ellipse, but it could also be one of the other conic sections.

Kepler studied the orbit of Mars all his life, and he deduced other important facts about its motion. For one thing, he found that the motion of Mars as a function of time was not uniform. Mars speeds up as it approaches the Sun, moves fastest at its point of nearest approach, then slows down as it moves away from the Sun. He expressed this fact in a quantitative way in his "Law of Areas" or, as we call it, Kepler's second law:

The radius vector of a planet sweeps out equal areas in equal times.

This is illustrated in Figure 10.12. If the planet moves from a to b in the same amount of time as to go from c to d then, according to Kepler's second law, the two shaded areas are equal. So another way to state Kepler's second law is, "The *areal* velocity of a planet is constant."

In point of fact, we already proved Kepler's second law, but you might not have noticed it because it was stated in quite a different way. We said, "For a central force, the angular momentum is constant." Are the two statements equivalent? Does the fact that the angular momentum is constant imply that the *areal velocity* is constant? The answer is yes. To prove that this is true, consider a planet located at vector position $\mathbf{r}$. After a time interval dt the planet will be at $\mathbf{r} + d\mathbf{r} = \mathbf{r} + \mathbf{v}dt$. See Figure 10.13. The shaded region is the area swept out. Recall that $|\mathbf{a} \times \mathbf{b}|$ is equal to the area of a parallelogram whose sides are the vectors $\mathbf{a}$ and $\mathbf{b}$. The shaded region in Figure 10.13 is one half of the parallelogram formed from $\mathbf{r}$ and $d\mathbf{r}$. Therefore, the area swept out by the planet's radius vector in time dt is

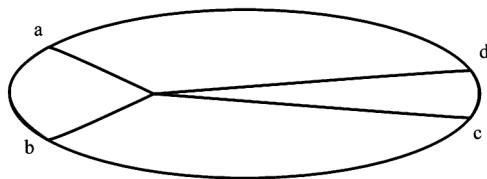


Figure 10.12 Kepler's second law states that equal areas are swept out in equal times.

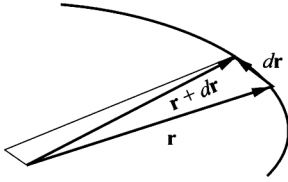


Figure 10.13 The area of the triangle is one half the area of the parallelogram. Therefore, $\Delta S = (1/2) |\mathbf{r} \times d\mathbf{r}| = (1/2) |\mathbf{r} \times \mathbf{v} \Delta t|$.

$$dS = \frac{1}{2} |\mathbf{r} \times \mathbf{v} dt| = \frac{dt}{2} |\mathbf{r} \times \mathbf{v}|.$$

But by the definition of angular momentum, $|\mathbf{r} \times \mathbf{v}| = l/m$. Therefore,

$$dS = \frac{1}{2} \frac{l}{m} dt,$$

or

$$\frac{dS}{dt} = \text{areal velocity} = \frac{l}{2m}.$$

The mass of a planet is constant, so the areal velocity is proportional to the angular momentum and $\frac{dS}{dt} = \text{constant}$ is completely equivalent to the relation $l = \text{constant}$. Kepler's second law is simply a consequence of the fact that in central force motion, the angular momentum is conserved.

Kepler's third law states:

The period of a planet squared is proportional to its semimajor axis cubed.

In mathematical terms, if τ is the time required for a planet to orbit the Sun and if a is the semimajor axis of the orbit, the third law can be written

$$\tau^2 = K a^3,$$

where K is some constant. Kepler discovered this law after years of studying planetary data. On the other hand, our analytical study of elliptical motion leads to it in a fairly simple way. We just note that

$$\frac{dS}{dt} = \frac{l}{2m} \Rightarrow \oint_{\text{orbit}} dS = \frac{l}{2m} \oint_{\text{orbit}} dt.$$

Let S equal the area of the ellipse. Then if τ is the period,

$$S = \frac{l}{2m} \tau.$$

But we have seen that $S = \pi ab = \pi a^2(1 - e^2)^{1/2}$ so

$$\frac{l}{2m} \tau = \pi a^2(1 - e^2)^{1/2}.$$

According to Equation (10.25), for an ellipse, $(1 - e^2) = l^2/GMm^2a$. Squaring we obtain

$$\frac{l^2}{4m^2} \tau^2 = \pi^2 a^4 \frac{l^2}{GMm^2 a},$$

or

$$\tau^2 = \frac{4\pi^2}{GM} a^3.$$

This relation states that the period squared is proportional to the semimajor axis cubed. Thus, Kepler's third law is proved. Note that the constant of proportionality depends only on the mass of the Sun, so the ratio τ^2/a^3 should be the same for all the planets. Therefore, determining the period of a planet or asteroid allows us to evaluate its semimajor axis. Actually, this result is not completely accurate. There is a very small correction that is due to the fact that the Sun is not really at rest at the origin. The Sun and the planet move around their common center of mass. This point is almost, but not quite, at the center of the Sun. This introduces a small correction into Kepler's third law so that the quantity M in the denominator should be replaced by $M + m$. Once again we appreciate the technique in physics of first solving an easier idealized problem and introducing the complications later. (Problem 10.11 asks you to carry out the appropriate calculations.)

Worked Example 10.4 A particle moves in an elliptical orbit with semimajor axis a and eccentricity e . The velocity of the particle is observed to vary from a minimum value v_1 to a maximum value v_2 . Determine the period.

Solution The angular momentum of the particle is constant:

$$l = mvr = \text{constant.}$$

Therefore, the maximum (and minimum) speeds occur when r is at the smallest (and greatest) distance from the force center. Call these distances r_1 and r_2 . By conservation of angular momentum,

$$mv_1r_1 = mv_2r_2.$$

The points at r_1 and r_2 as well as the force center all lie on the semimajor axis, and

$$r_1 + r_2 = 2a.$$

Hence

$$l = mv_1(2a - r_2) = mv_2r_2,$$

$$\therefore r_2 = \frac{2av_1}{v_1 + v_2},$$

so the angular momentum can be written

$$l = m \frac{2av_1v_2}{v_1 + v_2}.$$

The angular momentum is related to the areal velocity by

$$\frac{l}{2m} = \frac{dS}{dt} = \text{areal velocity.}$$

The period is equal to the area divided by the areal velocity

$$\tau = \frac{\text{area}}{\text{areal velocity}} = \frac{\pi ab}{(l/2m)} = \frac{\pi ab(2m)}{m \frac{2av_1v_2}{v_1+v_2}} = \frac{\pi b(v_1 + v_2)}{v_1v_2}.$$

By Exercise 10.12 the semiminor axis, b , is equal to $a(1 - e^2)^{1/2}$, so

$$\tau = \pi a(1 - e^2)^{1/2} \frac{v_1 + v_2}{v_1v_2}.$$

Worked Example 10.5 Suppose you were familiar with Kepler's work, so you knew that the orbits of planets are ellipses and that the angular momentum is constant. This suggests to you that the force on the planet is a central force, so $F = f(r)$, but you don't know any other properties of the force. Show that the force obeys the inverse square law, that is, show that $f(r) \propto 1/r^2$.

Solution In polar coordinates the radial equation of motion is

$$\ddot{r} - r\dot{\theta}^2 = -f(r)/m.$$

The angular equation of motion leads to

$$\dot{\theta} = l/mr^2.$$

Substituting,

$$\ddot{r} - \frac{l^2}{m^2 r^3} = -\frac{f(r)}{m}.$$

In terms of $u = 1/r$, recalling that $\ddot{r} = -\frac{l^2 u^2}{m^2} \frac{d^2 u}{d\theta^2}$, we obtain

$$\begin{aligned} -\frac{l^2 u^2}{m^2} \frac{d^2 u}{d\theta^2} - \frac{l^2 u^3}{m^2} &= -\frac{f(r)}{m}, \\ \frac{d^2 u}{d\theta^2} + u &= \frac{m}{l^2 u^2} f(r). \end{aligned} \quad (10.28)$$

Now the equation of an ellipse has the form

$$\begin{aligned} r &= \frac{p}{1 + e \cos \theta}, \\ \therefore u &= \frac{1}{p} + \frac{e}{p} \cos \theta. \end{aligned}$$

Taking the derivative with respect to θ twice yields

$$\frac{d^2 u}{d\theta^2} = -\frac{e}{p} \cos \theta,$$

and Equation (10.28) becomes

$$\begin{aligned} -\frac{e}{p} \cos \theta + \left(\frac{1}{p} + \frac{e}{p} \cos \theta \right) &= \frac{m}{l^2 u^2} f(r) \\ \frac{1}{p} &= \frac{m}{l^2 u^2} f(r) = \frac{m}{l^2} r^2 f(r). \end{aligned}$$

That is,

$$f(r) = \frac{l^2}{mp} \frac{1}{r^2} \propto \frac{1}{r^2}.$$

Q.E.D.

Exercise 10.15

A certain asteroid has a period of four years. What is its semimajor axis in AU? (Assume you do not know the gravitational constant G or the mass of the Sun, but you do know that the Earth is at a distance of 1 AU from the Sun.) Answer: 2.52 AU. ■

Exercise 10.16

(a) Using the form $\tau^2 = (4\pi^2/GM)a^3$ and assuming you do not know the mass of the Sun, determine the semimajor axis of Saturn in AU, given its period is 29.5 Earth years. (b) Using the correction $(M + m)$ and looking up the appropriate values, obtain a corrected value. Answers: 9.55 AU, 9.52 AU. ■

Exercise 10.17

For a planet in a circular orbit, $F = ma$ can be written as $GMm/r^2 = mv^2/r$. Use this to derive Kepler's third law for circular orbits. ■

Exercise 10.18

Halley's comet has an eccentricity of 0.967. Its perihelion distance is 8.81×10^{10} m. What is its period? Answer: 75.4 years. ■

10.10 Orbital Mechanics

The first artificial satellite (Sputnik 1) was launched by the USSR on October 4th, 1957. Since then, nearly 10 000 satellites have been launched and about half of them are still in orbit. Artificial earth satellites have become an integral part of modern life. The Global Positioning System (GPS) is a constellation of satellites that can specify a location on the Earth to within an accuracy of up to 30 cm. You probably use it nearly every day. Satellites are used for navigation, weather forecasting, communications, oceanography, atmospheric science, astronomy, and military purposes.

This section is a brief consideration of some of the physics used in analyzing the motion of artificial satellites. The two principal physical concepts are energy and angular momentum.

10.10.1 Energy and Angular Momentum of a Satellite

The “specific energy” of a satellite is usually expressed as

$$\mathcal{E} = \frac{1}{2}v^2 - \frac{\mu}{r},$$

where $\mathcal{E} = E/m$ is the energy per unit mass of the satellite and $\mu = GM$ is an abbreviation commonly used in celestial mechanics.¹¹ M is the mass of the Earth.

The angular momentum per unit mass of an object is denoted $\mathbf{h}$. By definition

$$\mathbf{h} = \mathbf{l}/m = \mathbf{r} \times \mathbf{v},$$

and is a constant of the motion.

Artificial satellites are placed in elliptical or circular orbits. The most important parameters of such orbits are the eccentricity (e) and the semimajor axis (a). Two other useful parameters are the distances from the center of the Earth to perigee (r_p) and to apogee (r_a). At these two points the velocity vector $\mathbf{v}$ and the position vector $\mathbf{r}$ are perpendicular to one another and we can write

$$h = r_a v_a = r_p v_p.$$

¹¹ I will use μ for GM in this section only.

We have seen that for an elliptical orbit,

$$r = \frac{a(1 - e^2)}{1 + e \cos \theta},$$

consequently, by Equation (10.25)

$$a(1 - e^2) = \frac{1}{GM} \left(\frac{l}{m} \right)^2 = \frac{h^2}{\mu}.$$

$$\therefore r = \frac{h^2/\mu}{1 + e \cos \theta}.$$

Measuring θ from perigee, we obtain

$$r_p = \frac{h^2/\mu}{1 + e} \quad \text{and} \quad r_a = \frac{h^2/\mu}{1 - e}.$$

The velocities at perigee and apogee are

$$v_p = \frac{h}{r_p} \quad \text{and} \quad v_a = \frac{h}{r_a}.$$

The energy (per unit mass) at perigee is

$$\mathcal{E} = \frac{1}{2}v_p^2 - \frac{\mu}{r_p} = \frac{h^2}{2r_p^2} - \frac{\mu}{r_p} = -\frac{\mu}{2a},$$

where the last step is obtained from Equation (10.27).

10.10.2 The Hohmann Transfer Orbit

Satellites are usually initially placed in a circular “parking orbit,” and then raised to their final orbit by means of a “transfer orbit.” We now consider a simple transfer orbit called the Hohmann transfer orbit. See Figure 10.14.

If a satellite in a circular orbit of radius r_c is given an impulse by turning on a rocket motor for a short burst, its velocity will increase by Δv and the orbit is transformed into an ellipse. The perigee of the ellipse will be $r_p = r_c$. The energy will have increased to $\mathcal{E}_t$. The elliptical orbit will have a semimajor axis given by $a = -\frac{\mu}{2\mathcal{E}_t}$, and the apogee will be at $r_a = 2a - r_p$. We can

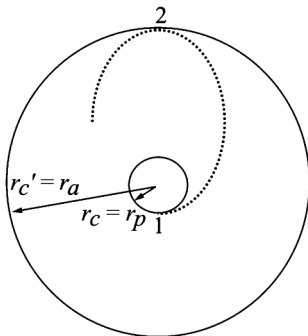


Figure 10.14 A Hohmann transfer orbit. The satellite is given an impulse at point 1 to place it in the elliptical orbit, and another impulse at 2 to place it in the circular orbit.

assume the position of the satellite has not changed during the “burn,” but now its velocity will be given by

$$v_1 = v_c + \Delta v_1 = \sqrt{2 \left(\frac{\mu}{r_p} + \mathcal{E}_t \right)},$$

where v_c is the velocity of the satellite in the initial circular orbit.

In a Hohmann transfer orbit, when the satellite reaches apogee it is given a second impulse to change the speed at apogee from $v_2 = v_a$ to that of a circular orbit of radius $r'_c = r_a$. Note that at the apogee of the elliptical transfer orbit, the speed is

$$v_2 = \sqrt{2 \left(\frac{\mu}{r_a} + \mathcal{E}_t \right)},$$

whereas the speed of a satellite in a circular orbit of radius r_a is

$$v'_c = \sqrt{2 \left(\frac{\mu}{r_a} - \frac{\mu}{2a} \right)} = \sqrt{\frac{\mu}{r_a}},$$

because $r_a = r'_c = a$. Recalling that the energy is a negative quantity, we appreciate that the satellite has to be given an additional impulse to increase its speed to that of the desired circular orbit. That is, $v'_c = v_2 + \Delta v_2$.

There are other orbit transfer schemes; the Hohmann transfer is the most economical (least amount of fuel burned) but the transfer takes a longer time than the other schemes.

Exercise 10.19

Show that $\mathcal{E} = \frac{\mu}{2a}$ implies that $r_p = a(1 - e)$. ■

Exercise 10.20

Show that the velocity of a satellite at perigee can be expressed as $v_p = \sqrt{\frac{1+e}{r_p}}$. ■

10.11 A Perturbed Circular Orbit

In this section we will study the *stability* of a circular orbit. When a physical system in an equilibrium state is perturbed by a small force, it will often oscillate about equilibrium. This is considered a *stable* equilibrium. On the other hand, if the perturbation causes the system to undergo a large change, the equilibrium is unstable. (The usual example of stable equilibrium is a marble at the bottom of a dip, and unstable equilibrium is a marble balanced on the tip of a bulge.) This section is both an application of the concepts studied in the previous sections and an introduction to the methods for dealing with small perturbations.

Consider a particle moving in a circular orbit under the action of a central force. For reasons that will eventually become clear, I will not require the force to obey the inverse square law. That is, although $\mathbf{F} = f(r)\hat{\mathbf{r}}$, the functional form of $f(r)$ will be left undetermined for now.

The equations of motion are given by generalizations of [Equations \(10.2\)](#):

$$\begin{aligned} m(\ddot{r} - r\dot{\theta}^2) &= f(r), \\ m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) &= 0. \end{aligned} \tag{10.29}$$

11

Harmonic Motion

This chapter is an in-depth study of harmonic motion. You are familiar with two examples of simple harmonic oscillators, namely, a pendulum and a mass on a spring. Now you will study several different kinds of harmonic motion, including underdamped, overdamped, and critically damped oscillatory motion, as well as forced harmonic motion and the motion of coupled harmonic oscillators. The techniques you will learn here are used in nearly every branch of physics.

11.1 Springs and Pendulums

A very simple oscillatory system consists of a mass m connected to a spring of constant k . To avoid complications due to forces such as gravity or friction, assume the mass is on a perfectly frictionless horizontal surface and can slide back and forth, as shown in Figure 11.1.

The kinetic energy of the mass is $T = \frac{1}{2}m\dot{x}^2$ and the potential energy of the spring is $V = \frac{1}{2}kx^2$. Therefore, as described in Section 4.3 the Lagrangian for the system is

$$L = T - V = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2.$$

The equation of motion is obtained from Lagrange's equation:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = 0,$$

which yields

$$\ddot{x} + \frac{k}{m}x = 0. \quad (11.1)$$

Another simple oscillatory system is a pendulum consisting of a particle of mass m (the bob) on an inextensible, massless string of length l . See Figure 11.2. The kinetic energy is

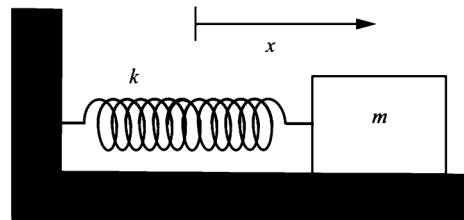


Figure 11.1 A mass on a frictionless surface connected to a spring. The distance x is measured from the equilibrium position to the center of mass of the block.

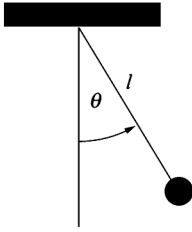


Figure 11.2 A simple pendulum.

$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2)$, where $x = l \sin \theta$ and $y = -l \cos \theta$ (assuming the origin is located at the point where the string is attached to the support).

$$T = \frac{1}{2}m(l^2\dot{\theta}^2 \cos^2 \theta + l^2\dot{\theta}^2 \sin^2 \theta) = \frac{1}{2}ml^2\dot{\theta}^2.$$

The potential energy is

$$V = -mgl \cos \theta.$$

Consequently, see Equation (4.4),

$$L = T - V = \frac{1}{2}ml^2\dot{\theta}^2 + mgl \cos \theta.$$

The Lagrange equation of motion is:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0.$$

That is,

$$\frac{d}{dt}(ml^2\dot{\theta}) + mgl \sin \theta = 0,$$

or

$$ml^2\ddot{\theta} + mgl \sin \theta = 0. \quad (11.2)$$

By restricting the motion to small angles ($\theta \lesssim 20^\circ$) you can replace $\sin \theta$ by θ (in radians). Then, dividing by ml^2 ,

$$\ddot{\theta} + \frac{g}{l}\theta = 0. \quad (11.3)$$

Note that Equations (11.1) and (11.3) have exactly the same *form*; they both tell us that the second derivative of the position variable is negatively proportional to the displacement from equilibrium, meaning the system is subjected to a restoring force that is always directed toward the equilibrium point. This form for the equation of motion is characteristic of oscillatory systems. Whenever you see it, you can be sure that you are faced with a case of simple harmonic motion. You can easily show by substitution that possible solutions to the two simple harmonic motion equations above are

$$x(t) = A \cos \left(\sqrt{\frac{k}{m}}t + \beta \right) \quad (11.4)$$

and

$$\theta(t) = A \cos \left(\sqrt{\frac{g}{l}}t + \beta \right). \quad (11.5)$$

These solutions describe a sinusoidal oscillation of amplitude A and angular frequency $\omega = \sqrt{k/m}$ for the spring and angular frequency $\omega = \sqrt{g/l}$ for the pendulum. At $t = 0$ the position is given by $x(0) = A \cos \beta$. The quantity β is called the phase constant and is related to the initial displacement.

Consider now the somewhat more complicated case of an oscillating system subjected to a *retarding force* such as air resistance. For the sake of a specific example, let the system be a mass on a spring acted upon by a resistive force that is proportional to the velocity.¹ Then the total force acting on the mass is not just $-kx$, but rather

$$F = -kx - b\dot{x},$$

where b is a constant of proportionality. The minus sign guarantees that the retarding force opposes the motion. The $-b\dot{x}$ term is called a *damping force*. The equation of motion will now be

$$m\ddot{x} + b\dot{x} + kx = 0. \quad (11.6)$$

This is the equation of motion for a *damped harmonic oscillator* (assuming the damping force is proportional to the first power of the velocity).

Finally, a harmonic oscillator may be subjected to some external force, F_e . In that situation we simply add the external force to the other forces (but put it on the other side of the equation) and write

$$m\ddot{x} + b\dot{x} + kx = F_e. \quad (11.7)$$

This is the equation for a *forced, damped, harmonic oscillator*.

Note that these equations of motion are all linear, ordinary, second-order differential equations with constant coefficients. You learned how to solve such equations in your course on differential equations. I will give a short review of the technique, but if you are uncertain as to any of the details please review your differential equations textbook.

Exercise 11.1

(a) Prove by substitution that Equation (11.4) and Equation (11.5) are solutions to their respective equations of motion. (b) Prove by substitution that

$$x = A \cos \sqrt{k/m}t + B \sin \sqrt{k/m}t$$

is a solution to Equation (11.1). This is one way to express the most general solution to Equation (11.1). ■

Exercise 11.2

A certain spring stretches 6 cm when a force of 6 N is applied to it. A 5 kg mass is attached to the spring which is stretched 3 cm and released with a push so that its initial speed is -6 cm/s. The system lies on a frictionless horizontal surface. Find an equation for the position of the mass as a function of time. (Hint: Use the general form given in Exercise 11.1.) Answer: $x = 0.030 \cos(4.47t) - 0.013 \sin(4.47t)$ m. ■

¹ Some 300 years ago, Charles Coulomb's experiments led him to conclude that kinetic friction is independent of velocity. Later more careful experiments showed that this is only approximately true. The velocity dependence of kinetic friction is still not well understood, but the frictional force for wetted surfaces is believed to be proportional to the velocity for low velocities.

Exercise 11.3

At time $t = 0$ a simple harmonic oscillator with angular frequency $\omega = 2$ rad/s is at $x = 3$ cm and has a speed of 1 cm/s. Obtain an expression for $x = x(t)$ and plot x vs. t . Answer: $x = 3 \cos 2t + (1/2) \sin 2t$ cm. ■

11.2 Solving the Differential Equation

An n th-order linear differential equation with constant coefficients has the form

$$a_n \frac{d^n x}{dt^n} + a_{n-1} \frac{d^{n-1} x}{dt^{n-1}} + \cdots + a_2 \frac{d^2 x}{dt^2} + a_1 \frac{dx}{dt} + a_0 x = f(t).$$

If the quantity $f(t)$ on the right-hand side is zero, the equation is called *homogeneous*. If $f(t) \neq 0$ the equation is called *inhomogeneous*. By “linear” we mean that the dependent variable (x) and its derivatives appear only to the first power. However, the independent variable (t) can be raised to any power. Thus $\frac{dx}{dt} = 3t^2$ is linear, but $\frac{dx}{dt} = 3x^2$ and $\left(\frac{dx}{dt}\right)^2 = 3x$ are not linear.²

11.2.1 Homogeneous Linear Differential Equations

A homogeneous differential equation of the form given above has a solution

$$x = e^{pt}.$$

The technique for obtaining the general solution is to plug $x = e^{pt}$ into the equation and carry out the indicated mathematical operations. This yields an equation for p in terms of the constants. I will illustrate this procedure in a moment. But first I want to mention that a first-order differential equation yields a single value for p , a second-order differential equation yields two values for p , and so on. Thus, depending on the order of the differential equation, you get several different solutions. Which one of these various solutions should you use? The answer is: all of them!

Rule 1. If $x_1(t)$ and $x_2(t)$ are both solutions of a linear homogeneous differential equation then $x_1(t) + x_2(t)$ is also a solution.

For simple harmonic motion we have a *second*-order differential equation so we will obtain *two* values for p . Let us call these two values p_1 and p_2 . The two solutions are:

$$x_1(t) = e^{p_1 t} \quad \text{and} \quad x_2(t) = e^{p_2 t}.$$

By Rule 1, the sum of these two solutions is also a solution. That is,

$$x(t) = e^{p_1 t} + e^{p_2 t}$$

is a solution.

We can generalize even further using another simple fact.

Rule 2. If $x(t)$ is a solution of a linear homogeneous differential equation, then $Cx(t)$, where C is an arbitrary constant, is also a solution.

Consequently,

$$x = C_1 e^{p_1 t} + C_2 e^{p_2 t}$$

is a solution to a second-order differential equation. In fact, it is the most general form for the solution of a homogeneous linear second-order differential equation with constant coefficients. Any other solution can be expressed as a linear combination of $e^{p_1 t}$ and $e^{p_2 t}$.

² The “degree” of a differential equation is the power of its highest-order derivative. Thus a linear differential equation is of the first degree. The “order” of a differential equation is the order of the highest derivative appearing in the equation.

Similarly, the general solution for a third-order differential equation is

$$x = C_1 e^{p_1 t} + C_2 e^{p_2 t} + C_3 e^{p_3 t},$$

and so on.

Note that the values of p ($p_1, p_2, \dots$) come from the parameters of the problem (such as the masses and spring constants) whereas the values of the constants ($C_1, C_2, \dots$) come from the initial conditions.

Let me apply these concepts to my favorite differential equation,

$$m\ddot{x} + kx = 0.$$

If $x = e^{pt}$ then $\ddot{x} = p^2 e^{pt}$. Plug in to obtain

$$mp^2 e^{pt} + ke^{pt} = 0.$$

Dividing by e^{pt} we obtain an equation for p ,

$$mp^2 + k = 0.$$

This is called the *auxiliary equation*. Solving for p gives

$$p = \pm \sqrt{-\frac{k}{m}} = \pm i \sqrt{\frac{k}{m}} = \pm i\omega,$$

where $i \equiv \sqrt{-1}$ and $\omega = \sqrt{k/m}$.

Thus we have two (linearly independent) solutions, namely,

$$x = e^{+i\omega t} \quad \text{and} \quad x = e^{-i\omega t}.$$

Because the sum of solutions is also a solution and because we can multiply each solution by an arbitrary constant, the general solution to the second-order differential equation is

$$x = C_1 e^{+i\omega t} + C_2 e^{-i\omega t}. \quad (11.8)$$

This equation is in a perfectly acceptable form. However, it can also be expressed in terms of sines and cosines, which may be preferable in some cases.

Using “Euler’s formula”

$$e^{\pm i\theta} = \cos \theta \pm i \sin \theta,$$

Equation (11.8) can be written

$$x = C_1(\cos \omega t + i \sin \omega t) + C_2(\cos \omega t - i \sin \omega t)$$

or

$$x = (C_1 + C_2) \cos \omega t + i(C_1 - C_2) \sin \omega t.$$

You might think that since x is a real quantity, you should discard the last term because it is imaginary; however, this would be unjustified because C_1 and C_2 are not yet determined and they could very well be complex numbers.

You now know basically everything you need for solving simple harmonic motion problems, except for one small detail: If the auxiliary equation for a second-order differential equation has two equal roots, then $p_1 = p_2 = p$ and the general solution is

$$x = C_1 e^{pt} + C_2 t e^{pt}. \quad (11.9)$$

This is easily verified by substitution. (Note the factor of t in the second term.)

Worked Example 11.1 Solve

$$\frac{d^2x}{dt^2} - 4x = 0,$$

given the initial conditions $x(t = 0) = 0$ m and $\dot{x}(t = 0) = 3$ m/s.

Solution Note that this is not the SHM equation because the coefficient of x is negative. To solve, substitute e^{pt} into the equation and obtain

$$p^2 - 4 = 0.$$

Therefore, $p = \pm 2$ and the solution is

$$x = C_1 e^{2t} + C_2 e^{-2t}.$$

The initial condition $x(t = 0) = 0$ leads immediately to $C_2 = -C_1$ so

$$x = C_1(e^{2t} - e^{-2t}) = 2C_1 \sinh(2t).$$

The initial condition $\dot{x} = 3$ leads to

$$\dot{x} = 2C_1(2) \cosh(2t) = 4C_1 \cosh(0) = 3.$$

$$\therefore C_1 = 3/4,$$

and

$$x = \frac{3}{2} \sinh(2t) \text{ m.}$$

Exercise 11.4

(a) Prove Rule 1 for a linear homogeneous second-order differential equation. (b) Prove Rule 2 for a linear homogeneous second-order differential equation. ■

Exercise 11.5

Verify that the relation $y = Ae^{ax} \cos bx + Be^{ax} \sin bx$ satisfies the equation $[(D - a)^2 + b^2]y = 0$, where $D \equiv \frac{d}{dx}$. ■

Exercise 11.6

Solve the equation

$$\frac{d^3x}{dt^3} - 4\frac{d^2x}{dt^2} + \frac{dx}{dt} + 6x = 0.$$

Answer: $x = C_1 e^{-t} + C_2 e^{2t} + C_3 e^{3t}$. ■

Exercise 11.7

Solve $(D^2 + 2D - 3)y = 0$, where $D \equiv \frac{d}{dx}$.

Answer: $y = C_1 e^x + C_2 e^{-3x}$. ■

Exercise 11.8

Solve $(D^2 - 6D + 9)y = 0$, where $D \equiv \frac{d}{dt}$. ■

Exercise 11.9

Obtain formulas for $\sin 2\theta$ and $\cos 2\theta$ in terms of $\sin \theta$ and $\cos \theta$. (Hint: Note that $e^{2i\theta} = (e^{i\theta})^2$.) ■

11.2.2 An Undamped Harmonic Oscillator

Consider once again a block of mass m attached to a spring of constant k . Suppose you pull the mass until the spring is stretched to x_0 . Now release it from rest. You are asked to determine the motion, $x = x(t)$ and to evaluate the amplitude and period of the oscillations.

As given by Equation (11.8), the solution has the form

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t},$$

with $\omega = \sqrt{k/m}$. The only thing you need to do now is determine the constants C_1 and C_2 . The initial conditions are

$$x(t = 0) = x_0,$$

and

$$\dot{x}(t = 0) = 0.$$

Plugging these into the solution you obtain

$$x(t = 0) = x_0 = C_1 + C_2,$$

and

$$\dot{x}(t = 0) = 0 = i\omega C_1 - i\omega C_2.$$

The second equation requires that $C_2 = C_1$. Then the first equation implies that $C_1 = C_2 = \frac{1}{2}x_0$. Therefore, the solution is

$$x = \frac{1}{2}x_0(e^{+i\omega t} + e^{-i\omega t}) = \frac{1}{2}x_0(2 \cos \omega t) = x_0 \cos \omega t.$$

This equation tells us that the mass oscillates back and forth between the values $+x_0$ and $-x_0$. The quantity x_0 is called the amplitude and is usually denoted by A . The motion takes place with a frequency

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}.$$

A plot of position as a function of time is presented in Figure 11.3. Note that the mass returns to its original position after a time t given by

$$t = \frac{2\pi}{\omega} = \frac{1}{f} = P = \text{period}.$$

The period (P) is the time for one complete oscillation. The frequency (f) is the number of oscillations per unit time and is equal to the inverse of the period. The quantity $\omega = 2\pi f$ is called the “angular frequency” but you should be aware that often it is carelessly referred to as the frequency.

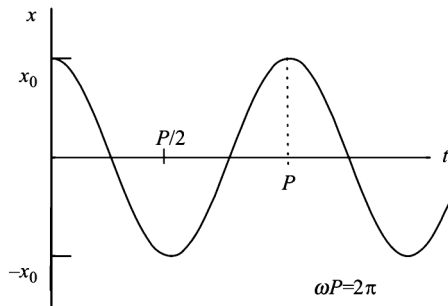


Figure 11.3 A plot of position as a function of time for a mass undergoing simple harmonic motion.

If the initial conditions are different, the form of the solution is slightly different. For example, suppose that you had been told that at $t = 0$ the block was at $x = 0$ and had a velocity v_0 . That is, $x(t = 0) = 0$ and $\dot{x}(t = 0) = v_0$. Then, using Equation (11.8) again you obtain

$$0 = C_1 + C_2,$$

and

$$v_0 = i\omega(C_1 - C_2).$$

Therefore,

$$C_1 = -C_2 \quad \text{and} \quad C_1 = v_0/2i\omega.$$

Consequently,

$$x = \frac{v_0}{2i\omega} (e^{i\omega t} - e^{-i\omega t}) = \frac{v_0}{2i\omega} (2i \sin \omega t) = \frac{v_0}{\omega} \sin \omega t. \quad (11.10)$$

The solution, $x = x(t)$, also gives information on the energy of the system. When $x = 0$, $\dot{x} = v_0$, and the potential energy is zero whereas the kinetic energy is a maximum. Also, when $x = x_0$ the potential energy is a maximum but the kinetic energy is zero. Since the total energy is constant,³

$$\frac{1}{2}mv_0^2 = \frac{1}{2}kx_0^2.$$

You can use this equation to relate x_0 and v_0 , thus:

$$x_0 = \pm \sqrt{\frac{m}{k}} v_0 = \pm \frac{v_0}{\omega}.$$

Therefore the result, Equation (11.10), can be written in the more familiar form

$$x = x_0 \sin \omega t.$$

The two examples you have just considered had very simple sets of initial conditions. In both cases C_1 and C_2 were replaced by a single constant. In general, however, the solution will contain two different constants. Then the general solution can be expressed in the form

$$x = x_0 \sin(\omega t + \beta).$$

³ Note that this result is in agreement with results from the virial theorem discussed in Worked Example 10.2. In the present problem, $n = 1$ so $\langle T \rangle = \frac{n+1}{2} \langle V \rangle$ yields $\langle T \rangle = \langle V \rangle$.

Energy of an Undamped Oscillator

The total energy ($E = T + V$) for an undamped oscillator is given by

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2,$$

where $x = A \sin(\omega t + \beta)$ and $\dot{x} = A\omega \cos(\omega t + \beta)$. Therefore,

$$\begin{aligned} E &= \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2, \\ &= \frac{1}{2}mA^2\omega^2\cos^2(\omega t + \beta) + \frac{1}{2}kA^2\sin^2(\omega t + \beta), \end{aligned}$$

but $\omega^2 = k/m$, so

$$\begin{aligned} E &= \frac{1}{2}kA^2(\cos^2(\omega t + \beta) + \sin^2(\omega t + \beta)), \\ &= \frac{1}{2}kA^2 = \frac{1}{2}m\omega^2A^2. \end{aligned}$$

Thus the total energy of a harmonic oscillator is proportional to the amplitude squared.

Worked Example 11.2 A mass of 0.25 kg is attached to a spring of force constant 1.0 N/m. The mass is displaced 0.15 m from its equilibrium point and released with zero initial velocity. Evaluate the total energy of the oscillator (kinetic plus potential). What is the maximum velocity? What is the period of the oscillation?

Solution The total energy is constant. The initial energy is $E = T + V = \frac{1}{2}mv_0^2 + \frac{1}{2}kx_0^2$. But $v_0 = 0$. $\therefore E = \frac{1}{2}kx_0^2 = \frac{1}{2}(1)(0.15)^2 = 1.13 \times 10^{-2} \text{ J}$.

The maximum velocity occurs when $x = 0$, so $E = \frac{1}{2}v_{\max}^2$. Consequently,

$$v_{\max}^2 = \frac{2E}{m} = \frac{(2)(1.13 \times 10^{-2})}{0.25},$$

and $v_{\max} = 0.3 \text{ m/s}$.

Since $\omega = \sqrt{k/m} = \sqrt{1/0.25} = 2$, the period is $P = 2\pi/\omega = \pi$.

Exercise 11.10

Show that $x = x_0 \sin(\omega t + \beta)$ is equivalent to $x = C_1 \cos \omega t + C_2 \sin \omega t$. ■

Exercise 11.11

A spring of constant 10 N/m is connected to a mass of 2 kg. It is initially at rest in an unstretched position. It is given an initial speed of 3 m/s. What is the amplitude of the oscillation? Answer: 1.34 m. ■

11.3 The Damped Harmonic Oscillator

Consider now an oscillator with a retarding force proportional to the velocity. This is a *damped* harmonic oscillator. See Figure 11.4. Common sense tells us that such an oscillator will lose energy and eventually stop.

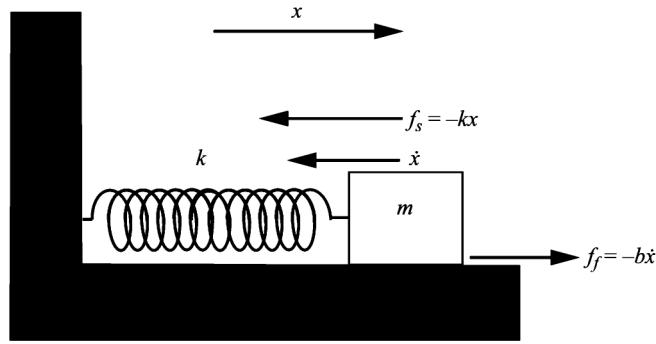


Figure 11.4 A damped harmonic oscillator. At the moment illustrated, the block is moving to the left, toward its equilibrium position. The spring force (f_s) is directed toward the left and the retarding force (f_f) is directed toward the right, opposite to the velocity.

The equation of motion of a damped harmonic oscillator is given by Equation (11.6) which I repeat here:

$$m\ddot{x} + b\dot{x} + kx = 0.$$

This is a second-order linear ordinary homogeneous differential equation with constant coefficients. As previously described, to solve it you make the substitution $x = e^{pt}$, giving

$$mp^2e^{pt} + bpe^{pt} + ke^{pt} = 0.$$

Dividing by e^{pt} yields the auxiliary equation:

$$p^2 + \frac{b}{m}p + \frac{k}{m} = 0.$$

This is a quadratic equation with two solutions:

$$p = -\frac{b}{2m} \pm \sqrt{\left(\frac{b}{2m}\right)^2 - \frac{k}{m}}. \quad (11.11)$$

Let us call the two solutions p_1 and p_2 . Since e^{p_1t} and e^{p_2t} are both solutions to the differential equation, the general solution is the sum of these:

$$x(t) = C_1e^{p_1t} + C_2e^{p_2t}, \quad (11.12)$$

where C_1 and C_2 are constants to be determined from the initial conditions.

If $\left(\frac{b}{2m}\right)^2 > \frac{k}{m}$, the quantity under the square root in Equation (11.11) is positive and the root is real. Note that the root will be smaller than $\frac{b}{2m}$ so p_1 and p_2 will be negative. Then, according to Equation (11.12), the quantity $x(t)$ will be exponentially decreasing.

On the other hand, if $\left(\frac{b}{2m}\right)^2 < \frac{k}{m}$ then the quantity under the square root is negative and p_1 and p_2 are complex. That is, they have an imaginary component (call it $i\omega$) and the solution contains terms of the form

$$e^{i\omega t} = \cos \omega t + i \sin \omega t.$$

We have reached an important conclusion. Let me repeat myself for emphasis. From the equation for p , Equation (11.11), you note that p is complex if and only if the quantity under the square root is negative, that is, iff

$$\left(\frac{b}{2m}\right)^2 < \frac{k}{m}.$$

In this case the system oscillates. On the other hand, if

$$\left(\frac{b}{2m}\right)^2 \geq \frac{k}{m},$$

then p is real and the motion dies out exponentially.

The relation of k/m to $(b/2m)^2$ characterizes the three types of motion for the damped oscillator. These are called “underdamped,” “overdamped,” and “critically damped.” They are described in the following table.

Name	Condition	Description
underdamped	$(b/2m)^2 < k/m$	oscillatory motion
critically damped	$(b/2m)^2 = k/m$	exponential decrease
overdamped	$(b/2m)^2 > k/m$	exponential decrease

The motion of the damped oscillator depends on the relative values of the parameters b, k, m . If $b/2m$ is large, the system is subjected to a considerable retarding force and the oscillations will die out quickly. On the other hand, if $b/2m$ is small, there is little damping and the oscillator will remain in motion for a long time. Furthermore, if k/m is large, the frequency of oscillation will be large, but if k/m is small, the system will oscillate slowly.

11.3.1 The Underdamped Oscillator

Let us begin by considering the underdamped oscillator. Then, $(b/2m)^2 < k/m$ and Equation (11.11) shows that p will be a complex quantity. It is convenient to change notation at this point. Because $\sqrt{k/m}$ is the “natural” frequency of the undamped oscillator, we shall denote k/m by ω_0^2 . Furthermore, $b/2m$ can be denoted by γ the “damping coefficient.” The new notation is :

$$\begin{aligned} \frac{k}{m} &= \omega_0^2 = \text{square of frequency of undamped oscillator,} \\ \frac{b}{2m} &= \gamma = \text{damping coefficient.} \end{aligned}$$

Using this notation, Equation (11.11) becomes

$$p = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}. \quad (11.13)$$

An underdamped oscillator has $\gamma^2 < \omega_0^2$ so the last term is the square root of a negative number, and p is complex. In that case it is convenient to write

$$\sqrt{(-1)(\omega_0^2 - \gamma^2)} = i\omega_1,$$

where $\omega_1 = \sqrt{\omega_0^2 - \gamma^2}$ is a real number. Then

$$p = -\gamma \pm i\omega_1,$$

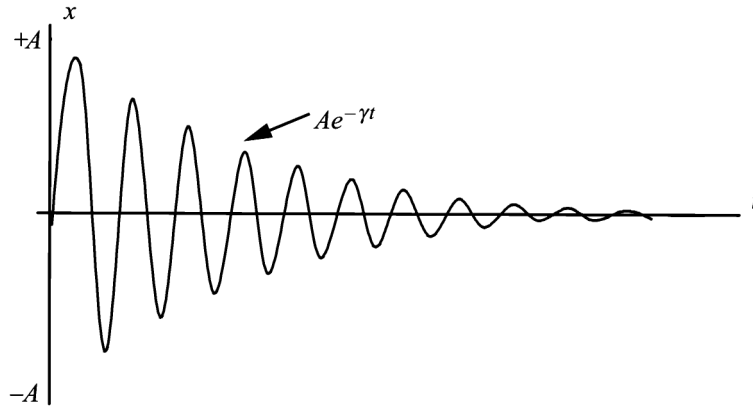


Figure 11.5 The motion of an underdamped oscillator consists of rapid oscillations (described by $\cos(\omega_1 t + \theta)$) modulated by an exponentially decreasing amplitude (described by $Ae^{-\gamma t}$).

a form that clearly shows that p is complex. Inserting the two values of p into the solution (11.12) gives

$$\begin{aligned} x(t) &= C_1 e^{(-\gamma + i\omega_1)t} + C_2 e^{(-\gamma - i\omega_1)t} \\ &= e^{-\gamma t} (C_1 e^{i\omega_1 t} + C_2 e^{-i\omega_1 t}). \end{aligned}$$

C_1 and C_2 are complex constants related to the initial conditions. These constants can be written in a variety of different ways. You obtain a very nice-looking expression if you write them in the form

$$C_1 = \frac{1}{2} A e^{i\theta} \text{ and } C_2 = \frac{1}{2} A e^{-i\theta}. \quad (11.14)$$

This basically replaces the two complex constants C_1 and C_2 by the real constants A and θ . Then

$$x(t) = e^{-\gamma t} \frac{1}{2} A (e^{i(\omega_1 t + \theta)} + e^{-i(\omega_1 t + \theta)}),$$

or

$$x(t) = A e^{-\gamma t} \cos(\omega_1 t + \theta). \quad (11.15)$$

Plotting x vs. t generates an oscillatory solution (due to the cosine term), but the amplitude of the oscillations is $A e^{-\gamma t}$, so the amplitude decreases exponentially with time. This behavior is illustrated in Figure 11.5.

Note that the oscillations have an angular frequency ω_1 , which is smaller than the natural frequency ω_0 . That is, the damped system oscillates more slowly than the undamped system. You probably expected this type of behavior.

Worked Example 11.3 An underdamped harmonic oscillator has $k = 2 \text{ N/m}$, $m = 1 \text{ kg}$, and $b = 0.1 \text{ kg/s}$. How many oscillations does the system make before the amplitude decreases to $1/e$ of its initial value?

Solution The time when the amplitude is $1/e$ of its initial value is obtained from

$$Ae^{-\gamma t} = \frac{1}{e}A \quad \therefore t = 1/\gamma = \frac{2m}{b} = 20 \text{ s.}$$

Then,

$$\begin{aligned}\omega_1 t &= \left(\sqrt{\omega_0^2 - \gamma^2} \right) t = \left(\sqrt{k/m - \gamma^2} \right) t \\ &= \left(\sqrt{2 - 0.05^2} \right) (20) = 28.26 \text{ rad,}\end{aligned}$$

and the number of oscillations is $28.26/2\pi = 4.5$.

Exercise 11.12

The complex constants C_1 and C_2 can be written in the form $C_1 = a_1 + ib_1$ and $C_2 = a_2 + ib_2$. But there can only be two independent constants, so a_1, b_1 and a_2, b_2 are not all independent. Using Equation (11.14) show that $a_2 = a_1$ and $b_2 = -b_1$. ■

Exercise 11.13

Obtain the general solution of the following differential equations: (a) $m\ddot{x} + b\dot{x} - kx = 0$, (b) $m\ddot{x} - b\dot{x} + kx = 0$. Assume m, b , and k are positive real quantities. ■

11.3.2 The Overdamped Oscillator

Now consider the *overdamped* oscillator, defined to be an oscillator for which

$$\left(\frac{b}{2m} \right)^2 > \frac{k}{m}.$$

According to Equation (11.11) or Equation (11.13), p will now have two *real* values denoted γ_1 and γ_2 . Using the notation of Equation (11.13), the solutions of the auxiliary equation are

$$p_1 = -\gamma_1 = -\gamma - \sqrt{\gamma^2 - \omega_0^2},$$

and

$$p_2 = -\gamma_2 = -\gamma + \sqrt{\gamma^2 - \omega_0^2}.$$

Note that both γ_1 and γ_2 are positive and that γ_1 is greater than γ_2 .

Plugging these expressions into the general solution of the differential equation (11.12) leads to

$$x(t) = C_1 e^{-\gamma_1 t} + C_2 e^{-\gamma_2 t}, \quad (11.16)$$

where γ_1 and γ_2 are real. Both terms are exponentially decreasing with time, as indicated in Figure 11.6. Because $\gamma_1 > \gamma_2$ we note that $e^{-\gamma_1 t}$ decays more rapidly than $e^{-\gamma_2 t}$.

The motion of the oscillator (that is, the value of x as a function of time) is, of course, the sum of two curves, as illustrated in Figure 11.6. The constants C_1 and C_2 depend on the initial conditions. For example, if you pull the block to some position x_0 and release it from rest ($v_0 = 0$), at $t = 0$, Equation (11.16) reduces to

$$x_0 = C_1 + C_2 \quad \Rightarrow \quad C_1 = x_0 - C_2.$$

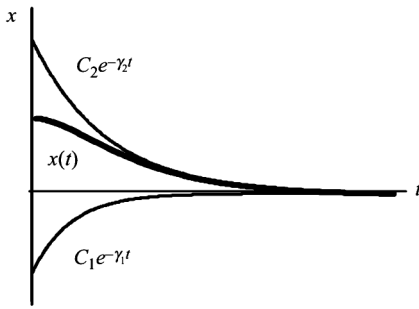


Figure 11.6 An overdamped harmonic oscillator. The motion is the sum of the two curves. The constants C_1 and C_2 depend on the initial conditions.

Since $\dot{x} = -\gamma_1 C_1 e^{-\gamma_1 t} - \gamma_2 C_2 e^{-\gamma_2 t}$,

$$v_0 = -\gamma_1 C_1 - \gamma_2 C_2.$$

But $v_0 = 0$, so $C_2 = -(\gamma_1 C_1 / \gamma_2)$, and hence,

$$C_1 = x_0 - C_2 = x_0 + (\gamma_1 / \gamma_2) C_1.$$

Therefore,

$$C_1 = -\frac{\gamma_2}{\gamma_1 - \gamma_2} x_0,$$

and

$$C_2 = \frac{\gamma_1}{\gamma_1 - \gamma_2} x_0.$$

Consequently,

$$x(t) = -\frac{\gamma_2}{\gamma_1 - \gamma_2} x_0 e^{-\gamma_1 t} + \frac{\gamma_1}{\gamma_1 - \gamma_2} x_0 e^{-\gamma_2 t}.$$

Depending on the initial conditions the oscillator will behave in one of three ways. The displacement $x(t)$ may rapidly decay to zero. Or $x(t)$ may increase for a time and then decay to zero. Finally, $x(t)$ could change sign, reach a local maximum or minimum, then approach zero.

Exercise 11.14

Consider the motion of an overdamped oscillator. Assume that initially $x(t = 0) = 1$ and $\dot{x}(t = 0) = 0$. Let $\gamma_1 = 3.414$ and $\gamma_2 = 0.586$. Plot $x(t)$. ■

Exercise 11.14

Obtain C_1 and C_2 for an overdamped oscillator if the initial conditions are $x(t = 0) = 0$ and $\dot{x}(t = 0) = v_0$. Answer: $C_1 = v_0 / (\gamma_2 - \gamma_1)$; $C_2 = -C_1$. ■

11.3.3 The Critically Damped Oscillator

Finally, consider the case of a critically damped harmonic oscillator for which

$$\left(\frac{b}{2m} \right)^2 = \frac{k}{m}.$$

Now the term under the root in Equation (11.11) is zero.

For convenience, define the quantity γ_c as follows:

$$\gamma_c = \frac{b}{2m} = \sqrt{\frac{k}{m}} = \omega_0.$$

Then Equation (11.13) yields a single value for p namely $p = -\gamma_c$. Recall that if p has only one value, the solution to a second-order differential equation has the form of Equation (11.9), that is

$$x(t) = C_1 e^{-\gamma_c t} + C_2 t e^{-\gamma_c t},$$

or

$$x(t) = (C_1 + C_2 t) e^{-\gamma_c t}.$$

For critical damping x decays quickly to zero.

Exercise 11.13

A critically damped harmonic oscillator is initially at its equilibrium position with velocity v_0 . Obtain expressions for C_1 and C_2 . Answer: $C_1 = 0$, $C_2 = v_0$. ■

Exercise 11.17

A critically damped harmonic oscillator has $b = 0.3$ kg/s and $k = 0.4$ N/m. (You are not given the mass.) It is released from rest at $x = 0.04$ m. Evaluate the constants C_1 and C_2 . Answer: $C_1 = 0.04$, $C_2 = 0.107$ (in meters). ■

11.4 The Forced Harmonic Oscillator

11.4.1 Statement of the Problem

A *forced* harmonic oscillator is an oscillator which is subjected to some sort of driving force. The external driving force is often periodic with the same frequency as the oscillator. When you push your little brother on a swing, you give the seat a shove each time it comes back to you at the top of its swing. The periodicity of your shoves is the same as the natural period of the swing. Similarly, the pendulum in a “grandfather” clock receives a small push on each swing. An old-fashioned pocket watch has an oscillating spring wheel that receives small periodic impulses from a spring and ratchet system.

The oscillator will, in general, be damped, so the equation of motion for such a system is

$$m\ddot{x} + b\dot{x} + kx = F(t), \quad (11.17)$$

where $F(t)$ is the driving force. The purpose of the driving force is to overcome the damping. If $F(t)$ is too small, the oscillations will gradually die out. On the other hand, if $F(t)$ is too large, the amplitude may grow unacceptably large. (If you push the swing too hard it will “go over the top,” leading to unpleasant consequences.) As you shall see shortly, the amplitude of the oscillations also depends on the *frequency* of the driving force. This is related to the phenomenon of resonance that we will consider later.

The equation of motion of the driven harmonic oscillator Equation (11.17) is an *inhomogeneous* second-order linear differential equation. I will now make a short aside to describe how to solve such an equation.

11.4.2 Inhomogeneous Linear Ordinary Differential Equations

The title of this section describes the kind of equation we want to solve. You should understand the implication of each of the many longish words in the description. The equation is, of course, a *differential* equation because there are derivatives in it. It is *linear* because the dependent variable (x) and its derivatives are not raised to a power larger than the first. We are dealing with an *ordinary* differential equation because Equation (11.17) has no partial derivatives in it.

You probably recall that differential equations are often expressed in terms of the *differential operator* $D \equiv \frac{d}{dt}$. Then $D^2 = \frac{d^2}{dt^2}$ and so on. A linear *homogeneous* differential equation is an equation which can be put in the form

$$a_n D^n x + a_{n-1} D^{n-1} x + \cdots + a_1 D x + a_0 x = 0.$$

Although the a_i s are usually constants, they can be functions of t . Note the zero on the right-hand side.

Of course, the equation need not have a zero on the right. There could be a function of x and t on the right-hand side. Thus,

$$a_n D^n x + a_{n-1} D^{n-1} x + \cdots + a_1 D x + a_0 x = f(x, t).$$

Such an equation is said to be *inhomogeneous*. The solution of an inhomogeneous differential equation can be determined using the following rule.

Rule 3. The solution of an *inhomogeneous* differential equation is the sum of the general solution of the corresponding *homogeneous* equation and a *particular solution* of the inhomogeneous equation.

Therefore, the first step in solving an inhomogeneous differential equation is to set the right-hand side equal to zero. This generates the corresponding homogeneous differential equation which we can solve by the method of Section 11.2. The general solution will, of course, contain as many arbitrary constants as the order of the equation. A second-order equation has a general solution with two arbitrary constants. We denote this general solution of the homogeneous equation by x_g . It is usually called the “complementary function.” Next we consider the *inhomogeneous* equation. We can sometimes figure out a solution by inspection. We denote this *particular* solution by x_p . The *general* solution of the inhomogeneous equation is $x_p + x_g$. It is easy to prove this assertion, and it is left as an exercise for you. We now return to the problem of the driven harmonic oscillator.

Exercise 11.18

Show that the general solution of an inhomogeneous differential equation is the sum of a particular solution and the general solution of the homogeneous differential equation. ■

11.4.3 Obtaining the Particular Solution

The equation of motion for the driven, damped harmonic oscillator is

$$m\ddot{x} + b\dot{x} + kx = F(t).$$

The forcing term, $F(t)$, is often periodic because that will keep the oscillator in motion, counteracting the effect of damping. Another common situation is an oscillator that has been subjected to an impulsive force. In general, the forcing term can be any function of time. It could even be a constant. In fact, for the sake of making the analysis simple, I will begin with a constant force. For example, a mass hanging from a spring in an external gravitational field (see Figure 11.7) has the equation of motion

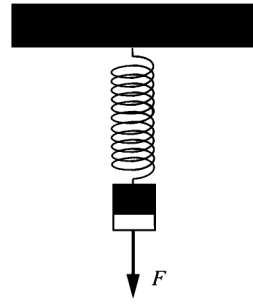


Figure 11.7 A mass hanging from a spring in a constant gravitational field.

$$m\ddot{x} + b\dot{x} + kx = mg, \quad (11.18)$$

where the damping could be due to air resistance, for example. The mass is only allowed to oscillate vertically; it does not simultaneously swing like a pendulum.

The corresponding homogeneous equation was solved in the previous section. If the oscillation is underdamped, for example, the general solution of the homogeneous equation is given by Equation (11.15) as:

$$x_g = x(t) = Ae^{-\gamma t} \cos(\omega_1 t + \theta).$$

Now you need to find a particular solution to Equation (11.18). That is, you need to find a function of x which satisfies (11.18). After a little creative staring at that equation you will realize that if x is constant, the first two terms are zero and you are left with $kx = mg$. Therefore, $x = mg/k$ is a possible solution to the equation. It is the *particular* solution, x_p . The general solution of the inhomogeneous equation is, therefore,

$$x(t) = x_g + x_p = Ae^{-\gamma t} \cos(\omega_1 t + \theta) + \frac{mg}{k}.$$

This tells you that the effect of the gravitational force is just to stretch the spring, giving a different (lower) value for the equilibrium point.

I will now go on to a more interesting problem, namely a driving force that varies sinusoidally.⁴ Let the driving frequency be denoted ω_d and assume $F(t) = F_0 \sin \omega_d t$. For simplicity, we shall begin with an *undamped* oscillator. Then the equation of motion is

$$m\ddot{x} + kx = F_0 \sin \omega_d t,$$

or

$$\ddot{x} + \omega_0^2 x = \frac{F_0}{m} \sin \omega_d t. \quad (11.19)$$

The general solution of the homogeneous equation is

$$x_g(t) = A \cos(\omega_0 t + \theta_0),$$

where A and θ_0 depend on the initial conditions. For example, $\theta_0 = 0$ corresponds to the oscillator at position $x = A$ at time $t = 0$. (For a pendulum this means that at time zero the bob was at the top of the swing, and for a mass on a spring, it means that at time zero the spring was fully extended.)

⁴ Recall that any periodic function can be expressed as a sum of sines and cosines (a “Fourier series”) so it is reasonable to consider a sinusoidal driving force.

Having a general solution to the homogeneous equation, you still need a particular solution to the inhomogeneous equation. Previously, we obtained a particular solution by inspection. This is usually not possible, so fortunately there are techniques available for determining a particular solution.⁵ One technique that always works for a second-order differential equation is to integrate the equation twice. To apply the method, first write the equation in the form

$$(a_2 D^2 + a_1 D + a_0)x = f(t). \quad (11.20)$$

This equation can be expressed as the product

$$(D - b_1)(D - b_2)x = f(t). \quad (11.21)$$

Let $u = (D - b_2)x$. Then the differential equation is

$$(D - b_1)u = f(t).$$

You now have a first-order linear differential equation that you can solve for $u(t)$. Next, going back to the definition of u , you solve

$$(D - b_2)x(t) = u(t)$$

for $x(t)$. This technique produces the general solution for the inhomogeneous equation, that is, it generates the complementary function as well as the particular solution. However, it may involve a considerable amount of mathematics, so I want to show you two other techniques which might be classified as “tricks” for finding the particular solution. (When a mathematical trick becomes very familiar we quit calling it a trick and start calling it a technique!)

Trick #1

The first trick (or technique) is used when the function on the right-hand side is of the form $f(t) \propto e^{at}$. Then, writing the second-order differential equation in the form

$$(D - b_1)(D - b_2)x = K e^{at}, \quad (11.22)$$

the particular solution is

$$\begin{aligned} x_p &= C e^{at} && \text{if } a \neq b_1 \text{ and } a \neq b_2 \text{ and } b_1 \neq b_2, \\ x_p &= C t e^{at} && \text{if } a = b_1 \text{ or } a = b_2 \text{ but } b_1 \neq b_2, \\ x_p &= C t^2 e^{at} && \text{if } a = b_1 = b_2, \end{aligned} \quad (11.23)$$

where C is determined by plugging x_p back into the differential equation.

Trick #2

The second trick is based on the first trick and we use it when $f(t)$ (the right-hand side) is either a sine or a cosine. In that case, write the equation in one of the two following forms:

$$(D - b_1)(D - b_2)\xi_p = \begin{Bmatrix} K \sin at \\ K \cos at \end{Bmatrix}, \quad (11.24)$$

where ξ_p is a complex function of t . Note that $K \sin at$ is the imaginary part of $K e^{iat}$ and $K \cos at$ is the real part of $K e^{iat}$. We then solve

$$(D - b_1)(D - b_2)\xi_p = K e^{iat}$$

using formulas (11.23) and take either the real or the imaginary part of the answer. That is, the particular solution x_p is either $\text{Re}(\xi_p)$ or $\text{Im}(\xi_p)$.

⁵ See, for example, Mary L. Boas, *Mathematical Methods in the Physical Sciences*, 3rd ed., John Wiley & Sons Inc., Hoboken, NJ, 2006, p. 417 ff.

Finally, we can generalize the first trick if the right-hand side is an exponential, e^{at} , times a polynomial of degree n , $P_n(t)$. Then the particular solution of

$$(D - b_1)(D - b_2)x = e^{at} P_n(t)$$

is

$$x_p = \begin{cases} e^{at} Q_n(t) & \text{if } a \neq b_1 \text{ and } a \neq b_2 \\ t e^{at} Q_n(t) & \text{if } a = b_1 \text{ or } a = b_2 \text{ and } b_1 \neq b_2 \\ t^2 e^{at} Q_n(t) & \text{if } a = b_1 = b_2 \end{cases},$$

where $Q_n(t)$ is a polynomial of degree n :

$$Q_n(t) = A + Bt + Ct^2 + \dots$$

To determine $A, B, C, \dots$ you plug x_p into the differential equation and equate the coefficients of the different powers of t .

Worked Example 11.4 Find the general solution for

$$(D^2 - 4)x = 2 - 8t$$

when $t = 0$, $x = 0$, and $\dot{x} = 5$.

Solution We use the method described by [Equation \(11.21\)](#) and the discussion following it. We write the differential equation as

$$(D - 2)(D + 2)x = 2 - 8t.$$

Let $u = (D + 2)x$ and write

$$(D - 2)u = 2 - 8t.$$

For a first-order differential equation such as

$$(D + f(t))x = g(t),$$

the integrating factor is $e^{\int f(t)dt} = e^{-2t}$. Multiplying by e^{-2t} we obtain

$$\begin{aligned} e^{-2t} \frac{du}{dt} - 2ue^{-2t} &= 2e^{-2t} - 8te^{-2t}, \\ \frac{d}{dt}(ue^{-2t}) &= 2e^{-2t} - 8te^{-2t}, \\ \int d(ue^{-2t}) &= \int 2e^{-2t} dt - \int 8te^{-2t} dt. \end{aligned}$$

Therefore,

$$u = C_1 e^{2t} + 4t + 1.$$

But since $u = (D + 2)x$ we have

$$\frac{dx}{dt} + 2x = C_1 e^{2t} + 4t + 1.$$

Now the integrating factor is e^{+2t} so we multiply through by this factor to get

$$\frac{d}{dt}(xe^{2t}) = C_1 e^{4t} + 4te^{2t} + e^{2t},$$

and consequently,

$$x(t) = C_2 e^{-2t} + \frac{1}{4} C_1 e^{2t} + 2t - \frac{1}{2}.$$

Inserting the initial conditions leads to $C_1 = 4$ and $C_2 = -1/2$, and finally

$$x(t) = e^{2t} - \frac{1}{2} e^{-2t} + 2t - \frac{1}{2}.$$

Note that this method yields the complete general solution, that is, it gives us the sum of the particular solution and the complementary function all at the same time.

Exercise 11.19

Obtain the general solution to $(D^2 - D - 6)x = 8$, where $D \equiv \frac{d}{dt}$. Answer: $x(t) = C_1 e^{3t} + C_2 e^{-2t} - \frac{4}{3}$. ■

Exercise 11.20

Obtain the general solution to $(D^2 - 9)x = 5e^{-2t}$ assuming that $x(t = 0) = 0$ and $\dot{x}(t = 0) = 1$ (in the appropriate set of units). Answer: $x(t) = \frac{1}{3}e^{3t} + \frac{2}{3}e^{-3t} - e^{-2t}$. ■

Exercise 11.21

Obtain the general solution of $(D^2 - 4)x = \sin t$. Answer: $x(t) = C_1 e^{2t} + C_2 e^{-2t} - \frac{1}{5} \sin t$. ■

11.4.4 The Forced Undamped Oscillator

Now that you have seen some methods for solving inhomogeneous second-order differential equations, let us return to the problem of the forced undamped harmonic oscillator.

The differential equation we set out to solve was [Equation \(11.19\)](#):

$$\ddot{x} + \omega_0^2 x = \frac{F_0}{m} \sin \omega_d t.$$

The right-hand side of this equation is proportional to $\sin \omega_d t$ so we use trick #2 and write

$$(D - i\omega_0)(D + i\omega_0)\xi_p(t) = \frac{F_0}{m} e^{i\omega_d t}. \quad (11.25)$$

According to trick #2 and [Equation \(11.23\)](#) the particular solution of [Equation \(11.25\)](#) is

$$\xi_p(t) = C e^{i\omega_d t}.$$

To obtain C , plug ξ_p back into the differential equation, thus:

$$\left(\frac{d^2}{dt^2} + \omega_0^2 \right) C e^{i\omega_d t} = \frac{F_0}{m} e^{i\omega_d t}.$$

Hence,

$$-\omega_d^2 C + \omega_0^2 C = \frac{F_0}{m},$$

and

$$C = \frac{F_0/m}{\omega_0^2 - \omega_d^2}.$$

Recall that x_p is the imaginary part of the complex function ξ_p , so the particular solution is

$$x_p(t) = \mathcal{I}m(\xi_p(t)) = \mathcal{I}m\left(\frac{F_0/m}{\omega_0^2 - \omega_d^2} e^{i\omega_d t}\right) = \frac{F_0/m}{\omega_0^2 - \omega_d^2} \sin \omega_d t,$$

and the general solution of Equation (11.19) is

$$x(t) = A \cos(\omega_0 t + \theta_0) + \frac{F_0/m}{\omega_0^2 - \omega_d^2} \sin \omega_d t.$$

The problem is now solved, except for determining A and θ_0 from the initial conditions.

Before leaving this problem, I want you to notice that there is something quite interesting about the particular solution,

$$x_p = \frac{F_0/m}{\omega_0^2 - \omega_d^2} \sin \omega_d t.$$

This expression indicates that as the driving frequency ω_d gets closer and closer to ω_0 , the natural frequency of oscillation, the amplitude of the oscillations get larger and larger. If $\omega_d = \omega_0$, the amplitude is infinite. This phenomenon is known as *resonance*. We will consider resonance further in the next section, refining our analysis and showing that the amplitude does not actually become infinite for a real physical system.

Exercise 11.22

A simple harmonic oscillator consists of a mass of 3 kg on a spring of force constant 0.15 N/m. Determine the frequency of the driving force which will cause the amplitude of oscillations to grow extremely large. Answer 0.22 rad/s. ■

11.4.5 The Forced Damped Oscillator

We now consider a *damped*, driven, harmonic oscillator and assume the driving force is, once again, $F_0 \sin \omega_d t$. The equation of motion is

$$\ddot{x} + \frac{b}{m} \dot{x} + \frac{k}{m} x = \frac{F_0}{m} \sin \omega_d t.$$

Note that $\sqrt{k/m} = \omega_0$ is the “natural” frequency of the undamped oscillator. For the sake of a specific example, we assume the oscillator is underdamped.

The technique for finding the particular solution is the same as before, so set

$$\xi_p = C e^{i\omega_d t},$$

and plug it into the differential equation. This gives

$$C(-\omega_d^2) e^{i\omega_d t} + \frac{b}{m} C(i\omega_d) e^{i\omega_d t} + \omega_0^2 C e^{i\omega_d t} = \frac{F_0}{m} e^{i\omega_d t},$$

where $\sin \omega_d t$ was replaced by $e^{i\omega_d t}$. (This means that at the end of the procedure we must only keep the imaginary part of the solution.)

Solving for C yields

$$C = \frac{F_0/m}{(\omega_0^2 - \omega_d^2) + i\omega_d b/m},$$

which is complex. To make this quantity easier to manipulate, the imaginary term is usually placed in the numerator. To achieve this, multiply top and bottom by the complex conjugate of the denominator and obtain

$$C = \frac{(F_0/m) [(\omega_0^2 - \omega_d^2) - i\omega_d b/m]}{[(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2]}.$$

You have already seen that the general solution of the homogeneous equation for the underdamped harmonic oscillator is given in Equation (11.15) as $x = Ae^{-\gamma t} \cos(\omega_1 t + \theta)$. This is the complementary function for the inhomogeneous equation. The general solution of the forced, underdamped harmonic oscillator is, then,

$$x(t) = Ae^{-\gamma t} \cos(\omega_1 t + \theta) + \mathcal{I}m \left(\frac{F_0/m [(\omega_0^2 - \omega_d^2) - i\omega_d b/m]}{[(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2]} e^{i\omega_d t} \right).$$

Eventually, the first term will die out. (This is, of course, also true for overdamped and critically damped oscillators.) Dropping the first term, you are left with

$$x(t) = \mathcal{I}m \left(\frac{F_0/m [(\omega_0^2 - \omega_d^2) - i\omega_d b/m]}{[(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2]} (\cos \omega_d t + i \sin \omega_d t) \right).$$

After a bit of algebra, this expression can be written

$$x(t) = \frac{F_0}{m} \frac{(\omega_0^2 - \omega_d^2) \sin \omega_d t - (\omega_d b/m) \cos \omega_d t}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2}. \quad (11.26)$$

At resonance, when $\omega_0 = \omega_d$, the denominator is minimized, but the damping term prevents it from going to zero. Therefore, at resonance you get a large amplitude, but it is not infinite. In real physical systems, there is always some damping that prevents the system from having infinite amplitude at resonance.

Equation (11.26) can be expressed in a more convenient way by using the trigonometric identity

$$\mathcal{A} \sin \omega t + \mathcal{B} \cos \omega t = \sqrt{\mathcal{A}^2 + \mathcal{B}^2} \cos(\omega t - \phi),$$

where $\phi = \tan^{-1}(\mathcal{A}/\mathcal{B})$. So,

$$\begin{aligned} x(t) &= \left[\frac{(F_0/m)^2 [(\omega_0^2 - \omega_d^2)^2 + \omega_d^2 b^2/m^2]}{[(\omega_0^2 - \omega_d^2)^2 + \omega_d^2 b^2/m^2]^2} \right]^{1/2} \cos(\omega_d t - \phi) \\ &= \frac{F_0}{[m^2(\omega_0^2 - \omega_d^2)^2 + \omega_d^2 b^2]^{1/2}} \cos(\omega_d t - \phi), \end{aligned} \quad (11.27)$$

where, from Equation (11.26),

$$\phi = \tan^{-1} \left[-\frac{\omega_0^2 - \omega_d^2}{\omega_d b/m} \right].$$

It is interesting to note that the amplitude of $x(t)$ is not maximized at $\omega_d = \omega_0$ but rather at a nearby frequency called the resonant frequency which you can determine by evaluating the derivative of the denominator and setting it equal to zero, thus:

$$\frac{d}{d\omega_d} \left[m^2(\omega_0^2 - \omega_d^2)^2 + \omega_d^2 b^2 \right]^{1/2} = 0.$$

This leads to the following condition for a minimum

$$\omega_d^2 = \omega_0^2 - \frac{b^2}{2m^2}. \quad (11.28)$$

The greatest amplitude occurs when the driving frequency is equal to the resonant frequency which we can denote by ω' .

$$\omega' = \sqrt{\omega_0^2 - \frac{b^2}{2m^2}}.$$

This is neither the frequency of the undamped oscillator (ω_0), nor the frequency of the damped oscillator (ω_1).

When $\omega_d = \omega'$ the amplitude of the oscillation is given by

$$A = \frac{F_0}{\left[m^2(\omega_0^2 - \omega'^2)^2 + \omega'^2 b^2 \right]^{1/2}}.$$

The resonance phenomenon is often associated with the transfer of energy. The average energy of a harmonic oscillator is proportional to the amplitude squared, so an important and useful parameter in studying resonance is A^2 , given by

$$A^2 = \frac{F_0^2}{m^2(\omega_0^2 - \omega'^2)^2 + \omega'^2 b^2}.$$

Worked Example 11.5 Show that the phase angle ϕ in [Equation \(11.27\)](#) is given by $\phi = \tan^{-1} \left[-\frac{\omega_0^2 - \omega_d^2}{\omega_d b/m} \right]$.

Solution We have shown in [Equation \(11.26\)](#) that

$$x = x(t) = \frac{F_0}{m} \frac{(\omega_0^2 - \omega_d^2) \sin \omega_d t - (\omega_d b/m) \cos \omega_d t}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2}.$$

Writing this in the form $\mathcal{A} \sin \omega t + \mathcal{B} \cos \omega t$ we have

$$\mathcal{A} = \frac{F_0}{m} \frac{(\omega_0^2 - \omega_d^2)}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2},$$

and

$$\mathcal{B} = \frac{F_0}{m} \frac{-(\omega_d b/m)}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2}.$$

So $\phi = \tan^{-1}(A/B)$ yields

$$\begin{aligned}\phi &= \tan^{-1} \frac{\frac{(\omega_0^2 - \omega_d^2)}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2}}{\frac{-(\omega_d b/m)}{(\omega_0^2 - \omega_d^2)^2 + (\omega_d b/m)^2}} = \tan^{-1} \frac{(\omega_0^2 - \omega_d^2)}{-(\omega_d b/m)} \\ &= \tan^{-1} \frac{-(\omega_d^2 - \omega_0^2)}{\omega_d b/m}.\end{aligned}$$

The phase shift ϕ describes the lag between the applied force $F(t)$ and the response $x(t)$.

Exercise 11.23

Fill in the missing steps in the derivation of Equation (11.28). ■

Exercise 11.23

Show that $A \sin \omega t + B \cos \omega t = \sqrt{A^2 + B^2} \cos(\omega t - \phi)$, where $\tan \phi = A/B$. ■

Exercise 11.25

A damped harmonic oscillator is driven by a force $F_0 e^{-at}$. Find the general solution. (Note: The particular solution can be expected to have the same time dependence as the driving force.) ■

The Q Factor

The Q factor is an important property of a damped driven harmonic oscillator. It is related to both the shape of the resonance curve and the energy loss rate of the oscillator.⁶

There are several related definitions of the Q factor. These yield similar but not identical results. You should be aware that in specific applications the Q factor may be expressed differently than the way we will describe here, but the basic ideas are always the same.

Let us begin with the definition of the Q factor in terms of the *shape* of the resonance curve. An appropriate definition of Q is the ratio of the resonant frequency (ω_0) to the width of the resonance curve at half maximum of the curve. ($\Delta\omega$ is denoted “full width at half maximum.”) That is, $Q = \frac{\omega_0}{\Delta\omega}$ which, as we shall show, can be written as

$$Q \equiv \frac{m}{b} \omega_0 = \sqrt{\frac{km^2}{b^2}}.$$

Note that the definition only involves the physical properties of the oscillator. As mentioned previously, a driven oscillator will have the largest amplitude when the frequency of the driving force is equal to the resonant frequency, which we have denoted ω' . For simplicity, we shall consider systems with small damping values, so $\omega' \doteq \omega_0 = \sqrt{k/m}$. A plot of the square

⁶ There are many applications of driven harmonic oscillators. For example, in musical instruments such as trumpets or French horns the vibration of a reed or of the musician's lips produces sounds with a wide range of frequencies. The instrument has a resonance at the desired frequency, thus picking out the desired note. In optical devices, such as the resonant cavity of lasers, the Q factor is an important parameter. Electromechanical resonators such as quartz crystals in digital watches allow for highly accurate timepieces.

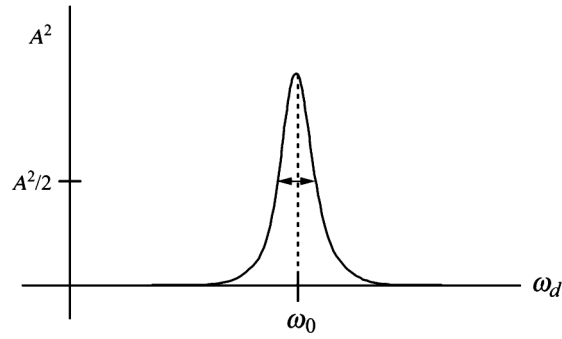


Figure 11.8 The resonance curve with full width at half maximum indicated.

of the amplitude as a function of driving frequency (ω_d) will look somewhat as illustrated in Figure 11.8.

To obtain a large response at a specific frequency the curve of Figure 11.8 should be very tall and very thin. In technical terms, it is desirable for the curve to have a small value for the width at half maximum (as shown by the double-headed arrow in the figure).

We have presented the resonance curve in terms of the amplitude squared rather than the amplitude. This is convenient because the energy of the oscillator is proportional to the square of the amplitude. Furthermore, a slightly different definition of Q is the ratio of the maximum energy stored in the oscillator to the energy lost per radian of motion. (We explore this alternative definition in Problem 11.16.)

The square of the amplitude is a maximum at $\omega_d = \omega_0$ and is given by

$$A_{\omega_0}^2 = \frac{F_0^2}{\omega_0 b^2}.$$

For some nearby driving frequency ω the amplitude is

$$A_{\omega}^2 = \frac{F_0^2/m^2}{[(\omega_0^2 - \omega^2)^2 + \omega^2 b^2/m^2]}.$$

If $A_{\omega}^2 = \frac{1}{2} A_{\omega_0}^2$ we have

$$\begin{aligned} 2\omega_0^2 b^2/m^2 &= (\omega_0^2 - \omega^2)^2 + \omega^2 b^2/m^2, \\ \frac{b^2}{m^2}(2\omega_0^2 - \omega^2) &= (\omega_0^2 - \omega^2)^2, \end{aligned}$$

and for ω close to ω_0 , we can write the approximate relation

$$\frac{b}{m}\omega_0 = \omega_0^2 - \omega^2.$$

Now $(\omega_0^2 - \omega^2) = (\omega_0 - \omega)(\omega_0 + \omega) \approx \Delta\omega_{1/2}(2\omega_0)$, where $\Delta\omega_{1/2} = \omega - \omega_0$ is the half width at half maximum. The full width at half maximum is $\Delta\omega = 2\Delta\omega_{1/2}$, so

$$\frac{b\omega_0}{m} = 2\Delta\omega_{1/2} \cdot \omega_0 = \Delta\omega\omega_0,$$

and

$$Q = \frac{\omega_0}{\Delta\omega} = \frac{\omega_0}{b/m} = \frac{m}{b}\omega_0.$$

You can appreciate that large Q means a small value of $\Delta\omega$, that is, a thin peak. It might be mentioned that a very large Q is not always desirable, as for example in a musical instrument it would make it harder to “hit the note.” On the other hand, for a radio to tune into one of two stations broadcasting at nearby frequencies, a high Q factor is highly desirable.

Another use of the Q factor is in determining the loss of energy of a damped oscillator. For simplicity let us consider an unforced oscillator. As you know, due to damping, the amplitude of the oscillations will decay exponentially as $e^{-\gamma t}$, where $\gamma = b/2m$; see Equation (11.15). Energy is proportional to the amplitude squared, so the decrease in energy in time t is given by $(e^{-\gamma t})^2 = e^{-2\gamma t}$. Therefore, the energy loss in the time the oscillator goes through one complete cycle is $e^{-2\gamma T}$, where T is the period ($T = \frac{2\pi}{\omega_0}$). Consequently, the ratio of final energy to initial energy is

$$\frac{E e^{-2\gamma T}}{E} = e^{-2\gamma T} = e^{-2 \frac{b}{2m} \frac{2\pi}{\omega_0}} = e^{-2\pi \frac{b}{m\omega_0}} = e^{-2\pi/Q}.$$

This tells us that an oscillator with large Q does not lose energy as quickly as an oscillator with small Q .

Resonance in Electrical Circuits (Optional)

Resonance is an interesting physical phenomenon of importance in such diverse fields as the design of electrical circuits and the motion of celestial bodies. For example, the RLC circuit you studied in your introductory electricity course is illustrated in Figure 11.9.

The charge on the capacitor plates is q . If the plates are initially charged, there will be a flow of charge (a current, I) through the inductor, setting up a back emf or voltage difference given by $L \frac{dI}{dt}$. This can also be written as $L \frac{d^2q}{dt^2}$. The voltage drop across the resistor is IR and this can be written as $R \frac{dq}{dt}$. Finally, the voltage drop across the capacitor is given by $\frac{q}{C}$. The sum of voltage drops around the circuit is zero, so

$$L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C}q = 0.$$

But this is exactly the equation for a damped harmonic oscillator! If you add a “driving force” by placing some source of emf in the circuit (such as an oscillator or power supply) you will have a driven oscillator. The natural frequency of the circuit is $\omega_0 = \sqrt{1/LC}$, as you can appreciate from the form of the equation if the damping term is removed. If the driving frequency is near or equal to the natural frequency, the amplitude of the current increases greatly. This is the principle behind tuning a radio or television receiver. The “driving” voltage is supplied by the antenna picking up electromagnetic radiation. Suppose a station you want to tune to is broadcasting at a frequency ω_d . The receiver contains a variable capacitor whose capacitance you change by turning a knob or pushing a button on your receiver. Changing C will, of course, change the value of ω_0 and hence the resonant frequency ω' . When ω' is equal to ω_d , the current through the resistor reaches a much higher value than for other frequencies. This is illustrated in Figure 11.10.

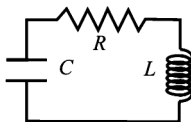
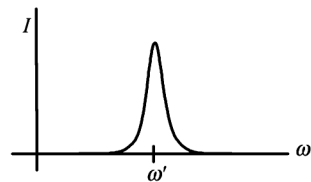


Figure 11.9 An RLC circuit.

Figure 11.10 A plot of the current (I) as a function of the angular frequency ω , illustrating the resonance phenomenon in an LRC circuit. When the frequency ω is nearly equal to the resonant frequency ω' , the current in the circuit increases greatly. If there were no resistance in the circuit, the current would become infinite at $\omega = \omega_0$.



Resonance in Astronomy

As mentioned in [Chapter 10](#), celestial bodies are affected by resonances. For example, the orbital period and the rotational period of the Moon are “locked” into a 1:1 resonance. In celestial mechanics, a resonance is defined as a system in which the frequency of the “driving” force (usually the effect of a perturbing body) and the “natural” frequency of the system are in the ratio of small numbers. Resonances explain why there are gaps in the rings of Saturn and in the asteroid belts between Mars and Jupiter, and why certain satellites (including the Moon) have rotation rates equal to their orbital rates. (The resonance phenomenon in celestial mechanics is often called a “commensurability.”)

Exercise 11.26

An RLC series circuit consists of a 10 ohm resistor, a 6 microfarad capacitor, and an 0.2 henry inductor. Determine the resonant frequency for the circuit. Answer: 913 rad/s. ■

11.5 Coupled Oscillators

In the real physical world, oscillators are seldom isolated from their environment and an oscillating body will set up oscillations in nearby bodies. A very good example of *coupled oscillators* is the motion of atoms in a crystal. A vibrating atom will cause neighboring atoms to start oscillating.

The analysis of coupled oscillators can be quite complicated. This section is a study of the simplest of such systems, namely two undamped oscillators connected by springs.

[Figure 11.11](#) illustrates two masses attached to rigid walls by springs of constants k_1 , k_2 , and k_3 . Let x_1 and x_2 be the displacements of masses m_1 and m_2 measured from their equilibrium positions. The equations of motion for the masses are easily determined from the Lagrangian or from Newton’s second law. We obtain the following *coupled* differential equations:⁷

$$\begin{aligned} m_1 \ddot{x}_1 &= -k_1 x_1 - k_3 (x_1 - x_2), \\ m_2 \ddot{x}_2 &= -k_2 x_2 - k_3 (x_2 - x_1). \end{aligned} \quad (11.29)$$

For simplicity, assume that $m_1 = m_2 = m$ and that $k_1 = k_2 = k$, but allow k_3 to be different. Then the equations of motion reduce to

$$\begin{aligned} m \ddot{x}_1 + (k + k_3)x_1 - k_3 x_2 &= 0, \\ m \ddot{x}_2 + (k + k_3)x_2 - k_3 x_1 &= 0. \end{aligned}$$

⁷ When applied to the atoms in a solid, we often assume all the masses are equal and all the “spring constants” are equal, yielding n equations of the form

$$m \ddot{x}_i = k(x_i - x_{i-1}) + k(x_{i+1} - x_i) \quad i = 1, \dots, n.$$

These relations reduce to [Equations \(11.29\)](#) for the situation pictured in [Figure 11.11](#).

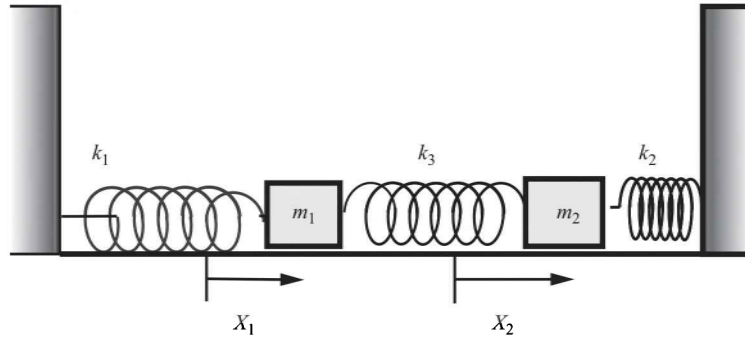


Figure 11.11 Coupled oscillators. Note that the positions of the two masses are measured from their equilibrium points.

Denote $k + k_3$ by k' . Dividing by m yields:

$$\begin{aligned}\ddot{x}_1 + (k'/m)x_1 - (k_3/m)x_2 &= 0, \\ \ddot{x}_2 + (k'/m)x_2 - (k_3/m)x_1 &= 0.\end{aligned}$$

Now replace k'/m by ω_0^2 . The reason for this notation is that ω_0 is the frequency of oscillation of either one of the masses if the other mass is held at rest. Furthermore, let us denote k_3/m by $\Delta\omega^2$, for reasons that will soon become clear.

Using this new notation the equations become

$$\begin{aligned}\ddot{x}_1 + \omega_0^2 x_1 - \Delta\omega^2 x_2 &= 0, \\ \ddot{x}_2 + \omega_0^2 x_2 - \Delta\omega^2 x_1 &= 0.\end{aligned}\tag{11.30}$$

This system of *coupled* linear differential equations can be solved by assuming the solutions

$$\begin{aligned}x_1 &= Ae^{i\omega t}, \\ x_2 &= Be^{i\omega t}.\end{aligned}$$

We assume oscillatory solutions (that is why the exponent is imaginary) with both of the variables x_1 and x_2 having the same angular frequency ω . (Except for a slight change in the notation, this is the same technique previously used in assuming solutions of the form e^{pt} .) Following our usual procedure we insert the expressions for x_1 and x_2 into the coupled equations of motion to determine the “auxiliary equations” that will give us ω . We obtain

$$\begin{aligned}A(i\omega)^2 e^{i\omega t} + \omega_0^2 A e^{i\omega t} - \Delta\omega^2 B e^{i\omega t} &= 0, \\ B(i\omega)^2 e^{i\omega t} + \omega_0^2 B e^{i\omega t} - \Delta\omega^2 A e^{i\omega t} &= 0,\end{aligned}$$

or

$$\begin{aligned}(-\omega^2 + \omega_0^2)A - \Delta\omega^2 B &= 0, \\ -\Delta\omega^2 A + (-\omega^2 + \omega_0^2)B &= 0.\end{aligned}\tag{11.31}$$

These two equations are a system of homogeneous coupled linear *algebraic* equations in the unknowns A and B . (Solving them will indirectly lead to a solution for ω .)

The technique for solving systems of coupled algebraic equations is to apply Cramer’s rule. I believe you are familiar with this rule, but I will review it briefly anyway.

Consider the pair of linear algebraic equations

$$\begin{aligned}ax + by &= e, \\ cx + dy &= f.\end{aligned}$$

To solve, first evaluate the determinant of the coefficients, thus

$$D = \begin{vmatrix} a & b \\ c & d \end{vmatrix}.$$

The first column of this determinant is formed from the coefficients of x and the second column contains the coefficients of y . Then, according to Cramer's rule, the solutions are:

$$x = \frac{\begin{vmatrix} e & b \\ f & d \end{vmatrix}}{D} \quad \text{and} \quad y = \frac{\begin{vmatrix} a & e \\ c & f \end{vmatrix}}{D}.$$

Note that the determinant in the numerator of the equation for x is obtained from D by replacing the column of the coefficients of x with the constants on the right-hand side of the equations. Similarly y is given by the same equation but with the column of coefficients of y replaced by the right-hand constants. (You may recall using Cramer's rule to determine the currents in an electric circuit when applying Kirchhoff's laws.)

Unfortunately, when Cramer's rule is applied to *homogenous* linear equations such as Equations (11.31), the zeros on the right-hand side mean that the determinants in the numerators have a column of zeros. This leads to the *trivial solutions* $x = 0$ and $y = 0$. However, if the *denominator* is also zero (if $D = 0$), then $x = \frac{0}{0}$ and $y = \frac{0}{0}$, both of which are undefined. You may think this is not much of an advantage, but it actually allows you to find the solution! In fact it is a general rule that coupled homogeneous algebraic equations have nontrivial solutions *if and only if* the determinant of the coefficients is zero.

Going back to Equations (11.31), the condition for a nontrivial solution is that the determinant of the coefficients is equal to zero:

$$D = \begin{vmatrix} -\omega^2 + \omega_0^2 & -\Delta\omega^2 \\ -\Delta\omega^2 & -\omega^2 + \omega_0^2 \end{vmatrix} = 0,$$

or

$$(-\omega^2 + \omega_0^2)^2 - (\Delta\omega^2)^2 = 0,$$

$$\omega^2 = +\omega_0^2 \mp \Delta\omega^2.$$

That is,

$$\omega = \pm(\omega_0^2 \mp \Delta\omega^2)^{\frac{1}{2}}.$$

This yields four possible values for ω . Two of them are

$$\omega_1 = \pm(\omega_0^2 - \Delta\omega^2)^{\frac{1}{2}},$$

and the other two are

$$\omega_2 = \pm(\omega_0^2 + \Delta\omega^2)^{\frac{1}{2}}. \quad (11.32)$$

The general solutions for x_1 and x_2 are the sum of all possible solutions:

$$\begin{aligned} x_1 &= A_1 e^{i\omega_1 t} + A_{-1} e^{-i\omega_1 t} + A_2 e^{i\omega_2 t} + A_{-2} e^{-i\omega_2 t}, \\ x_2 &= B_1 e^{i\omega_1 t} + B_{-1} e^{-i\omega_1 t} + B_2 e^{i\omega_2 t} + B_{-2} e^{-i\omega_2 t}. \end{aligned} \quad (11.33)$$

These two expressions involve eight constants. But solving two second-order differential equations should lead to only four constants. Therefore, not all of the eight constants are independent. We now determine the four independent constant amplitudes.

Note that each term in our expression for x_1 satisfies the differential equation. Thus, for example, a possible solution involves oscillations of frequency ω_1 . Then,

13 Waves

A wave is an oscillation of a medium, such as a water wave in the ocean or a wave propagating down a stretched slinky or a taut string. In general, the medium as a whole does not translate. For example, a wave in the ocean can be thought of as water molecules moving vertically up and down while the wave itself moves horizontally. A strange and interesting thing about waves is that although there is no transport of mass, the wave does transport energy and momentum.

A *transverse* wave, such as a wave in a string, is one in which the material particles move perpendicular to the direction of the wave. On the other hand, a sound wave in air or the wave set up in a metal rod when you hit one end with a hammer, are examples of *longitudinal* waves. In these waves the material particles move back and forth parallel to the direction of the wave.

This chapter is an introduction to wave motion and is limited to an analysis of waves in strings.² However, the wave equation and the wave properties derived here are valid for all types of waves. The mathematical techniques considered in this chapter include expressing a periodic function as a Fourier series and solving the wave equation by separation of variables. You will find the material of this chapter to be particularly useful when you study electromagnetic waves.

13.1 A Wave in a Stretched String

Consider a long string. If you shake one end of the string up and down, you can set up a *traveling wave*. A single shake will generate a pulse that moves along the string; continuous shaking will produce a wave train which is often sinusoidal in form. If you have a shorter string, such as one of the strings in your guitar, and you pluck the string, you can set up a *standing wave*.

A wave is characterized by a number of physical parameters. The main ones are the following.

Amplitude (A): The maximum displacement from equilibrium.

Wavelength (λ): The distance from peak to peak or between any two corresponding points on the wave.

Period (τ): The time it takes for a point on the wave to go through one complete oscillation.

Frequency (f): The number of oscillations per unit time. Note that $f = 1/\tau$.

Speed (v): How fast the waveform is displaced in the direction of its motion, $v = \lambda/\tau = \lambda f$.

Angular frequency (ω): Another measure of frequency, defined by $\omega = 2\pi f = 2\pi/\tau$.

¹ This is not quite true. A molecule of water also has a small horizontal oscillation and traces out an elliptical path – but you get the general idea.

² An ideal string is perfectly flexible and linearly elastic. This means the tension is everywhere the same and always directed tangentially. Linear elasticity means the tension depends linearly on the amount the string is stretched (Hooke's law).

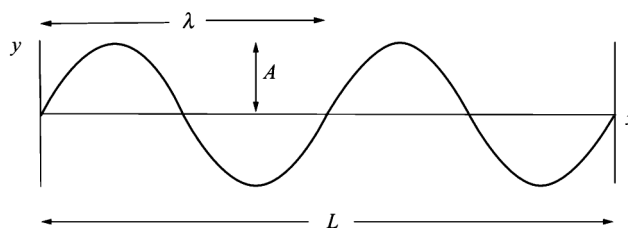


Figure 13.1 A wave in a string of length L . The wavelength and amplitude are indicated on the figure. The displacement from equilibrium is $y = y(x)$.

Wave number (k): A parameter related to wavelength, defined by $k = 2\pi/\lambda$. Because $\lambda = v\tau$ the wave number can also be expressed as $k = \omega/v$. Note that the wave number is analogous to the angular frequency: one of them is an inverse time and the other is an inverse length.

Consider a string of length L with fixed end points. Suppose you generated a wave in the string and took a photograph of the wave. In your photograph the string might look somewhat like the sketch in [Figure 13.1](#). The vertical displacement of a particle in the string is y . This displacement depends on the horizontal position, x , so $y = y(x)$. (There is no time dependence yet because this is just a description of the instantaneous photograph of the wave.) The curve repeats after a distance $x = \lambda$. Suppose you measured the vertical displacement (y) along the string and found it satisfied the following sinusoidal curve³

$$y(x) = A \sin\left(\frac{2\pi x}{\lambda}\right) = A \sin(kx).$$

When x is any multiple of $\lambda/2$ the quantity in parentheses is a multiple of π and there is zero displacement at that point. Such points are called the *nodes* of the waveform.

The wave in [Figure 13.1](#) has a wavelength equal to one half the length of the string, L . The only waves that will “fit” onto the string are those with wavelengths $\lambda = 2L/n$ where $n = 1, 2, 3, \dots$

Now imagine that instead of an instantaneous photograph you have a video of the oscillating string. If you look at a particular point on the string you will see it moving up and down. Thus the displacement y is also a function of the time t . As you might suspect, the wave can be expressed mathematically by an expression involving sines and cosines of the quantities kx and ωt .

To derive the expression for $y = y(x, t)$ we need to determine the equation of motion and then integrate it. Consider a string of length L under tension F . The two ends of the string are at $x = 0$ and $x = L$. Ignore gravity. If you pluck the string you can set up a “standing wave” as in a guitar string. If you look at a plucked guitar string you will notice that the amplitude of the wave is much smaller than the length of the string. That is, y is relatively a small quantity. Furthermore, the slope dy/dx is also small. The tension in the string is the same at every point.

[Figure 13.2](#) shows an infinitesimal portion (dx) of the string as a pulse propagates along it.

The portion of the string extends from x to $x + dx$. The mass of this element is $dm = \rho dx$, where ρ is the mass per unit length. The two ends are displaced vertically by $y(x, t)$ and $y(x + dx, t)$. The element of the string is subjected to a tension $F(x)$ acting to the left and down at x (at angle θ) and a tension $F(x + dx)$ acting to the right and up at $x + dx$ (at angle ϕ). The horizontal components of these two tensions must cancel because the element of string is not accelerating to

³ As you will see, a wave can generally be expressed as a sum of sines and cosines. For simplicity, at the moment, we assume a single sinusoid.

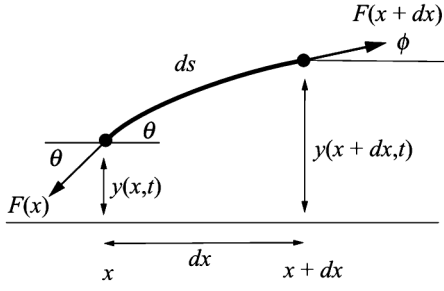


Figure 13.2 A small portion of a vibrating string.

the right or the left. (The tension in the string is the same at all points so there actually is a small oscillating horizontal force that we shall consider in [Section 13.7](#).) There is, however, a vertical component of force bringing the string back down to its equilibrium position.

The magnitude of the tension is the same at both ends, that is,

$$F(x + dx) = F(x) = F.$$

The vertical acceleration is $a_y = \partial^2 y / \partial t^2$, so applying Newton's second law gives⁴

$$\begin{aligned} \rho dx \frac{\partial^2 y(x, t)}{\partial t^2} &= F(x + dx) \sin \phi - F(x) \sin \theta, \\ &= F(\sin \phi - \sin \theta). \end{aligned}$$

Assuming the angles θ and ϕ are small (because both y and dy/dx are small)

$$\begin{aligned} \sin \theta &\doteq \tan \theta = \frac{\partial y(x, t)}{\partial x}, \\ \sin \phi &\doteq \tan \phi = \frac{\partial y(x + dx, t)}{\partial x}. \end{aligned}$$

So,

$$\rho dx \frac{\partial^2 y(x, t)}{\partial t^2} = F \left[\frac{\partial y(x + dx, t)}{\partial x} - \frac{\partial y(x, t)}{\partial x} \right].$$

Dividing both sides by ∂x and taking the limit as $\partial x \rightarrow 0$ leads to

$$\rho \frac{\partial^2 y}{\partial t^2} = F \frac{\partial}{\partial x} \left[\frac{\partial y}{\partial x} \right],$$

or

$$\frac{\partial^2 y}{\partial x^2} = \frac{\rho}{F} \frac{\partial^2 y}{\partial t^2}.$$

If this last equation looks familiar to you it is because it has the form of the well-known relation called the “wave equation” which is usually expressed as

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}, \quad (13.1)$$

where v is the speed of the wave.

⁴ We use partial derivatives because the acceleration depends on x as well as t .

Thus we have not only obtained the equation of motion for the wave in a taut string, we have also shown that the wave propagates in the x direction with speed

$$v = \sqrt{F/\rho}.$$

The wave equation (13.1) is a very important, fundamental relation. It is a second-order partial differential equation giving the displacement of the string as a function of time and position. I strongly suggest you memorize it.

Exercise 13.1

A string of length 6 m is fixed at both ends. Its mass is 0.1 kg and the tension in the string is 50 N. (a) What is the wavelength of the longest possible standing wave in this string? (b) What is the frequency of that wave? Answer: (a) 12 m, (b) 4.56 Hz. ■

Exercise 13.2

If n is the number of half wavelengths, show that the frequency of a standing wave in a string is $f = nv/2L$. ■

13.2 Direct Solution of the Wave Equation

Our next task is to solve Equation (13.1), the wave equation, to obtain an expression for $y = y(x, t)$. Many second-order partial differential equations (including this one) can be solved by a technique called *separation of variables*.

The basic idea behind the separation of variables method is to express the unknown function (in our case $y = y(x, t)$) as the product of two functions, each depending on only one variable. That is, you *assume* that you can write $y(x, t)$ in the following form:

$$y(x, t) = X(x)T(t),$$

where X is a function only of x and T is a function only of t . Substitute this expression into the differential equation (13.1) to get

$$\begin{aligned}\frac{\partial^2 X(x)T(t)}{\partial t^2} &= v^2 \frac{\partial^2 X(x)T(t)}{\partial x^2}, \\ X \frac{\partial^2 T}{\partial t^2} &= v^2 T \frac{\partial^2 X}{\partial x^2}, \\ \frac{1}{T} \frac{\partial^2 T}{\partial t^2} &= \frac{v^2}{X} \frac{\partial^2 X}{\partial x^2}.\end{aligned}$$

Now the left-hand side depends only on t and the right-hand side depends only on x . Since x and t can be varied independently, the only way the equation can be satisfied is if the right-hand side and the left-hand side are equal to the same constant. Call it $-\omega^2$. The partial differential equation has then been *separated* into the following two *ordinary* differential equations:

$$\begin{aligned}\frac{v^2}{X} \frac{d^2 X}{dx^2} &= -\omega^2, \\ \frac{1}{T} \frac{d^2 T}{dt^2} &= -\omega^2,\end{aligned}$$

or

$$\frac{d^2 X}{dx^2} + \frac{\omega^2}{v^2} X = 0,$$

$$\frac{d^2 T}{dt^2} + \omega^2 T = 0.$$

Both of these equations have the form of the simple harmonic oscillator equation. You have seen the SHM equation many times, so I will simply write down the general solutions

$$X(x) = A \cos \frac{\omega}{v} x + B \sin \frac{\omega}{v} x = A \cos kx + B \sin kx,$$

$$T(t) = C \cos \omega t + D \sin \omega t.$$

The solution for $y = y(x, t)$ is the product of these two functions, thus:

$$y(x, t) = X(x)T(t) = (A \cos kx + B \sin kx) (C \cos \omega t + D \sin \omega t). \quad (13.2)$$

Note that the partial differential [equation \(13.1\)](#) is satisfied by the four expressions

$$y(x, t) \propto \begin{cases} \cos kx \cos \omega t \\ \cos kx \sin \omega t \\ \sin kx \cos \omega t \\ \sin kx \sin \omega t \end{cases}.$$

The general solution, [Equation \(13.2\)](#), is the sum of all of these. The coefficients, A, B, C, D are determined from the boundary conditions.

Exercise 13.3

Show that an expression of the form

$$y = A \sin kx + B \cos kx$$

can be expressed as

$$y = C \cos(kx + \alpha).$$

Express C and α in terms of A and B . (The quantity α is called the “phase constant.”) Answer: $C = \sqrt{A^2 + B^2}$, $\alpha = \tan^{-1}(-A/B)$. ■

13.2.1 Fourier Series (Optional)

You are probably familiar with Fourier series from your mathematics courses, so this discussion is intended to refresh your memory. If you have not been exposed to Fourier series previously, I strongly suggest you read about them in a “Math Methods” text.⁵

Any periodic function (such as a wave) or a portion of a periodic function (such as a standing wave) can be represented as a sum of sines and cosines. Consider, for example, the portion of a “square wave” shown in [Figure 13.3](#).

The function we are approximating with a Fourier series is $f(x) = -\pi/2$ for $-\pi < x < 0$, $f(x) = +\pi/2$ for $0 < x < +\pi$. This is a portion of the square wave which alternates between plus and minus $\pi/2$ with a periodicity of π . In the middle panel of [Figure 13.3](#), I plotted $2 \sin x$

⁵ An excellent source is [Chapter 7](#) in *Mathematical Methods in the Physical Sciences* by Mary L. Boas (Wiley and Sons, 2006).

Figure 13.3 The top panel shows the square wave $f(x) = -\pi/2$ for $-\pi < x < 0$, $f(x) = +\pi/2$ for $0 < x < +\pi$. The middle panel also shows $f(x) = 2 \sin x$, a rather poor approximation to the square wave. The bottom panel shows $f(x) = 2 \sin x + \frac{2}{3} \sin 3x$, a somewhat better approximation.

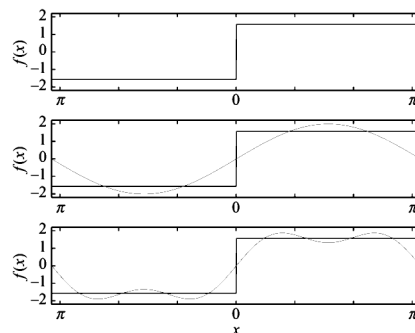
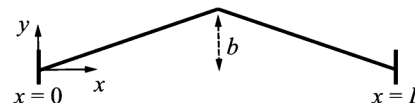


Figure 13.4 A string plucked at its center. The displacement at $L/2$ is b .



which has the appropriate periodicity but does not approximate the original function very well. In the bottom panel the additional term $\frac{2}{3} \sin 3x$ has been added to our highly truncated Fourier series, and you can appreciate that the approximation is getting better. The next approximation (not plotted) is $f(x) = 2 \sin x + \frac{2}{3} \sin 3x + \frac{2}{5} \sin 5x$. The Fourier series more and more closely approximates the square wave as more and more terms are included.

Before getting into the details of how to obtain a Fourier series, let us consider why they are useful. That is, if we already know the function $f(x)$, why would we want to express it as an infinite sum of sines and cosines? To answer this question let us note that the solution to the wave equation (13.2) is expressed in terms of sines and cosines which are multiplied by constant coefficients which we do not know. These coefficients will be determined from the initial and/or boundary conditions. In the problem of an oscillating mass on a spring or a pendulum this was easily done because the initial conditions were the position and the velocity of what was essentially a point mass. But the initial conditions for a vibrating string involve the positions and velocities of all the points of the string. Suppose that we know the initial configuration of a plucked string, and it looks like Figure 13.4, which shows us two straight lines. How can we match this initial condition to the solution that looks like Equation (13.2), i.e., a sum of sines and cosines? The answer is to express a function such as that of Figure 13.4 as a Fourier series and thus determine the coefficients in the solution.

In general a function $f(x)$ defined in the interval $-\pi < x < +\pi$ can be expressed as a Fourier series having the form

$$f(x) = \frac{1}{2}a_0 + a_1 \cos x + a_2 \cos 2x + \cdots + b_1 \sin x + b_2 \sin(2x) + \cdots$$

or

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx. \quad (13.3)$$

The problem now reduces to determining the coefficients a_n and b_n . This is accomplished by using the orthogonality properties of sines and cosines which can be expressed by the following three integrals:

$$\begin{aligned}\int_{-\pi}^{+\pi} \sin nx \cos mx dx &= 0 \text{ for all } m \text{ and } n \\ \int_{-\pi}^{+\pi} \sin nx \sin mx dx &= \pi \text{ for } n = m, \text{ and } 0 \text{ for } n \neq m \\ \int_{-\pi}^{+\pi} \cos nx \cos mx dx &= \pi \text{ for } n = m, \text{ and } 0 \text{ for } n \neq m\end{aligned}$$

To determine the coefficients a_n and b_n we multiply Equation (13.3) by $\cos mx$ and integrate from $-\pi$ to $+\pi$:

$$\begin{aligned}\int_{-\pi}^{+\pi} f(x) \cos mx dx &= \frac{a_0}{2} \int_{-\pi}^{+\pi} \cos mx dx \\ &+ \int_{-\pi}^{+\pi} \sum_{n=1}^{\infty} a_n \cos nx \cos mx dx \\ &+ \int_{-\pi}^{+\pi} \sum_{n=1}^{\infty} b_n \sin nx \cos mx dx.\end{aligned}$$

For $m = 0$ this reduces to $\int_{-\pi}^{+\pi} f(x) dx = \int_{-\pi}^{+\pi} 1 dx = \frac{a_0}{2}(\pi + \pi)$, or

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{+\pi} f(x) dx$$

and for $m \neq 0$

$$\begin{aligned}\int_{-\pi}^{+\pi} f(x) \cos mx dx &= \frac{a_0}{2} \int_{-\pi}^{+\pi} \cos mx dx + \sum_{n=1}^{\infty} a_n \pi \delta_{mn} + \sum_{n=1}^{\infty} b_n \int_{-\pi}^{+\pi} \sin nx \cos mx dx \\ &= 0 + a_m \pi + 0.\end{aligned}$$

Therefore,

$$a_m = \frac{1}{\pi} \int_{-\pi}^{+\pi} f(x) \cos mx dx.$$

Similarly, if we multiply Equation (13.3) by $\sin mx$ and integrate we obtain

$$b_m = \frac{1}{\pi} \int_{-\pi}^{+\pi} f(x) \sin mx dx.$$

We assumed the function $f(x)$ was defined in the interval $-\pi$ to $+\pi$. But this assumption was only made so that we could keep the notation simple. For a function defined on the arbitrary interval $-l$ to $+l$ we simply change the variable from x to $\frac{\pi x}{l}$ and obtain the coefficients from

$$\begin{aligned}a_m &= \frac{1}{l} \int_{-l}^{+l} f(x) \cos \frac{m\pi}{l} x dx \quad m = 0, 1, 2, 3, \dots \\ b_m &= \frac{1}{l} \int_{-l}^{+l} f(x) \sin \frac{m\pi}{l} x dx \quad m = 1, 2, 3, \dots\end{aligned}$$

A labor saving fact is that if $f(x)$ is an odd function of x ($f(x) = -f(-x)$), then only sine terms appear in the Fourier series, and if $f(x)$ is an even function of x ($f(x) = f(-x)$), then only cosine terms appear in the series.

Exercise 13.4

Show that if $f(x)$ is defined in the interval $0 < x < l$, the coefficients in the Fourier expansion are

$$a_m = \frac{2}{l} \int_0^l f(x) \cos \frac{m\pi}{l} x dx,$$

$$b_m = \frac{2}{l} \int_0^l f(x) \sin \frac{m\pi}{l} x dx.$$

■

13.3 Standing Waves

As an example of how to determine the coefficients in the general solution, [Equation \(13.2\)](#), consider a standing wave, as in the string illustrated in [Figure 13.1](#). The string is tied down at the end points so

$$y(x=0) = y(x=L) = 0.$$

Then

$$y(0, t) = 0 = (A \cos 0 + B \sin 0) (C \cos \omega t + D \sin \omega t),$$

or

$$0 = A(C \cos \omega t + D \sin \omega t).$$

This requires that $A = 0$. The general equation has now been reduced to

$$y(x, t) = B \sin(kx) (C \cos \omega t + D \sin \omega t).$$

The boundary condition at $x = L$ leads to

$$y(L, t) = 0 = B \sin(kL) (C \cos \omega t + D \sin \omega t).$$

This condition is satisfied if $B = 0$, but in that case $y = 0$ everywhere and at all times! Although it satisfies the differential equation, this is obviously not a satisfactory solution. (It is the “trivial” solution.) However, there is another way to ensure that this expression is zero; if

$$\sin(kL) = 0,$$

the condition is satisfied. This means that kL must be an integer multiple of π because if $\sin \theta = 0$, then $\theta = 0, \pi, 2\pi, 3\pi, \dots$. That is,

$$kL = n\pi, \quad n = 1, 2, 3, \dots$$

Note that $n = 0$ is not included because that would require $k = 0$ and again $y(x, t)$ would be zero for all values of x . If k can take on the values $n\pi/L$, we must also require that ω take on a set of allowed values because $\omega = kv = k\sqrt{\frac{F}{\rho}}$. Thus, ω has the values

$$\omega_n = \frac{n\pi}{L} \sqrt{\frac{F}{\rho}} = \frac{n\pi v}{L},$$

and [Equation \(13.2\)](#) reduces to

$$y(x, t) = B \sin \frac{n\pi x}{L} (C \cos \omega_n t + D \sin \omega_n t).$$

Finally, combining the constants BC and BD and calling them A_n and B_n we write

$$y(x, t) = (A_n \cos \omega_n t + B_n \sin \omega_n t) \sin \frac{n\pi x}{L}. \quad (13.4)$$

The differential equation and the boundary conditions are satisfied for any integer value of n . But note that the coefficients A_n and B_n are generally different for different values of n . The general solution to the differential equation is, of course, the sum of all possible solutions, that is,

$$\begin{aligned} y(x, t) &= \sum_{n=1}^{\infty} (A_n \cos \omega_n t + B_n \sin \omega_n t) \sin \frac{n\pi x}{L}, \\ &= \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi v}{L} t + B_n \sin \frac{n\pi v}{L} t \right) \sin \frac{n\pi x}{L}. \end{aligned} \quad (13.5)$$

To determine the coefficients A_n and B_n we use the initial conditions, as illustrated in the following example.

Worked Example 13.1 A string of length L is pulled up at its midpoint a distance b and then released. The displacement of the string at any position ($0 \leq x \leq L$) and time ($t \geq 0$) is given by Equation (13.5). Determine the coefficients A_n and B_n . See Figure 13.4.

Solution The waveform at $t = 0$ is given by the two straight lines, $y = (2b/L)x$ for $0 \leq x \leq L/2$ and $y = (-2b/L)x + 2b$ for $L/2 \leq x \leq L$.

At time $t = 0$ the string is not moving, so $\partial y / \partial t = 0$. From Equation (13.5)

$$\begin{aligned} 0 &= \frac{\partial}{\partial t} \left[\sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{L} \cos \omega_n t + \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L} \sin \omega_n t \right]_{t=0}, \\ 0 &= \left[\sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{L} (-\omega_n \sin \omega_n t) + \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L} (\omega_n \cos \omega_n t) \right]_{t=0}, \\ 0 &= \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L}. \end{aligned}$$

This is zero for all values of x iff $B_n = 0$ for all n . So we are left with

$$y(x, t) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{L} \cos \omega_n t,$$

$$y(x, 0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{L}.$$

But $y(x, 0)$ has the shape illustrated in Figure 13.4. As you will show in Problem 13.1, this can be expressed as the following Fourier series:

$$y(x, 0) = \frac{8b}{\pi^2} \left(\sin \frac{\pi x}{L} - \frac{1}{3^2} \sin \frac{3\pi x}{L} + \frac{1}{5^2} \sin \frac{5\pi x}{L} \cdots \right). \quad (13.6)$$

That is,

$$\begin{aligned} y(x, 0) &= \frac{8b}{\pi^2} \left(\sin \frac{\pi x}{L} - \frac{1}{3^2} \sin \frac{3\pi x}{L} + \frac{1}{5^2} \sin \frac{5\pi x}{L} \cdots \right) \\ &= \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{L}, \end{aligned}$$

where

$$A_n = \frac{8b}{n^2\pi^2} \sin \frac{n\pi}{2} \quad n = 1, 3, 5, \dots,$$

and

$$A_n = 0 \quad n = 2, 4, 6, \dots$$

The factor $\sin(n\pi/2)$ is included to give the sign of the odd n terms and to delete all the even n terms. Finally, the waveform of the plucked string is

$$y(x, t) = \sum_{n=\text{odd}}^{\infty} \left(\frac{8b}{n^2\pi^2} \sin \frac{n\pi}{2} \right) \cos \omega_n t \sin \frac{n\pi x}{L}.$$

Exercise 13.5

Equation (13.6) is a Fourier series. Plotting the terms in this series, we find the integer n is equal to the number of *loops* in the standing wave when only the y_n term is excited. The various modes, $n = 1, 2, 3, \dots$ are called the first, second, third $\dots$ harmonics. Plot the first four harmonics for a string of length L . ■

13.4 Traveling Waves

In the previous section we considered a standing wave. Now consider a traveling wave, such as a pulse traveling down an infinite string (or an electromagnetic wave propagating in empty space). Imagine a very long (perhaps infinite) string that is plucked at some point, or perhaps shaken at one end. Your experience with ropes and slinkies tells you that a pulse will move down the string, as pictured in Figure 13.5. If there is no loss of energy, the shape will be repeated at a later time at a further position, as shown in the lower panel of the figure. Suppose the top panel gives the waveform at time $t = 0$, so the plot is $y = y(x, 0)$. The bottom panel shows the wave at a later time t . But $y(x, t) = y(x - vt, 0)$, that is, the displacement at point x at time t is the same as the displacement at $x - vt$ at the earlier time. Note that during the time interval t , the wave has traveled a distance vt . Therefore, v is the speed at which the waveform propagates. This is called the *phase speed*. (Recall that we identified v with the quantity $\sqrt{F/\rho}$.)

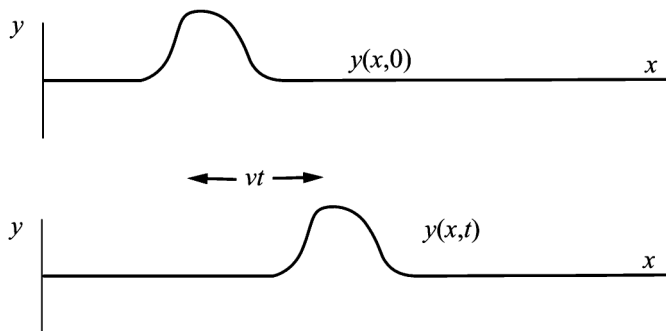


Figure 13.5 A pulse in a string moving to the right. It does not change shape, so the displacement obeys the condition $y(x, t) = y(x + dx, t + dt)$.

The fact that the displacement at time t is the same as the displacement at $x - vt$ at an earlier time suggests that the waveform for a wave traveling at speed v will have the functional form

$$y(x, t) = f(x - vt).$$

In Figure 13.5, the lower panel represents a later time, so the wave is moving toward the right, that is, toward positive x . A wave traveling in the opposite direction would have the functional form

$$y(x, t) = g(x + vt).$$

For example, a sinusoidal traveling wave moving toward the right can be represented as

$$y(x, t) = A \sin k(x - vt).$$

Here I introduced k simply as a constant to make the argument of the sine dimensionless; however, if you think about it, you will realize that k is actually the wave number, so the expression becomes

$$y(x, t) = A \sin(kx - \omega t).$$

You might plot this expression for a few nearby values of time and verify for yourself that this indeed is a representation of a traveling sine wave moving in the positive x -direction.

Although the expression

$$y = f(x - vt)$$

seems a reasonable way to describe a traveling wave, it remains to be shown that it satisfies the wave equation for any function f that depends on x and t in the combination $x - vt$. For example, a number of different possible forms of f would be

$$y(x, t) = A \cos k(x - vt)$$

$$y(x, t) = Ae^{ik(x-vt)}$$

$$y(x, t) = A \sin(k^2(x - vt)^2).$$

The assumption that any function of $(x - vt)$ satisfies the wave equation can be proved by plugging $y = f(x - vt)$ into the equation and noting that if $u = x - vt$, then $\frac{\partial f}{\partial t} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial t} = -v \frac{\partial f}{\partial u}$, so

$$\begin{aligned} \frac{\partial^2 y}{\partial t^2} &= \frac{\partial}{\partial t} \left[\frac{\partial f(u)}{\partial t} \right] = -v \frac{\partial}{\partial t} \left[\frac{\partial f}{\partial u} \right] \\ &= -v \left[\left(\frac{\partial}{\partial u} \frac{\partial f}{\partial u} \right) \frac{\partial u}{\partial t} \right] = +v^2 \frac{\partial^2 f}{\partial u^2}, \end{aligned}$$

and since $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} = \frac{\partial f}{\partial u}(1)$, we can write

$$\begin{aligned} v^2 \frac{\partial^2 y}{\partial x^2} &= v^2 \frac{\partial}{\partial x} \left[\frac{\partial f}{\partial x} \right] = v^2 \frac{\partial}{\partial x} \left[\frac{\partial f}{\partial u} \right] \\ &= v^2 \frac{\partial}{\partial u} \left[\frac{\partial f}{\partial u} \right] = v^2 \frac{\partial^2 f}{\partial u^2}. \end{aligned}$$

The right-hand sides of the two expressions are equal, proving the wave equation is satisfied for any function of $x - vt$.

A particularly useful function of $x - vt$ that is often used to represent a traveling wave is

$$y(x, t) = Ae^{ik(x-vt)} = Ae^{i(kx-\omega t)}.$$

Note that this can be expressed in terms of sines and cosines using the Euler relation:

$$y(x, t) = Ae^{i(kx-\omega t)} = A [\cos(kx - \omega t) + i \sin(kx - \omega t)].$$

Although this expression satisfies the wave equation, it is not obvious how to interpret the imaginary part. Therefore, we avoid the problem by writing

$$y(x, t) = \text{Re} [Ae^{i(kx-\omega t)}],$$

that is, only taking the real part of the expression. This formulation is particularly useful when studying electromagnetic waves.

Exercise 13.6

Argue that the constant k introduced above as a proportionality constant must actually be the wave number. ■

Exercise 13.7

Plot $y = A \sin(kx - \omega t)$ for four nearby values of time to show that this does indeed represent a wave traveling to the right. (Select any reasonable values for the parameters, but make sure that the four values of t are less than a quarter of the period.) ■

13.5 Standing Waves as a Special Case of Traveling Waves

You have just seen that the solution to the wave equation is any function of $(kx \pm \omega t)$. But the standing wave of [Figure 13.1](#) that was considered earlier was also a solution of the wave equation and it had the form

$$y(x, t) = A \sin kx \cos \omega t + B \sin kx \sin \omega t. \quad (13.7)$$

This expression is a very slightly modified version of [Equation \(13.4\)](#). It does not look much like a function of $(kx \pm \omega t)$! Are the two expressions equivalent? The answer is yes, and the underlying reason is that a standing wave can be considered to be the sum of two traveling waves moving in opposite directions. We now show that this is indeed true.

To be a bit less general but to make the analysis easier to follow, assume the traveling waves are sinusoidal. The general form of two sinusoidal waves traveling in opposite directions is a combination of sine and cosine terms, thus,

$$y(x, t) = A \sin(kx - \omega t) + B \sin(kx + \omega t) + C \cos(kx - \omega t) + D \cos(kx + \omega t).$$

The boundary conditions require that $y = 0$ at $x = 0$ and at $x = L$. If $x = 0$

$$\begin{aligned} 0 &= -A \sin \omega t + B \sin \omega t + C \cos \omega t + D \cos \omega t, \\ &= (B - A) \sin \omega t + (C + D) \cos \omega t. \end{aligned}$$

This condition holds at all times if

$$\begin{aligned} B &= A, \\ C &= -D. \end{aligned}$$

and the solution reduces to

$$y = A [\sin(kx - \omega t) + \sin(kx + \omega t)] + C [\cos(kx - \omega t) - \cos(kx + \omega t)].$$

Imposing the second boundary condition ($y = 0$ at $x = L$) gives

$$0 = A [\sin(kL - \omega t) + \sin(kL + \omega t)] + C [\cos(kL - \omega t) - \cos(kL + \omega t)].$$

This expression is valid at all times, including $t = 0$, in which case it is easy to appreciate that k is equal to any of the values

$$k = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots$$

(Recall that the standing wave satisfied this condition.)

The sum of the two traveling waves (with $y = 0$ at $x = 0, L$) is, therefore,

$$y(x, t) = A [\sin(kx - \omega t) + \sin(kx + \omega t)] + C [\cos(kx - \omega t) - \cos(kx + \omega t)],$$

with $k = n\pi/L$ and $\omega = kv = n\pi v/L$.

Using the trigonometric identities for the sums and differences of angles,⁶

$$y(x, t) = [2A \sin(kx)] \cos \omega t + [2C \sin(kx)] \sin \omega t.$$

Recalling that the standing wave was given by

$$y(x, t) = A \sin(kx) \cos \omega t + B \sin(kx) \sin \omega t,$$

we see that the two expressions are identical except for the notation of the coefficients. Thus, we have shown that the standing wave can be considered the superposition of two traveling waves moving in opposite directions.

Exercise 13.8

Show that $y = 0$ when $x = L$. ■

13.6 Energy

Let us determine the energy in a wave. As an example, consider a standing wave in a string of length L .

The kinetic energy of the mass element ρdx as it oscillates up and down is

$$dT = \frac{1}{2} \rho dx \left(\frac{dy}{dt} \right)^2.$$

To determine the potential energy note that the element of unstretched length dx has stretched length ds , so the displacement produces a net stretch in the string of $ds - dx$. (See [Figure 13.2](#).) The tension in the string is F , so the work done to stretch the string (which is equal to the increase in its potential energy) is

$$dV = F(ds - dx).$$

⁶ These are:

$$\begin{aligned} \sin \alpha + \sin \beta &= 2 \sin \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta) \\ \cos \alpha - \cos \beta &= -2 \sin \frac{1}{2}(\alpha + \beta) \sin \frac{1}{2}(\alpha - \beta). \end{aligned}$$

But

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx,$$

so

$$dV = F \left[\sqrt{1 + \left(\frac{dy}{dx}\right)^2} - 1 \right] dx.$$

The binomial expansion of the square root leads to

$$dV = F dx \left[1 + \frac{1}{2} \left(\frac{dy}{dx}\right)^2 + \dots - 1 \right] \doteq F \frac{1}{2} \left(\frac{dy}{dx}\right)^2 dx.$$

Consequently, the total energy in a vibrating string of length L is

$$E = T + V = \int_0^L (dT + dV) = \int_0^L \left[\frac{1}{2} \rho \left(\frac{\partial y}{\partial t}\right)^2 + \frac{1}{2} F \left(\frac{\partial y}{\partial x}\right)^2 \right] dx.$$

For example, if the wave is described by

$$y(x, t) = A \sin \frac{\pi x}{L} \cos \omega t,$$

then

$$\frac{\partial y}{\partial t} = -\omega A \sin \frac{\pi x}{L} \sin \omega t,$$

and

$$\frac{\partial y}{\partial x} = \frac{\pi}{L} A \cos \frac{\pi x}{L} \cos \omega t.$$

Consequently,

$$\begin{aligned} E &= \int_0^L \left[\frac{1}{2} \rho \omega^2 A^2 \left(\sin^2 \frac{\pi x}{L} \sin^2 \omega t \right) + \frac{1}{2} F \frac{\pi^2}{L^2} A^2 \left(\cos^2 \frac{\pi x}{L} \cos^2 \omega t \right) \right] dx, \\ &= \frac{1}{2} A^2 \left[\rho \omega^2 \sin^2 \omega t \int_0^L \sin^2 \frac{\pi x}{L} dx \right] + \frac{1}{2} A^2 \left[F \frac{\pi^2}{L^2} \cos^2 \omega t \int_0^L \cos^2 \frac{\pi x}{L} dx \right], \\ &= \frac{1}{2} A^2 \frac{L}{2} \left[\rho \omega^2 \sin^2 \omega t + F \frac{\pi^2}{L^2} \cos^2 \omega t \right]. \end{aligned}$$

Now recall that $F/\rho = v^2$ and $\omega = \pi v/L$, so $F\pi^2/L^2 = \rho v^2 \pi^2 / (\pi^2 v^2 / \omega^2) = \rho \omega^2$ and

$$\begin{aligned} E &= \frac{1}{2} A^2 \frac{L}{2} \rho \omega^2 (\sin^2 \omega t + \cos^2 \omega t) \\ &= A^2 \rho \omega^2 \frac{L}{4}. \end{aligned}$$

Note that the energy is proportional to the amplitude squared.

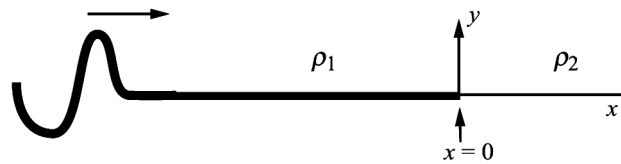


Figure 13.6 Two strings of different linear densities are connected at $x = 0$.

13.6.1 Energy Flow

Consider a wave passing from one medium into another. An example is the sunlight incident on the window of your room. As you know, some of the incident light will be transmitted and some will be reflected. Reflection is a characteristic property of any kind of wave when it is incident upon the interface of two materials in which the wave travels at different speeds. For the light wave, glass has a larger index of refraction than air, and the wave travels more slowly in glass. For a wave in a string, the speed depends on the mass per unit length of the string (ρ).

An analogy to a light wave incident on glass would be a traveling wave in a string of linear density ρ_1 incident on a second string of different linear density ρ_2 . This is illustrated in Figure 13.6.

Imagine a wave coming in from the left (from $x = -\infty$). Call this the *incident* wave. At the junction of the two strings (at $x = 0$), the incident wave is split into two waves, the *reflected* wave that moves back toward $x = -\infty$, and the *transmitted* wave that continues on toward $x = +\infty$. Let us denote these three waves by y_i , y_r , and y_t . These are all traveling waves, so they can be represented as follows:

$$\begin{aligned} y_i &= A_i \cos(k_1 x - \omega t), \\ y_r &= A_r \cos(k_1 x + \omega t), \\ y_t &= A_t \cos(k_2 x - \omega t). \end{aligned} \tag{13.8}$$

Note the sign on the ωt term in the expression for y_r and the subscript on k for y_t .

The speed of the wave is different in the two mediums. Because $v = \omega/k$, a change in the speed could be due to a change in either ω or k (or both). But it turns out that the waves have the same value of ω . You may wonder why k changes but ω stays the same. The reason is that the frequency of the wave is determined by the physical mechanism that is generating the wave. If you are standing at one end of the string and shaking it up and down, you are controlling the frequency of the wave by how quickly (or slowly) you move the end of the rope. This you can control. However, you have no control over the wavelength. The wavelength depends on the properties of the medium. When the properties of the medium change the wavelength changes, and so the wave number $k = 2\pi/\lambda$ also changes.

The energy of a wave is proportional to the square of its amplitude. An interesting and important question is how much energy is reflected at an interface and how much is transmitted into the second medium. The question can be answered by determining the amplitudes of the reflected and transmitted waves relative to the incident amplitude. In other words, it is desirable to determine the ratios A_r/A_i and A_t/A_i . This can be done by realizing that both y and dy/dx must be continuous at the junction between the two strings. The value of y is continuous because the strings are attached to one another. The reason why dy/dx is the same on both sides of the junction is that the tension is the same on both sides. These two boundary conditions can be expressed as

$$\begin{aligned} [y_i + y_r]_{x=0} &= [y_t]_{x=0}, \\ \left[\frac{\partial y_i}{\partial x} + \frac{\partial y_r}{\partial x} \right]_{x=0} &= \left[\frac{\partial y_t}{\partial x} \right]_{x=0}. \end{aligned}$$

Substituting the expressions from [Equations \(13.8\)](#) we obtain

$$\begin{aligned} A_i \cos(-\omega t) + A_r \cos(+\omega t) &= A_i \cos(-\omega t), \\ A_i + A_r &= A_t, \end{aligned}$$

and

$$k_1(A_i - A_r) = k_2 A_t,$$

or

$$\frac{1}{v_1}(A_i - A_r) = \frac{1}{v_2}A_t.$$

Therefore,

$$\begin{aligned} \frac{A_r}{A_i} &= \frac{v_2 - v_1}{v_1 + v_2}, \\ \frac{A_t}{A_i} &= \frac{2v_2}{v_1 + v_2}. \end{aligned}$$

Note that, since $v = \sqrt{F/\rho}$, if the second string is heavier than the first, $\rho_2 > \rho_1$ and $v_2 < v_1$ and A_r is negative. This means the reflected wave is inverted (“upside down”). Going to the extreme case, if the second string is infinitely massive, $v_2 = 0$ and $A_t = 0$. This is equivalent to string 1 being “nailed down” at $x = 0$.

To determine the energy flow through the junction consider the rate at which work is done by a particle on one side of $x = 0$ on a particle on the other side. Assume the particle on the left (at $x = 0^-$) exerts a downward force on the particle at $x = 0^+$, given by

$$F_y = -F \sin \theta \doteq -F \tan \theta = -F \frac{\partial y}{\partial x}.$$

The rate at which work is done is the power and is given by $P = \mathbf{F} \cdot \mathbf{v}$. Therefore,

$$\begin{aligned} \left[\frac{dW}{dt} \right]_{x=0^+} &= \left[F_y \frac{\partial y}{\partial t} \right]_{x=0} \\ &= \left[\left(-F \frac{\partial y_t}{\partial x} \right) \frac{\partial y_t}{\partial t} \right]_{x=0} \\ &= -F [(k_2 A_t \sin(k_2 x - \omega t)) (A_t (-\omega) \sin(k_2 x - \omega t))]_{x=0} \\ &= F A_t^2 k_2 \omega \sin^2(\omega t) = F A_t^2 (\omega^2 / v_2) \sin^2(\omega t), \end{aligned}$$

where we used $k = \omega/v$. The average of $\sin^2(\omega t)$ over one period is $1/2$. The average energy flow into string number 2 is equal to the average work done, so

$$\left\langle \frac{dE}{dt} \right\rangle_{x=0^+} = \frac{F A_t^2 \omega^2}{2v_2}.$$

An equivalent analysis for the energy leaving string number 1 gives

$$\left\langle \frac{dE}{dt} \right\rangle_{x=0^-} = \frac{F(A_i^2 - A_r^2)\omega^2}{2v_1}.$$

These expressions must be equal if there is no energy loss at the junction.

Worked Example 13.2 Obtain an expression for the reflection coefficient R defined as the ratio of the intensity of the reflected wave to the intensity of the incident wave ($R = I_r/I_i$).

Solution The intensity of a wave is defined as the energy transported across a normal unit area per unit time, that is, power per unit area. Because energy is proportional to the amplitude squared

$$R = \frac{I_r}{I_i} = \frac{A_r^2/(\text{area} \times \text{time})}{A_i^2/(\text{area} \times \text{time})} = \frac{A_r^2}{A_i^2} = \left(\frac{v_2 - v_1}{v_2 + v_1} \right)^2$$

$$= \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2.$$

Exercise 13.8

Show that if the tension is continuous at the junction, the slope dy/dx is continuous. ■

Exercise 13.10

Obtain the appropriate boundary condition for two strings joined by a knot of mass m . Answer:

$$F \left(\left. \frac{\partial y}{\partial x} \right|_{0+} - \left. \frac{\partial y}{\partial x} \right|_{0-} \right) = m \left. \frac{\partial^2 y}{\partial t^2} \right|_0.$$

13.7 Momentum (Optional)

The particles in a transverse wave move perpendicular to the motion of the wave. Thus, it would seem that a transverse wave would not transport momentum longitudinally. However, it is a known fact that waves in strings *do* transport momentum. This means that our assumption of purely transverse motion for a particle in a string must be wrong. There must be some longitudinal motion as well as transverse motion.⁷ This can be easily appreciated conceptually by (mentally) replacing the string with a series of particles of mass m connected by massless springs of constant k , as shown in Figure 13.7. As indicated in the figure, if one particle is plucked so that it is displaced vertically, it causes the particles on either side to be displaced both vertically *and* horizontally.

When the plucked particle is released, it sets up a (large amplitude) transverse wave as well as a (small amplitude) longitudinal wave. Rowland and Pask⁸ carried out numerical studies using

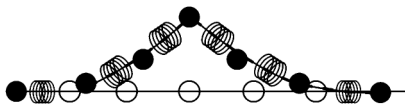


Figure 13.7 A string modeled as a series of particles connected by springs. The open circles represent the equilibrium positions of the particles. The filled circles represent their positions when a particle is displaced vertically.

⁷ Electromagnetic waves in a vacuum are transverse. The momentum density associated with these waves is $\epsilon_0 \mathbf{E} \times \mathbf{B}$, which is perpendicular to the transverse oscillations of $\mathbf{E}$ and $\mathbf{B}$. For a string, however, perfectly transverse waves would not have a longitudinal component of momentum.

⁸ David R. Rowland and Colin Pask, “The Missing Wave Momentum Mystery,” *American Journal of Physics*, **67**, 378–388 (1999) and David R. Rowland, “Comment on ‘What Happens to Energy and Momentum When Two Oppositely-Moving Wave Pulses Overlap?’ by N. Gauthier,” *American Journal of Physics*, **72**, 1425–1429 (2004). It is interesting to note that the theory of a vibrating string was first developed by Lagrange in 1759 but there are still

15

Accelerated Reference Frames

By now you have heard me say many times that Newton's laws of motion are applicable only in inertial (nonaccelerated) frames of reference. The question arises: How do we deal with motion in an *accelerated* reference frame? After all, most real reference frames are accelerating and, for many of them, the acceleration cannot be neglected. Since we live on the surface of a large rotating sphere, it is important for us to be able to solve physics problems in noninertial systems.

In this chapter you will learn how to express Newton's second law in a noninertial (accelerating) reference frame. We are, of course, particularly interested in rotating coordinate systems, which are a prime example of accelerating reference frames. Our study of these systems will introduce you to "fictitious" forces such as the centrifugal force and the Coriolis force. The Coriolis force, as you may know, is responsible for a number of important geophysical processes such as hurricanes and ocean currents. Finally, we will consider the motion of a very long simple pendulum and show that the plane of this "Foucault" pendulum precesses due to the rotation of the Earth.

15.1 A Linearly Accelerating Reference Frame

Although the most interesting noninertial reference frame for our purposes is a rotating reference frame such as the Earth, I would like to begin by looking at a reference frame that is accelerating *linearly* relative to an inertial reference frame. Figure 15.1 shows a reference frame O that is at rest "with respect to the fixed stars" and a second reference frame, O' , that is accelerating at a constant acceleration $\ddot{\mathbf{r}}$ relative to O .¹ In the figure, the position of a particle of mass m with respect to the inertial frame is $\mathbf{r}_O$ and the position of the particle with respect to the accelerating frame is $\mathbf{r}_{O'}$. The position of the origin O' with respect to O is $\mathbf{r}$. By tip to tail addition of vectors,

$$\mathbf{r}_O = \mathbf{r} + \mathbf{r}_{O'}. \quad (15.1)$$

Because O is an inertial coordinate system, Newton's second law holds in that frame and we can write

$$m\ddot{\mathbf{r}}_O = \mathbf{F}, \quad (15.2)$$

where $\mathbf{F}$ is the force acting on the particle. The acceleration of the particle relative to the accelerating frame is $\ddot{\mathbf{r}}_{O'}$. The force on the particle and the mass of the particle are assumed independent of the reference frame. (I am specifically excluding relativistic effects.)

¹ Please ignore the annoying fact that eventually O' will be moving at the speed of light!

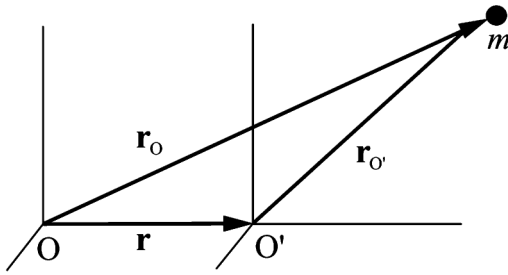


Figure 15.1 Coordinate system O' is accelerating at a constant rate $\ddot{\mathbf{r}}$ with respect to coordinate system O .

Taking the second time derivative of Equation (15.1) yields

$$\ddot{\mathbf{r}}_O = \ddot{\mathbf{r}}_{O'} + \ddot{\mathbf{r}},$$

where $\ddot{\mathbf{r}}$ is the acceleration of reference frame O' relative to the inertial frame O . Newton's law (15.2) then reads

$$\mathbf{F} = m\ddot{\mathbf{r}}_O = m(\ddot{\mathbf{r}}_{O'} + \ddot{\mathbf{r}}) = m\ddot{\mathbf{r}}_{O'} + m\ddot{\mathbf{r}}.$$

Therefore, from the point of view of the observer in the *accelerated reference frame*, Newton's second law has the form

$$m\ddot{\mathbf{r}}_{O'} = \mathbf{F} - m\ddot{\mathbf{r}}.$$

That is, there appears to be an *additional* force $\mathbf{f} = -m\ddot{\mathbf{r}}$ acting on the particle. If asked to determine the force on the particle, the observer in the accelerated reference frame would answer $\mathbf{F} + \mathbf{f}$.

Thus the observer in the accelerated system will believe that an additional force is acting on the particle. This is called a *fictitious force* because it is really just a mass times acceleration term which has been moved to the other side of the equation. Well-known fictitious forces include the centrifugal force and the Coriolis force.

15.2 A Rotating Coordinate Frame

Consider an ant standing still on the surface of a basketball. If the ball is rotating, the ant is moving in a circle around the axis of rotation. (The ant may *think* it is at rest. You may *think* you are sitting still in this room, but you are actually moving around in a large circle at over 600 miles per hour.) An object moving in a circle has a continuously changing velocity. Therefore, it is accelerating. The acceleration is directed towards the center of the circle. A coordinate system attached to a rotating sphere is an accelerated coordinate system. A particle at rest relative to this coordinate system is accelerating relative to inertial space.

As a very important special case, consider a coordinate system (x', y', z') rigidly connected to the rotating Earth. Suppose the origin of this coordinate system is at the center of the Earth, as shown in Figure 15.2. The Earth and the coordinate system rotate together about the z' -axis at an angular velocity $\boldsymbol{\Omega}$ (assumed constant). Figure 15.2 also shows an inertial coordinate system (x, y, z) that is at rest with respect to the fixed stars. (I am ignoring the motion of the Earth around the Sun and the motion of the Sun around the center of the Galaxy.)

The common origin of those two coordinate systems is the center of the Earth. The z - and z' -axes coincide and lie along the axis of rotation of the Earth. (You can imagine these axes pointing from the center of the Earth out through the North Pole.) The x, y and x', y' axes all lie in the

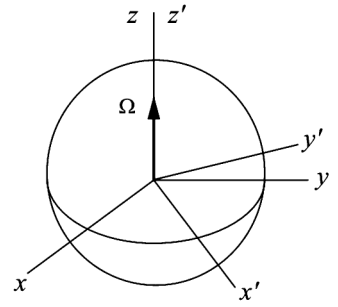


Figure 15.2 A rotating coordinate system (x', y', z') fixed in the Earth and an inertial system (x, y, z) at rest with respect to the fixed stars.

equatorial plane. Again, the coordinate axes (x, y, z) are fixed in space and the coordinate axes (x', y', z') are rigidly attached to the Earth and rotating with it.

A person standing still on the surface of the Earth is at rest with respect to the rotating coordinate system. The position of this person is given by the vector $\mathbf{r}$ from the origin to the point P where the person is standing. The components of $\mathbf{r}$ can be specified in either the inertial or the rotating coordinate system. Because P is at rest in the rotating (Earth or primed) system,

$$\left[\frac{d\mathbf{r}}{dt} \right]_{\text{rot}} = 0.$$

On the other hand, according to an observer in the inertial system, the point P is moving with angular velocity $\boldsymbol{\Omega}$. The linear velocity of the point in the inertial system is

$$\left[\frac{d\mathbf{r}}{dt} \right]_{\text{inertial}} = \boldsymbol{\Omega} \times \mathbf{r}.$$

Next assume the person has a velocity $\mathbf{v}_r$ with respect to the Earth, so that

$$\left[\frac{d\mathbf{r}}{dt} \right]_{\text{rot}} = \mathbf{v}_r.$$

Then, from the point of view of the inertial frame, the person's velocity is

$$\left[\frac{d\mathbf{r}}{dt} \right]_{\text{inertial}} = \mathbf{v}_r + \boldsymbol{\Omega} \times \mathbf{r}.$$

This last equation can be written

$$\left[\frac{d\mathbf{r}}{dt} \right]_{\text{inertial}} = \left[\frac{d\mathbf{r}}{dt} \right]_{\text{rot}} + \boldsymbol{\Omega} \times \mathbf{r}.$$

It is easy to generalize this relationship to the time derivative of any vector $\mathbf{U}$. The relation between the rate of change of $\mathbf{U}$ in the inertial system and the rate of change of $\mathbf{U}$ in the rotating system is

$$\left[\frac{d\mathbf{U}}{dt} \right]_{\text{inertial}} = \left[\frac{d\mathbf{U}}{dt} \right]_{\text{rot}} + \boldsymbol{\Omega} \times \mathbf{U}. \quad (15.3)$$

This can be expressed as an *operator equation*

$$\left[\frac{d}{dt} \right]_{\text{inertial}} = \left[\frac{d}{dt} \right]_{\text{rot}} + \boldsymbol{\Omega} \times, \quad (15.4)$$

where it is understood that the various terms are all operating on the same vector.

Exercise 15.1

Somewhere out in space there is an inertial reference frame and (strange to say) in it there is a merry-go-round that is rotating at angular velocity $\mathbf{\Omega}$. A child is sitting on a wooden horse a distance R from the axis of the merry-go-round. (a) What is the velocity of the child in the rotating frame? (b) What is the velocity of the child in the inertial frame? (c) What is the apparent acceleration of the child in the rotating frame? (d) Determine the acceleration of the child in the inertial frame by elementary methods, that is, by using the definition of centripetal acceleration. (e) Determine the acceleration of the child in the inertial frame using Equation (15.3). Answers (a) zero, (b) $\mathbf{\Omega}R\hat{\phi}$, (c) zero, (d) $-\mathbf{\Omega}^2R\hat{\rho}$. ■

Exercise 15.2

Prove that the rate of change of the angular velocity of the rotating system is the same in both the inertial and the rotating reference frame. ■

15.3 Fictitious Forces

Let us continue ignoring the motion of the Earth around the Sun and imagine the Earth to be a rotating sphere isolated in space. It is rotating at a nearly constant angular velocity of 2π rad/day. The relationship between the time rate of change of the position vector ($\mathbf{r}$) of a particle in the inertial reference frame and its time rate of change in the rotating reference frame is, by Equation (15.3),

$$\left[\frac{d\mathbf{r}}{dt}\right]_{\text{inertial}} = \left[\frac{d\mathbf{r}}{dt}\right]_{\text{rot}} + \mathbf{\Omega} \times \mathbf{r}.$$

The velocity of the particle with respect to inertial space is $\left[\frac{d\mathbf{r}}{dt}\right]_{\text{inertial}}$. Call it $\mathbf{v}_i$. Also, $\left[\frac{d\mathbf{r}}{dt}\right]_{\text{rot}}$ is the velocity of the particle with respect to the rotating frame. Call it $\mathbf{v}_r$. Consequently,

$$\mathbf{v}_i = \mathbf{v}_r + \mathbf{\Omega} \times \mathbf{r}.$$

This equation indicates that if a person on the Earth perceives a particle to be at rest, an observer in inertial space will perceive the particle to have a velocity $\mathbf{v}_i = \mathbf{\Omega} \times \mathbf{r}$.

Acceleration is defined as the time rate of change of velocity. Hence,

$$\mathbf{a}_{\text{inertial}} = \left[\frac{d\mathbf{v}_i}{dt}\right]_i.$$

This is the acceleration in the inertial reference frame and is, of course, the acceleration that is expressed in Newton's second law. However, we live in a rotating reference frame, so we want to express the second law in terms of quantities we can measure, that is, $\mathbf{a}_r$ and $\mathbf{v}_r$.

Applying operator equation (15.4) yields

$$\begin{aligned} \mathbf{a}_i &= \left[\frac{d}{dt}(\mathbf{v}_r + \mathbf{\Omega} \times \mathbf{r})\right]_{\text{inertial}} \\ &= \left[\frac{d}{dt}(\mathbf{v}_r + \mathbf{\Omega} \times \mathbf{r})\right]_{\text{rot}} + \mathbf{\Omega} \times (\mathbf{v}_r + \mathbf{\Omega} \times \mathbf{r}), \\ &= \left[\frac{d}{dt}\mathbf{v}_r\right]_{\text{rot}} + \left[\frac{d}{dt}(\mathbf{\Omega} \times \mathbf{r})\right]_{\text{rot}} + \mathbf{\Omega} \times \mathbf{v}_r + \mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{r}), \\ &= \mathbf{a}_r + \frac{d\mathbf{\Omega}}{dt} \times \mathbf{r} + \left[\left(\mathbf{\Omega} \times \frac{d\mathbf{r}}{dt}\right)\right]_{\text{rot}} + \mathbf{\Omega} \times \mathbf{v}_r + \mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{r}). \end{aligned}$$

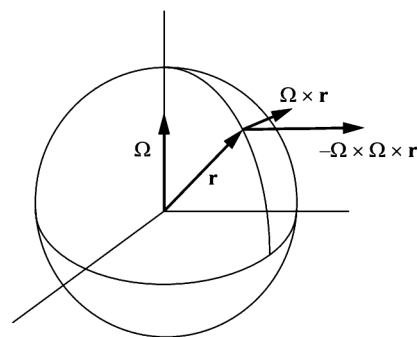


Figure 15.3 The centrifugal acceleration, $-\Omega \times (\Omega \times \mathbf{r})$, points away from the axis of rotation.

If the Earth's rotation rate is assumed constant, $\frac{d\Omega}{dt} = 0$. Using the notation $\left[\frac{d\mathbf{r}}{dt}\right]_{\text{rot}} = \mathbf{v}_r$, the inertial acceleration is given by

$$\mathbf{a}_i = \mathbf{a}_r + 2\Omega \times \mathbf{v}_r + \Omega \times (\Omega \times \mathbf{r}).$$

Newton's second law is valid in the inertial frame, so $\mathbf{a}_i = \mathbf{F}/m$, where $\mathbf{F}$ is the net external ("physical") force acting on the particle. Replacing $\mathbf{a}_i$ and rearranging leads to

$$\mathbf{a}_r = \mathbf{F}/m - 2\Omega \times \mathbf{v}_r - \Omega \times (\Omega \times \mathbf{r}). \quad (15.5)$$

This equation shows that the acceleration *as measured in the rotating reference frame* will depend not only on the force acting on the particle, but also on two other terms. These additional terms are a consequence of considering the motion in a noninertial reference frame.

One often writes Equation (15.5) in the suggestive form

$$\mathbf{F} - 2m\Omega \times \mathbf{v}_r - m[\Omega \times (\Omega \times \mathbf{r})] = m\mathbf{a}_r. \quad (15.6)$$

This equation *looks* like $F = ma$. The additional terms on the left-hand side of the equation have the units of force and they are referred to as "fictitious forces." Keep in mind, however, that they really are just mass times acceleration terms that have been brought over from the other side of the equation. A "real" force is an interaction between bodies. A "fictitious" force is a consequence of the acceleration of the coordinate system.

The term

$$-2m\Omega \times \mathbf{v}_r,$$

is called the "Coriolis force" and the term

$$-m\Omega \times (\Omega \times \mathbf{r}),$$

is called the "centrifugal force." The reason for the name is that this "force" is directed away from axis of rotation, as illustrated in Figure 15.3

Worked Example 15.1 An old-fashioned record player is rotating with an angular velocity $\Omega_0 = \Omega_0 \hat{\mathbf{k}}$. A bug is crawling with constant speed v_b relative to the record along a groove in the record, a distance r from the axis of rotation. Assume the bug is moving in a circular path, so there is no radial component to its velocity. (a) Determine the velocity of the bug relative to the room (that is the "inertial space"). (b) Determine the acceleration of the bug by elementary means, using the definition of centripetal acceleration. (c) Determine the acceleration of the bug using Equation (15.4) and compare your answer to part (b). (Comment: This example will help you learn the mechanics of using Equation (15.4).) In particular, you will appreciate that the same vector must be used on both sides of that equation. That is, if you are calculating

$\left[\frac{d\mathbf{v}_i}{dt}\right]_{\text{inertial}}$ you get the wrong answer if you put $\left[\frac{d\mathbf{v}_{\text{rot}}}{dt}\right]_{\text{rot}}$ on the other side. This is fairly obvious, but it is an easy error to make.)

Solution Let $\hat{\rho}, \hat{\phi}, \hat{\mathbf{k}}$ be cylindrical unit vectors in the rotating frame. (a) The velocity of the bug in the rotating coordinate frame is

$$\left[\frac{d\mathbf{r}}{dt}\right]_{\text{rot}} = v_b \hat{\phi}.$$

The velocity of the bug in inertial space by Equation (15.4) is

$$\begin{aligned} \left[\frac{d\mathbf{r}}{dt}\right]_{\text{inertial}} &= \left[\frac{d\mathbf{r}}{dt}\right]_{\text{rot}} + \boldsymbol{\Omega}_0 \times \mathbf{r} \\ \mathbf{v}_i &= \mathbf{v}_{\text{rot}} + \boldsymbol{\Omega}_0 \times \mathbf{r} \\ &= v_b \hat{\phi} + \Omega_0 r (\hat{\mathbf{k}} \times \hat{\rho}) \\ &= v_b \hat{\phi} + \Omega_0 r \hat{\phi}. \end{aligned}$$

For convenience, let us write $v_b \hat{\phi}$ as $\Omega_b r \hat{\phi}$. Then

$$\mathbf{v}_i = (\Omega_b + \Omega_0) r \hat{\phi}.$$

(b) By the definition of centripetal acceleration

$$\begin{aligned} \mathbf{a}_i &= \frac{v_i^2}{r} (-\hat{\rho}) = \frac{-\hat{\rho}}{r} [(\Omega_b + \Omega_0)r]^2 \\ &= -\hat{\rho} r (\Omega_b + \Omega_0)^2. \end{aligned}$$

(c) Using Equation (15.4) to determine $\mathbf{a}_i$ we have

$$\begin{aligned} \mathbf{a}_i &= \left[\frac{d\mathbf{v}_i}{dt}\right]_{\text{inertial}} = \left[\frac{d}{dt} (\Omega_b + \Omega_0) r \hat{\phi}\right]_{\text{inertial}} \\ &= \left[\frac{d}{dt} (\Omega_b + \Omega_0) r \hat{\phi}\right]_{\text{rot}} + \boldsymbol{\Omega}_0 \times (\Omega_b + \Omega_0) r \hat{\phi}. \end{aligned}$$

Since $v_b = \Omega_b r$ and because the rate of change of $\hat{\phi}$ in the rotating frame is only due to the motion of the bug, we have $\left[\frac{d\hat{\phi}}{dt}\right]_{\text{rot}} = -\Omega_b \hat{\rho}$. Furthermore, keeping in mind that $|\mathbf{r}| = r = \text{constant}$, we obtain

$$\begin{aligned} \mathbf{a}_i &= (\Omega_b + \Omega_0) r (-\Omega_b) \hat{\rho} + \Omega_0 (\Omega_0 + \Omega_b) r (\hat{\mathbf{k}} \times \hat{\phi}) \\ &= -\hat{\rho} r [(\Omega_b + \Omega_0)\Omega_b + \Omega_0(\Omega_b + \Omega_0)] \\ &= -\hat{\rho} r [\Omega_b^2 + 2\Omega_b\Omega_0 + \Omega_0^2] \\ &= -\hat{\rho} r (\Omega_b + \Omega_0)^2, \end{aligned}$$

in agreement with the result of part (b).

15.4 Centrifugal Force and the Plumb Bob

Assume the Earth is a sphere. Most people probably assume that a hanging plumb bob points directly towards the center of the Earth. However, that would only be true if the Earth were not

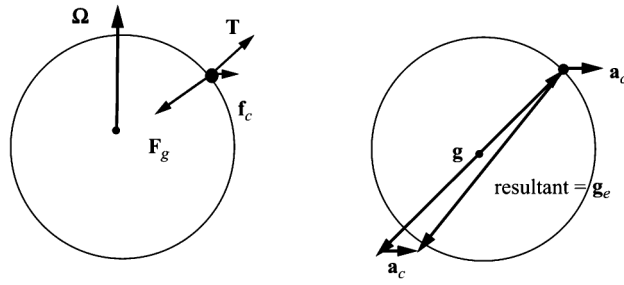


Figure 15.4 A plumb bob in the Northern Hemisphere points below the center of a spherical Earth. The sketch on the left shows the “forces” acting on the bob. The vector $\mathbf{f}_c$ represents the centrifugal force. The sketch on the right illustrates the vector sum of $\mathbf{g}$ and $\mathbf{a}_c = -\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$.

rotating. On a rotating Earth, the centrifugal force causes the plumb bob to be deflected slightly away from the line to the center. The proof is fairly simple. You may want to try it on your own before reading my derivation.

As shown in [Figure 15.4](#), the “real” forces acting on the bob are the tension in the string and the gravitational attraction of the Earth. That is,

$$\mathbf{F} = \mathbf{T} + m\mathbf{g}.$$

Inserting this expression for the force into the “ $F = ma$ ” equation in the form of Equation (15.6) we get

$$\mathbf{T} + m\mathbf{g} - 2m\boldsymbol{\Omega} \times \mathbf{v}_r - m[\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})] = m\mathbf{a}_r,$$

or

$$\mathbf{T} + m\mathbf{g} - m[\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})] = 0,$$

where I used the fact that the velocity and acceleration of the plumb bob in the rotating frame are zero.

Now $\mathbf{g}$ points towards the center of the Earth, but the equation for the plumb bob written in the form

$$\mathbf{T} + m\mathbf{g} = m[\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})],$$

indicates that the tension is not parallel to $m\mathbf{g}$ and therefore it is not directed along the line to the center of the Earth.

It is usually convenient to include the centrifugal acceleration into the gravitational acceleration. The “effective” gravitational acceleration is defined by

$$\mathbf{g}_e = \mathbf{g} - \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}),$$

and the equation reduces to

$$\mathbf{T} + m\mathbf{g}_e = 0.$$

That is, the tension is directed opposite to $\mathbf{g}_e$ rather than opposite to $\mathbf{g}$. See [Figure 15.4](#). In the Northern Hemisphere $\mathbf{g}_e = \mathbf{g} - \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$ points slightly below the center of the Earth while in the Southern Hemisphere it points slightly above the center.

The centrifugal force is responsible for the fact that the Earth is not a sphere; it has a “bulge” at the equator. To see how this comes about, consider a particle on the surface of a perfectly spherical Earth. This particle is subjected to the gravitational force, a normal force and the centrifugal

force. If you draw a force diagram for this situation, you will find that the centrifugal force is an unbalanced force. Resolving the centrifugal force into components parallel and perpendicular to the Earth's surface, you will find that the perpendicular component simply adds to the normal force, but the unbalanced parallel component points toward the equator. Thus, the centrifugal force is responsible for the fact that the Earth is an oblate spheroid rather than a sphere. You can convince yourself that if the Earth's equatorial bulge is taken into consideration there will be no unbalanced forces on a particle on the surface. This argument also tells you that although a plumb line does not point towards the center of the Earth, it is perpendicular to the Earth's *surface*, so it does define a local perpendicular.

Exercise 15.3

Draw the force diagram for a particle on the surface of a perfectly smooth, spherical, rotating planet and show that the particle accelerates towards the equator. Do this for both the Northern and Southern Hemispheres. Draw the same sort of sketch for a planet with an equatorial bulge and demonstrate that in this case there is no net force on the particle. ■

15.5 The Coriolis Force

It is well known (at least to physicists) that a projectile fired near the surface of the Earth will veer towards the right in the Northern Hemisphere and towards the left in the Southern Hemisphere. This is due to the so-called Coriolis force, which is also responsible for oceanic circulation, cyclones, hurricanes, and other geophysical phenomena.

The Coriolis force is the term $-2m\boldsymbol{\Omega} \times \mathbf{v}_r$ in Equation (15.6). Note that the direction of this fictitious force depends on the direction of the velocity vector.

When studying the effect of the Coriolis force on objects moving near the surface of the Earth it is convenient to introduce a Cartesian coordinate system in which the z -axis points up from the surface (i.e., opposite to the direction of the effective gravitational acceleration $\mathbf{g}_e$), the x -axis points East, and the y -axis points North.² See Figure 15.5. Note that the angular velocity of the Earth ($\boldsymbol{\Omega}$) has components along the y -axis (North) and along the $+z$ -axis (up). Thus:

$$\boldsymbol{\Omega} = \Omega \cos \lambda \hat{\mathbf{k}} + \Omega \sin \lambda \hat{\mathbf{j}},$$

where λ is the “colatitude.”

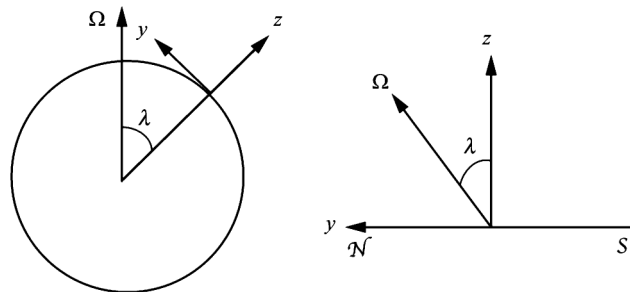


Figure 15.5 A reference frame fixed to the rotating Earth. The z -axis is along the local vertical, the y -axis points North and the x -axis points East (into the page). The angle λ is the colatitude.

² Some books use a different convention with the x -axis pointing South. The equations for the motion will be slightly different, so be careful when comparing equations from different books.

Exercise 15.4

Draw force diagrams to determine the direction of the Coriolis force on the following objects. (a) A falling stone. (b) A projectile fired due North in the Northern Hemisphere. (c) A projectile fired due East in the Northern Hemisphere. (d) A projectile fired due East in the Southern Hemisphere. ■

15.5.1 A Falling Body

An object of mass m is dropped from a height h . Where does it land? It turns out that the object hits the ground a tiny bit to the East of the spot directly below the point where it was released. The reason is that the falling object has a downward velocity so there is a Coriolis force acting on it, given by $-2m\boldsymbol{\Omega} \times \mathbf{v}_r$. (In the rotating reference frame of the Earth, the forces acting on the particle are the gravitational force, the Coriolis force, and the centrifugal force.)

For simplicity, we incorporate the centrifugal acceleration into $\mathbf{g}$ and write $\mathbf{g}_e$ for the sum of these two terms. The equation of motion of the falling body is, then

$$m\mathbf{a}_r = m\mathbf{g}_e - 2m\boldsymbol{\Omega} \times \mathbf{v}_r.$$

If you think about it, you will soon realize we cannot solve this equation for $\mathbf{a}_r$ because we do not know $\mathbf{v}_r$, and we cannot determine $\mathbf{v}_r$ until we solve the equation! It would seem that we are faced with an unsolvable problem.

The way out of this dilemma is to use the fact that the Coriolis acceleration is much smaller than the gravitational acceleration, so $\mathbf{v}_r$ is almost (but not quite) equal to the velocity of the falling body in the absence of the Coriolis force. That is, the Coriolis effect can be treated as a *perturbation*. The technique is as follows: First solve the unperturbed problem and determine the unperturbed velocity and time for the object to fall. This is called the “zeroth-order approximation.” Then use the unperturbed velocity in the expression for the Coriolis acceleration to obtain the “first-order approximation.” This process can be repeated as many times as desired to get better and better approximations to the correct answer, but usually the first- or second-order approximation is quite adequate.³

Let us begin by solving the problem for the *unperturbed* velocity and position of the particle as a function of time (the zeroth-order approximation). The only force acting on the body is the (effective) gravitational force. Consequently,

$$\mathbf{a}_r = -g_e \hat{\mathbf{k}}.$$

Integrating once,

$$\mathbf{v}_r = -g_e t \hat{\mathbf{k}},$$

and integrating again,

$$\mathbf{z} = h\hat{\mathbf{k}} - \frac{1}{2}g_e t^2 \hat{\mathbf{k}}.$$

The time required for the particle to reach the ground, t_f , is the value of t when $z = 0$, that is,

$$t_f = \sqrt{2h/g_e}.$$

Note that none of the above relations include the Coriolis effect.

Now include the Coriolis acceleration, given by:

$$\mathbf{a}_c = -2\boldsymbol{\Omega} \times \mathbf{v}_r.$$

³ This technique is known as the method of successive approximations.

You cannot evaluate $\mathbf{a}_c$ exactly because you do not know the value of $\mathbf{v}_r$ which itself depends on the Coriolis acceleration. However, you know the zeroth-order approximation to the velocity. It is $\mathbf{v}_r = -g_e t \hat{\mathbf{k}}$. The *first-order* correction is obtained by including the Coriolis term in the acceleration but using the zeroth-order velocity. That is, $\mathbf{a} = g_e \hat{\mathbf{k}} + \mathbf{a}_c$ where

$$\begin{aligned}\mathbf{a}_c &= -2\boldsymbol{\Omega} \times \mathbf{v}_r = -2 \left[\Omega \cos \lambda \hat{\mathbf{k}} + \Omega \sin \lambda \hat{\mathbf{j}} \right] \times \left[-g_e t \hat{\mathbf{k}} \right] \\ &= 2g_e t \Omega \sin \lambda \hat{\mathbf{i}}.\end{aligned}$$

Note that for the falling body, the Coriolis acceleration is in the positive x -direction, that is, towards the East. The eastward component of the velocity (v_x) is initially zero and at any later time it is obtained by integrating the expression above. Therefore,

$$v_x = g_e t^2 \Omega \sin \lambda.$$

The eastward deflection (x) is obtained by integrating again.

$$x = \frac{1}{3} g_e t^3 \Omega \sin \lambda.$$

To determine the total eastward deflection when the object is dropped from height h , substitute the time of flight ($t = \sqrt{2h/g_e}$) into this last expression.

The results obtained are the first-order solution. As mentioned above, these results can be plugged back into the equation of motion to obtain the second-order solution, and so on.

Students are sometimes mystified by this result. The eastward deflection is not due to the Earth rotating under the falling body – anyway, that would give a westward deflection! A physical explanation for the horizontal (eastward) deflection of a dropped body comes from the conservation of angular momentum. Let $\mathbf{l}$ be the angular momentum of the particle in the inertial frame. There is no torque acting on it, so $\mathbf{l}$ is constant. Initially the object is at position $\mathbf{r}$ (where $\mathbf{r}$ is a vector from the center of the Earth). In the inertial frame the body has an eastward linear velocity $\mathbf{v}$. Because the body is at rest with respect to the surface of the Earth, this initial velocity is $\mathbf{v} = \boldsymbol{\Omega} \times \mathbf{r} \doteq \Omega R_e \sin \lambda \hat{\mathbf{i}}$. The angular momentum of the particle is

$$\mathbf{l} = m \mathbf{r} \times \mathbf{v}.$$

For $\mathbf{l}$ to remain constant while $\mathbf{r}$ decreases (as the particle falls), the velocity $\mathbf{v}$ must increase. Therefore, in the Earth-based reference frame, the object acquires a velocity towards the East.⁴

Another strange and interesting effect of Coriolis forces is the fact that if you throw an object straight up, it does not land at the point from which it started. (You will prove this in [Problem 15.6](#).)

Worked Example 15.2 To first order, a falling particle is not deflected towards the South. However, there is a southward deflection in the second-order approximation. Estimate this deviation and compare it with the first-order deviation towards the East.

Solution To first order, the velocity is

$$\mathbf{v}_r = v_z \hat{\mathbf{k}} + v_x \hat{\mathbf{i}} = -g_e t \hat{\mathbf{k}} + g_e t^2 \Omega \sin \lambda \hat{\mathbf{i}}.$$

⁴ A slightly different explanation for the eastward deflection of a dropped object is based on the fact that if you hold an object above a point on the Earth's surface, both the object and the surface point are moving towards the East with the same angular velocity (Ω). But because $v = \Omega r$ the object has a slightly greater eastward velocity than the surface point. This greater eastward velocity is not affected by the fall and therefore the object lands to the East of the surface point.

The Coriolis acceleration is

$$\begin{aligned}\mathbf{a}_c &= -2\boldsymbol{\Omega} \times \mathbf{v}_r \\ &= -2 \left[\Omega \cos \lambda \hat{\mathbf{k}} + \Omega \sin \lambda \hat{\mathbf{j}} \right] \times \left[-g_e t \hat{\mathbf{k}} + g_e t^2 \Omega \sin \lambda \hat{\mathbf{i}} \right].\end{aligned}$$

That is,

$$\begin{aligned}a_x &= 2 (\Omega \sin \lambda) g_e t, \\ a_y &= -2 \left(\Omega^2 \cos \lambda \sin \lambda \right) g_e t^2, \\ a_z &= 2 \left(\Omega^2 \sin^2 \lambda \right) g_e t^2 - g_e.\end{aligned}$$

Consequently,

$$v_y = \int a_y dt = -\frac{2}{3} \left(\Omega^2 \cos \lambda \sin \lambda \right) g_e t^3,$$

and

$$y = -\frac{1}{6} \left(\Omega^2 \cos \lambda \sin \lambda \right) g_e t^4.$$

Using the first-order value for the time of flight we obtain

$$y = -\frac{2}{3} \frac{h^2}{g_e} \Omega^2 \cos \lambda \sin \lambda.$$

Note that the time of flight will actually be slightly longer because the Coriolis acceleration opposes the gravitational acceleration.

Comparing the southward and eastward deflections we have

$$\frac{y}{x} = \frac{-(1/6) (\Omega^2 \cos \lambda \sin \lambda) g_e t^4}{(1/3) (\Omega \sin \lambda) g_e t^3} = -\frac{1}{2} (\Omega \cos \lambda) t = -\Omega \cos \lambda \sqrt{h/2g_e}.$$

Since Ω is about 10^{-5} and $\cos \lambda$ is near unity and the time of flight will be in the range of 1–10 s, we appreciate that the southward deflection is significantly smaller than the eastward deflection.

Exercise 15.5

A stone is dropped from a height of 50 m. How far is it deflected towards the East? Assume the latitude is 60° N. Answer: 0.39 cm. ■

15.5.2 A Projectile

To determine the motion of a projectile fired from the surface of the Earth with arbitrary initial speed and direction, let us return to the equation for the acceleration in a rotating coordinate frame. A slight modification of Equation (15.5) gives

$$\mathbf{a} = \mathbf{g} - 2(\boldsymbol{\Omega} \times \mathbf{v}).$$

I dropped the subscript “e” on $\mathbf{g}$, but keep in mind that the centrifugal force is incorporated into $\mathbf{g}$. I also dropped the subscript $\mathbf{r}$ on $\mathbf{v}_r$ and on $\mathbf{a}_r$. The velocity $\mathbf{v}$ at any time can be expressed as

$$\mathbf{v} = v_x \hat{\mathbf{i}} + v_y \hat{\mathbf{j}} + v_z \hat{\mathbf{k}},$$

and

$$\mathbf{\Omega} = \Omega \sin \lambda \hat{\mathbf{j}} + \Omega \cos \lambda \hat{\mathbf{k}}.$$

Hence,

$$\mathbf{a} = -g\hat{\mathbf{k}} + 2(v_y\Omega \cos \lambda - v_z\Omega \sin \lambda)\hat{\mathbf{i}} - 2v_x\Omega \cos \lambda \hat{\mathbf{j}} + 2v_x\Omega \sin \lambda \hat{\mathbf{k}}, \quad (15.7)$$

or

$$a_x = 2v_y\Omega \cos \lambda - 2v_z\Omega \sin \lambda,$$

$$a_y = -2v_x\Omega \cos \lambda,$$

$$a_z = 2v_x\Omega \sin \lambda - g.$$

Recall that you cannot determine the motion by integrating the acceleration terms twice because the accelerations depend on the velocity which is unknown. So once again it is necessary to use the method of successive approximations. That is, you obtain the first-order approximation by using the velocity components of the zeroth-order approximation, and similarly calculating the higher-order approximations. However, even without doing any calculations, the functional dependence of the acceleration components give a good qualitative idea of the motion. For example, in the Northern Hemisphere, if the projectile is fired due North, then v_x (the eastward component of the velocity) is initially zero and there is no Coriolis acceleration in the y - or z -directions. There is, however, an *eastward* component of the acceleration and the projectile will be deflected towards the East (to the right of its direction of motion). To actually evaluate this deflection you must incorporate the zeroth-order approximations of v_y and v_z into the expression for a_x and then integrate.

Similarly, the expressions for the acceleration components indicate that a projectile fired due East (x -direction) is deflected towards the South. The velocity component v_x slightly changes a_z , making it less negative, therefore slightly increasing the time of flight of the projectile resulting in an increase in range. Furthermore, a_x is nonzero, which also affects the range of the projectile. Although the vertical velocity v_z is positive during the first part of the trajectory and negative during the second part, you cannot assume that the second term in a_x averages out to zero.

In solving for the motion of a projectile, you must use perturbation techniques, but fortunately a reasonably accurate solution is usually given by the lowest order approximation that includes the Coriolis force.

You should convince yourself that in the Northern Hemisphere a projectile will be deflected toward its right and in the Southern Hemisphere it will be deflected toward its left.

Worked Example 15.3 A large cannon is located at colatitude λ in the Northern Hemisphere. It fires a shell with velocity $\mathbf{v}_0$. The cannon is aimed at an angle ϕ East of North and has an elevation angle θ . (a) Write expressions for the initial (unperturbed) velocity components. (Call them v_{0x}, v_{0y}, v_{0z} .) (b) Write expressions for a_x, a_y, a_z in terms of the unperturbed velocity components. (c) Obtain first-order expressions for the velocity components as functions of time and the unperturbed values of the velocity. Ignore air resistance. (The results of this example will be found useful when solving the problems at the end of the chapter.)

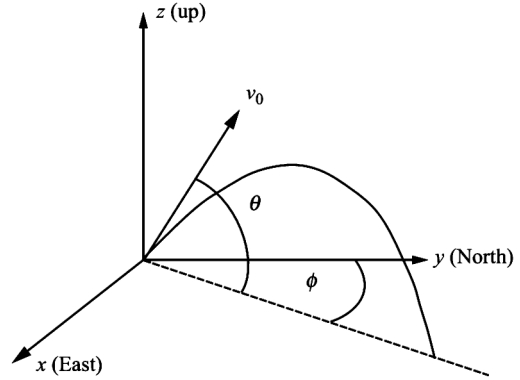
Solution (a) By inspection of [Figure 15.6](#) the initial velocity components are

$$v_{0x} = v_0 \cos \theta \sin \phi,$$

$$v_{0y} = v_0 \cos \theta \cos \phi,$$

$$v_{0z} = v_0 \sin \theta.$$

Figure 15.6 A projectile fired East of North. Note that ϕ is measured from North. The elevation angle θ is measured from the horizontal plane.



(b) The Coriolis acceleration is

$$\begin{aligned}\mathbf{a}_c &= -2\boldsymbol{\Omega} \times \mathbf{v} = -2 \left[\Omega \sin \lambda \hat{\mathbf{j}} + \Omega \cos \lambda \hat{\mathbf{k}} \right] \times \left[v_x \hat{\mathbf{i}} + v_y \hat{\mathbf{j}} + v_z \hat{\mathbf{k}} \right], \\ &= -2 \left[-v_x \Omega \sin \lambda \hat{\mathbf{k}} + v_z \Omega \sin \lambda \hat{\mathbf{i}} + v_x \Omega \cos \lambda \hat{\mathbf{j}} - v_y \Omega \cos \lambda \hat{\mathbf{i}} \right].\end{aligned}$$

Since $v_x = v_{0x}$, $v_y = v_{0y}$ and $v_z = v_{0z} - gt$ we obtain

$$\begin{aligned}a_x &= 2v_0 \Omega \cos \lambda \cos \theta \cos \phi - 2\Omega \sin \lambda (v_0 \sin \theta - gt), \\ a_y &= -2v_0 \Omega \cos \lambda \cos \theta \sin \phi, \\ a_z &= 2v_0 \Omega \sin \lambda \cos \theta \sin \phi - g.\end{aligned}$$

(c) The velocity is determined by integrating the acceleration components to obtain:

$$\begin{aligned}v_x &= v_{0x} + \int a_x dt \\ &= v_0 \cos \theta \sin \phi + (2v_0 \Omega \cos \lambda \cos \theta \cos \phi) t \\ &\quad - (2v_0 \Omega \sin \lambda \sin \theta) t + g t^2 \Omega \sin \lambda, \\ v_y &= v_{0y} + \int a_y dt = v_0 \cos \theta \cos \phi - (2v_0 \Omega \cos \lambda \cos \theta \sin \phi) t, \\ v_z &= v_{0z} + \int a_z dt = v_0 \sin \theta - gt + (2v_0 \Omega \sin \lambda \cos \theta \sin \phi) t.\end{aligned}$$

Exercise 15.6

Carry out the operation $\boldsymbol{\Omega} \times \mathbf{v}$ to verify Equation (15.7). ■

Exercise 15.7

A projectile located at 60°N is fired due East with an initial velocity of 300 m/s at an angle 25° above the horizontal. Evaluate the initial values of the components of the Coriolis acceleration. (Note: $\lambda = 30^\circ$.) Answer: $a_x(t = 0) = -9.22 \times 10^{-3} \text{ m/s}^2$. ■

15.6 The Foucault Pendulum

The Foucault pendulum is an important example of the effect of a rotating coordinate system on the dynamics of a physical system. This pendulum consists of a massive bob suspended by a very

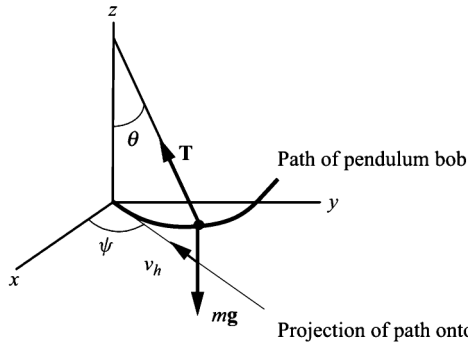


Figure 15.7 The Foucault pendulum.

long string or thin wire. Foucault pendulums range from ten to sixty meters in length. When set in motion the bob swings back and forth and the plane of motion of the pendulum slowly precesses. The precession rate is $\Omega \cos \lambda$, where Ω is the angular velocity of the Earth (2π rad/day) and λ is the colatitude. A long pendulum goes through many oscillations before the motion is damped out, and the precession is quite noticeable. When Foucault suspended a pendulum inside the Pantheon in Paris large crowds showed up to “observe the rotation of the Earth.”

The behavior of the Foucault pendulum is easy to understand if you imagine a pendulum suspended above the North Pole. The Earth rotates under the pendulum and an observer standing on the surface sees the plane of motion precessing at a rate of one revolution per day. On the other hand, at the equator, a pendulum does not precess at all. The motion at an arbitrary latitude is not intuitively obvious and requires an analysis to understand it.

Consider a long pendulum at an arbitrary latitude. Let the z -axis be oriented along the local vertical. See Figure 15.7, which shows the projection of the path of the pendulum onto the xy -plane. The projected path makes an angle ψ with the x -axis. The precession of the pendulum causes ψ to change as a function of time. The basic problem consists in demonstrating that the rate of change of ψ is

$$\frac{d\psi}{dt} = \Omega \cos \lambda.$$

The pendulum bob is acted upon by two real forces: the tension in the string ($\mathbf{T}$) and gravity ($m\mathbf{g}$). It is also acted upon by two fictitious forces: the Coriolis force ($-2m\boldsymbol{\Omega} \times \mathbf{v}$) and the centrifugal force. For convenience assume the centrifugal force is absorbed into $m\mathbf{g}$.

Therefore,

$$m\mathbf{a} = \mathbf{T} + m\mathbf{g} - 2m(\boldsymbol{\Omega} \times \mathbf{v}).$$

In component form this relationship becomes

$$\begin{aligned} m a_x \hat{\mathbf{i}} + m a_y \hat{\mathbf{j}} + m a_z \hat{\mathbf{k}} = & -T \sin \theta \cos \psi \hat{\mathbf{i}} - T \sin \theta \sin \psi \hat{\mathbf{j}} + T \cos \theta \hat{\mathbf{k}} - m g \hat{\mathbf{k}} \\ & - 2m \left[\Omega \cos \lambda \hat{\mathbf{k}} + \Omega \sin \lambda \hat{\mathbf{j}} \right] \times \left[v_x \hat{\mathbf{i}} + v_y \hat{\mathbf{j}} + v_z \hat{\mathbf{k}} \right]. \end{aligned}$$

That is,

$$-T \sin \theta \cos \psi + 2m v_y \Omega \cos \lambda - 2m v_z \Omega \sin \lambda = m a_x, \quad (15.8)$$

$$-T \sin \theta \sin \psi - 2m v_x \Omega \cos \lambda = m a_y, \quad (15.9)$$

$$T \cos \theta - m g + 2m v_x \Omega \sin \lambda = m a_z. \quad (15.10)$$

Consider Equations (15.8) and (15.9). Because the vertical displacement is small, the vertical velocity v_z is much smaller than the horizontal velocity. Therefore, we are justified in ignoring the term containing v_z in Equation (15.8), leading to the set of coupled differential equations

$$\begin{aligned} -T \sin \theta \cos \psi + 2m v_y \Omega \cos \lambda &= m a_x, \\ -T \sin \theta \sin \psi - 2m v_x \Omega \cos \lambda &= m a_y. \end{aligned}$$

The pendulum is very long so the angle θ is quite small and the string is essentially vertical at all times (Figure 15.7 is significantly out of scale). To a high degree of accuracy the tension in the string is equal to mg . Furthermore, $x = l \sin \theta \cos \psi$ and $y = l \sin \theta \sin \psi$. Therefore, the coupled equations above can be formulated as:

$$\begin{aligned} -\left(\frac{g}{l}\right)x + 2v_y \Omega \cos \lambda &= a_x, \\ -\left(\frac{g}{l}\right)y - 2v_x \Omega \cos \lambda &= a_y. \end{aligned}$$

Noting that $g/l = \omega^2$, where ω is the oscillatory frequency of the pendulum, and letting $K = \Omega \cos \lambda$, these equations become

$$\begin{aligned} \ddot{x} - 2K\dot{y} + \omega^2 x &= 0, \\ \ddot{y} + 2K\dot{x} + \omega^2 y &= 0. \end{aligned} \tag{15.11}$$

The problem now is to solve these coupled linear differential equations. I will do this in three ways. First I will use a “trick” then I will solve the equations in a straightforward, but more tedious manner. Then, in Worked Example 15.5, I will show you a third way to find the solution. The reason for doing this is because I want you to see several techniques for solving coupled differential equations.

A Neat Trick

The set of coupled equations (15.11) can be solved very nicely by introducing the complex quantity

$$\zeta = x + iy.$$

Multiplying the second equation by i ($= \sqrt{-1}$) and adding the two equations gives

$$\ddot{\zeta} + 2iK\dot{\zeta} + \omega^2 \zeta = 0,$$

which resembles the equation of motion for a damped harmonic oscillator. This linear differential equation is solved in the usual way, assuming $\zeta = e^{pt}$ and obtaining the “secular” or “characteristic” equation

$$p^2 + 2iKp + \omega^2 = 0,$$

with solution

$$p = -iK \pm i\sqrt{K^2 + \omega^2}.$$

Now $K = \Omega \cos \lambda$, where Ω is the rotation rate of the Earth, whereas $\omega = \sqrt{g/l}$ is the angular frequency of the pendulum. So $K \ll \omega$ and the K^2 under the root can be neglected. There are two possible values for p , so the solution is

$$\zeta = Ae^{-i(K+\omega)t} + Be^{-i(K-\omega)t}, \tag{15.12}$$

where A and B are complex constants. Writing the complex constants in the form $(a + ib)$ and $(c + id)$ the solution is

$$\zeta(t) = (a + ib)e^{-i(K+\omega)t} + (c + id)e^{-i(K-\omega)t}. \quad (15.13)$$

This complex solution is expressed in terms of real quantities, but it is still not obvious what the motion looks like. The following example will help you to visualize the motion described by Equation (15.13).

Worked Example 15.4 At time $t = 0$ the Foucault pendulum is released from rest at position $x = \mathcal{A}$, $y = 0$. Obtain expressions for $x(t)$ and $y(t)$.

Solution Let us write the constants in Equation (15.12) as $\frac{1}{2}Ce^{\pm i\theta}$, thus

$$\begin{aligned} \zeta &= \frac{1}{2}Ce^{-i\theta}e^{-i(\omega+K)t} + \frac{1}{2}Ce^{i\theta}e^{i(\omega-K)t} \\ &= \frac{1}{2}Ce^{-iKt} \left[e^{-i(\omega t + \theta)} + e^{+i(\omega t + \theta)} \right] \\ &= \frac{1}{2}C [\cos Kt - i \sin Kt] [2 \cos(\omega t + \theta)] \\ &= C [\cos Kt \cos(\omega t + \theta) - i \sin Kt \cos(\omega t + \theta)]. \end{aligned}$$

Because $x = \text{Re}(\zeta)$ and $y = \text{Im}(\zeta)$ we conclude that

$$\begin{aligned} x &= C \cos(t\Omega \cos \lambda) \cos(\omega t + \theta), \\ y &= -C \sin(t\Omega \cos \lambda) \cos(\omega t + \theta). \end{aligned}$$

The constants C and θ are now easily obtained from the initial conditions, yielding $C = \mathcal{A}$ and $\theta = 0$, so

$$\begin{aligned} x(t) &= [\mathcal{A} \cos(t\Omega \cos \lambda)] \cos \omega t, \\ y(t) &= -[\mathcal{A} \sin(t\Omega \cos \lambda)] \cos \omega t. \end{aligned}$$

The motion is a rapid oscillation of frequency ω , modulated by a low-frequency variation in amplitude ($\Omega \cos \lambda$). The modulated x and y oscillations have time varying amplitudes that are 180° out of phase. The resultant motion is a rotation of the plane of motion with a frequency $\Omega \cos \lambda$. This is illustrated in Figure 15.8.

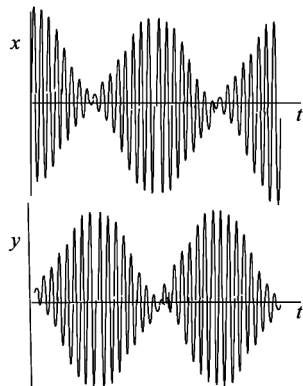


Figure 15.8 Amplitudes of oscillations in the x and y directions for the Foucault pendulum.

Exercise 15.8

A Foucault pendulum is observed to have a velocity v_0 in the $+x$ -direction at time $t = 0$ when it passes through the origin. Obtain expressions for x and y as functions of t . Answer: $x = \operatorname{Re}(\zeta) = -\frac{v_0}{K} \sin \omega t \cos Kt$, $y = \operatorname{Im}(\zeta) = \frac{v_0}{K} \sin \omega t \sin Kt$. ■

A More General Way to Solve the Problem

Return again to [Equations \(15.11\)](#). They are a pair of coupled second-order linear differential equations whose solution is desired. By this time you are very familiar with the technique of assuming a solution of the form Ae^{pt} . In this case there are two unknowns and the procedure is to assume two solutions, but to let both of them have the same time dependence.⁵ In other words, assume the solutions

$$\begin{aligned}x &= Ae^{pt}, \\y &= Be^{pt}.\end{aligned}$$

Plugging these solutions into [Equations \(15.11\)](#) leads to the following set of secular equations

$$\begin{aligned}p^2 A - 2KpB + \omega^2 A &= 0, \\p^2 B + 2KpA + \omega^2 B &= 0.\end{aligned}$$

or

$$\begin{aligned}A(p^2 + \omega^2) - B(2Kp) &= 0, \\A(2Kp) + B(p^2 + \omega^2) &= 0.\end{aligned}$$

This is a system of homogeneous linear algebraic equations in the unknowns A and B . Recall that the condition for a nontrivial solution is

$$\begin{vmatrix} p^2 + \omega^2 & -2Kp \\ 2Kp & p^2 + \omega^2 \end{vmatrix} = 0.$$

This leads to

$$p = -iK \pm i\sqrt{K^2 + \omega^2}. \quad (15.14)$$

The rest of the solution is straightforward.⁶

Worked Example 15.5 Solve for the precession rate of a Foucault pendulum by considering the problem in a rotating reference frame in which the pendulum does not precess.

Solution Using a rotating reference frame makes the problem easier to solve. Imagine looking down on the pendulum from a coordinate system rotating at an angular velocity $K = \Omega \cos \lambda$, equal to the precession rate. In this coordinate frame the bob moves back and forth in a straight line. Let the axes in this coordinate frame be x^* and y^* and define the x^* -axis to lie along the path of the bob. Then y^* is always zero. Denoting the distance from the origin to the bob along the x^* -axis by s , [Figure 15.9](#) shows that the Earth-based coordinates x and y are

$$x = s \cos Kt$$

⁵ You probably remember that this is the procedure we used to solve the coupled oscillators problem. (See [Section 11.5](#).)

⁶ It is interesting to note that the Foucault pendulum problem has many parallels to a seemingly completely unrelated problem, namely the violation of CP symmetry in the decay of an elementary particle called the neutral kaon. To explore further, see the paper “Classical Illustration of CP Violation in Kaon Decays” by Jonathan L. Rosner and Scott A. Slezak, *American Journal of Physics*, **69**, 44–49 (2001).

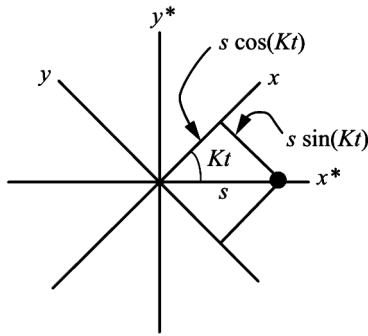


Figure 15.9 The starred coordinate system rotates at an angular velocity $K = \Omega \cos \lambda$ relative to the unstarred system. The motion of the pendulum bob is along the x^* -axis.

and

$$y = -s \sin Kt.$$

The sum of the equations of motion, [Equations \(15.11\)](#), is

$$\ddot{x} + \ddot{y} + 2K(\dot{x} - \dot{y}) + \omega^2(x + y) = 0 \quad (15.15)$$

Plugging in the expressions for x and y in terms of s gives

$$\ddot{s} + (K^2 + \omega^2)s = 0. \quad (15.16)$$

Again, $K \ll \omega$ so, to an excellent approximation,

$$\ddot{s} + \omega^2 s = 0.$$

This is just the SHM equation. Thus, in the rotating coordinate system the Foucault pendulum simply oscillates back and forth, or, from the point of view of the observer on the surface of the Earth, the plane of the pendulum rotates at a constant rate $K = \Omega \cos \lambda$.

Exercise 15.9

Fill in the missing steps between [Equations \(15.15\)](#) and [\(15.16\)](#). ■

Exercise 15.10

Show that [Equation \(15.14\)](#) leads to the same result as obtained in Worked [Example 15.5](#). ■

Exercise 15.11

Suppose that a 20 m Foucault pendulum is mounted at San Francisco, California (latitude 37.77°). Determine the precession rate of its plane of motion. Answer: $11.9^\circ/\text{h}$. ■

15.7 Application: The Tidal Force (Optional)

The tidal force is an interesting and important astronomical phenomenon. Tides are responsible for the formation of rings around large planets, tidal locking (or resonances) between bodies, the stretching of bodies falling into black holes (“spaghettization”), the frictional heating of Jupiter’s

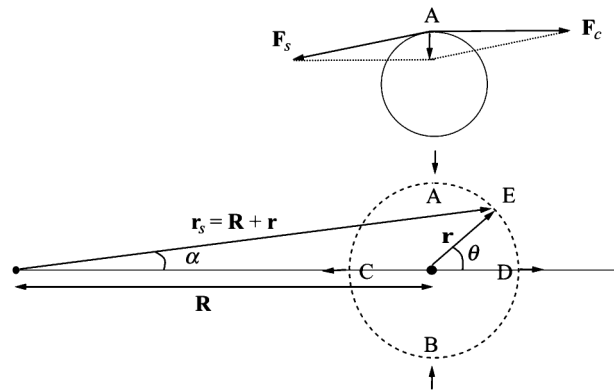


Figure 15.10 The lower figure gives the geometrical parameters. The upper figure shows that forces at point A do not cancel.

moon Io, and moonquakes, to mention a few tidal effects. In [Chapter 7](#) we noted that the ocean tides on the Earth, due primarily to the Moon, are slowing down the rotation rate of the Earth.

A particularly interesting aspect of tidal forces is that they are due to the inhomogeneity of the gravitational field acting on a body and consequently are given by an inverse cube force law.

To analyze the tidal force, we shall consider the tides raised by the Sun on the Earth's oceans. Although the lunar tides are nearly twice as big as the solar tides, the solar tides are easier to analyze (because the center of mass of the system is essentially at the center of the Sun) and the physical principles are the same. The equations we will derive are valid for both situations, although the numerical values will be different.

We begin by assuming a solid rigid Earth that is not rotating about its axis, but it is revolving about the Sun in a circle at angular velocity Ω . Thus a coordinate system fixed in the Earth has its axes pointing towards the same fixed stars as it orbits the Sun. This has been described as the same motion a cook uses in moving a frying pan in a circle while holding the handle. Of course, the Earth really is rotating and we will include that fact a little later, but the picture of the nonrotating Earth can be helpful in understanding why there are two tidal bulges. The geometry of the system is shown in the bottom part of [Figure 15.10](#).

In this coordinate system the forces acting on a particle *at the center of the Earth* are the attractive force of the Sun ($\mathbf{F}_s = -G \frac{Mm}{R^2} \hat{\mathbf{R}}$) and the (fictitious) centrifugal force ($\mathbf{F}_c = -m \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{R})$), where $\mathbf{R}$ is the vector from Sun to Earth, M is the mass of the Sun and m is the mass of the particle. These forces are equal and opposite.

The Earth is assumed to be perfectly rigid so all points in the Earth are accelerating at the same rate. That is, every part of the Earth feels the same centrifugal force. But not all points feel the same gravitational attraction toward the Sun. As indicated in the top part of [Figure 15.10](#), at point A the two forces are not perfectly antiparallel and there is a small unbalanced component directed toward the center of the globe. Such forces, due to the difference between $\mathbf{F}_s$ and $\mathbf{F}_c$ are called “tidal forces.” Similarly, at point B there is a net force toward the center. Point C is nearer to the Sun than the center of the planet, so the solar attraction is greater than the centrifugal force, resulting in a small unbalanced (tidal) component directed *away* from the center. At point D the centrifugal force is greater than the solar force so once again the tidal force is away from the center. In other words, at points A and B there is a compression and at points C and D the tidal force exerts a stretching force on the Earth. (It might be helpful to consider the behavior of a spherical gas cloud falling towards the Sun. The same forces are acting on the gas cloud, but it is

not rigid so the cloud is squeezed and stretched into a long thin shape. This would be an example of spaghettization.)

Let us now determine the tidal force at an arbitrary point on Earth's surface such as point E in the bottom part of Figure 15.10. Because the centrifugal force at the center of the Earth is equal and opposite to the Sun's gravitational attraction we can write

$$\mathbf{F}_c = \frac{GMm}{R^3} \mathbf{R}.$$

At point E the gravitational force depends on the distance to the Sun, $\mathbf{r}_s$.

$$\mathbf{F}_s = -\frac{GMm}{r_s^3} \mathbf{r}_s.$$

Therefore, the tidal force, being the difference between these, is

$$\mathbf{F}_{Tide} = \mathbf{F}_c - \mathbf{F}_s = -GMm \left(\frac{\mathbf{r}_s}{r_s^3} - \frac{\mathbf{R}}{R^3} \right).$$

Let us now determine r_s^{-3} .

Note that $r_s^2 = (\mathbf{R} + \mathbf{r}) \cdot (\mathbf{R} + \mathbf{r}) = R^2 + r^2 + 2(\mathbf{R} \cdot \mathbf{r})$. Since $R^2 \gg r^2$ we can write

$$\begin{aligned} r_s^2 &\approx R^2 \left(1 + 2 \frac{\mathbf{R} \cdot \mathbf{r}}{R^2} \right) \\ \therefore 1/r_s^3 &= (r_s^2)^{-3/2} = \left[R^2 \left(1 + 2 \frac{\mathbf{R} \cdot \mathbf{r}}{R^2} \right) \right]^{-3/2} \\ &\doteq \frac{1}{R^3} \left(1 - 3 \frac{\mathbf{R} \cdot \mathbf{r}}{R^2} \right), \end{aligned}$$

so

$$\begin{aligned} \mathbf{F}_{Tide} &= -GMm \left[\frac{\mathbf{r}_s}{R^3} \left(1 - 3 \frac{\mathbf{R} \cdot \mathbf{r}}{R^2} \right) - \frac{\mathbf{R}}{R^3} \right] \\ &= -\frac{GMm}{R^3} \left[(\mathbf{R} + \mathbf{r}) \left(1 - 3 \frac{\mathbf{R} \cdot \mathbf{r}}{R^2} \right) - \mathbf{R} \right] \\ &= -\frac{GMm}{R^3} \left(\mathbf{R} + \mathbf{r} - 3 \frac{\mathbf{R} \cdot \mathbf{r}}{R^2} (\mathbf{R} + \mathbf{r}) - \mathbf{R} \right). \end{aligned}$$

But $\mathbf{R} + \mathbf{r} \approx \mathbf{R}$, so

$$\mathbf{F}_{Tide} = -\frac{GMm}{R^3} \left(\mathbf{r} - 3\mathbf{R} \frac{\mathbf{R} \cdot \mathbf{r}}{R^2} \right).$$

Observe that the main contributions of $\mathbf{F}_s$ and $\mathbf{F}_c$ have cancelled, leaving only their difference. At points A and B we have $\mathbf{R} \perp \mathbf{r}$ so $\mathbf{F}_{Tide} = -\frac{GMm}{R^3} \mathbf{r}$ which is pointing towards the center of the Earth, i.e., a compression. At point C the angle θ is π so and we obtain $\mathbf{F}_{Tide} = +2\frac{GMm}{R^3} \mathbf{r}$ which points away from the surface. A similar expression is obtained at D.

The tidal forces at various points on the surface of the globe are shown in Figure 15.11.

At a general point, such as E in Figure 15.11, the tidal force can be resolved into components perpendicular to the surface (along $\hat{\mathbf{r}}$) and parallel to the surface (along $\hat{\boldsymbol{\theta}}$). The components perpendicular to the surface (such as at ABCD) simply add to the Earth's gravitational force and are not of interest. But the tangential components can affect particles on the Earth's surface that are not rigidly attached to it and these are the forces that produce the tides in the ocean and atmosphere.